

Woldia University

College Of business and economics

Department of economics

Labor economics

Course code:-Econ 3101

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Chapter one: Introduction

- *Labour* is the amount of physical, mental, and social effort used to produce goods and services in an economy.
- *Labour* is the measure of the work done by human beings.
- *Labor economics*:-examines how labor markets function.
- *Labor economics* is the application of economic theory and methods to the study of labor markets.
- *Labour economics* is the study of the market for one particular commodity in the economy: *Labour Services*

Cont....

- *More specifically, labour economics is primarily concerned with the behaviour of employers and employees in response to the general incentives of :-wages, prices, profits, and non pecuniary .*
- *Labour economics yields information and develops analytical tools that may be useful in making personal and managerial decisions relevant to labour markets*

Labor Market and the Economy

- *In economics **market** is a concept, rather than a **physical entity**.*
- *It is the interaction of **buyers and sellers**, in which a price is determined and a product or service is exchanged.*
- ***Product markets** where the output of firms sold to households*
- ***Factor markets** where the inputs to production are sold by households to firms.*
- *How ever ,The market that has the responsibility of allocating workers to jobs and coordinating employment decisions is the **labour market**.*

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- *Labor market. :- A place where worker and employer interact each other*
- *Employer compete to hire the best.*
- *While workers compete for the best satisfying job.*
- *As workers search for jobs and firms search for workers, these conflicting desires are "balanced out" and the labour market reaches an equilibrium.*

The actors (participants) in the labor market

Households (workers)

- When to enter the labour market
- How much education or training
- Which occupation or industry
- Number of hours to work
- Whether to quit or retire

Firms choose

- Number of workers to hire
- Hours of work
- When to layoff or close plant
- Pension or retirement policy

❖ *Government establishes the environment by:*

- *providing training*
 - *unemployment insurance*
 - *workers compensation pensions*
 - *laws to protect workers and firms*
- *The interactions between these players in the labour market determines:*
- ✓ *The equilibrium price: which is the which is the **wage that workers receive.***
 - ✓ *The equilibrium quantity: which is the **amount of work that people do in the economy.***

Determinants labour supply and demand

- *The determinants of the labor supply*
 - *Demographics*
 - *Immigration (in) and emigration policies (out) (perhaps a brain drain?),*
 - *The labor force participation decision,*
 - *The hours of work decision (including overtime and moonlighting),*
 - *Education and training (human capital decisions), and*
 - *The disincentive effects of income maintenance and*
 - *Unemployment insurance policies.*

Cont..

- *The determinants of the labor demand :-*

- ✓ *Cost of hiring labour*
- ✓ *Wages / salaries*
- ✓ *National Insurance contributions*
- ✓ *Pension contributions*
- ✓ *Administration costs associated with tax payments and adhering to employment laws and regulations*

❖ *Since labour demand is a **derived demand** (derived from the demand for the firm's output), it is also influenced by factors such as **free trade, global competition and technological change.***

Basic concepts

- *Labour union* :-an organization of workers formed for the purpose of advancing its member of interest in respect to wage, benefits, working condition under a collective bargaining agreement with employer.
- *Wage*: a compensation paid to employees for work by organization during a period of time.

Characteristics of labor

- *Labour is rented and not bought and sold
(intangible)*
- *Non monetary aspects*
- *Labor is perishable commodity*
- *Labor demand is a derived demand*
- *Labor is an active factor of production*

The Importance of the Labour Market

- *The lion share of the national income flows to workers in the form of wages and salaries.*
- *The obvious implication is that quantitatively, labour is our most important economic resource.*
- *Helps to understand the distribution of income in society – who earns what and why- we need to understand labour markets, how wages are determined, etc.*

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- *Like every commodity labour has a number of special features that must be taken into account in a more in depth analysis.*
- *Each of these features has important implications .*
- *Example:-wage reflects*
 - ✓ *Investment in education*
 - ✓ *Compensation for risk*

Scope of labour economics

- ✓ *Manpower planning :-the total supply of persons available and fitted for service.*
- ✓ *Labour organization*
- ✓ *Wage and employment theory*
- ✓ *Collective bargaining theory*
- ✓ *Practice of social security and welfare*
- ✓ *Size and composition of the labour force and labour market.*

Chapter two

Labour supply

Definition of Labour supply

- *Labor supply is the term used to describe the human input into the production process*
- *The **supply of labour** is defined as the **amount of labour**, measured in **person-hours**, offered for hire during a given **time-period**.*
- *The subject matter of labour supply is how individuals earn a living by selling labour services through the labour market.*
- *Utility maximizing workers sell labour in exchange of compensation*

Measuring Labour Supply

- *1st Head count” or the number of people are sometimes used to measure labor supply.*
 - *It does not convey the notion of engagement in economically productive activities.*
 - *It might include even under-age children.*
- *2nd working age category ; - Counting only people of working age category improves the measurement of labor supply.*
 - *The working age category excludes under age children, retired people and over-age adults.*

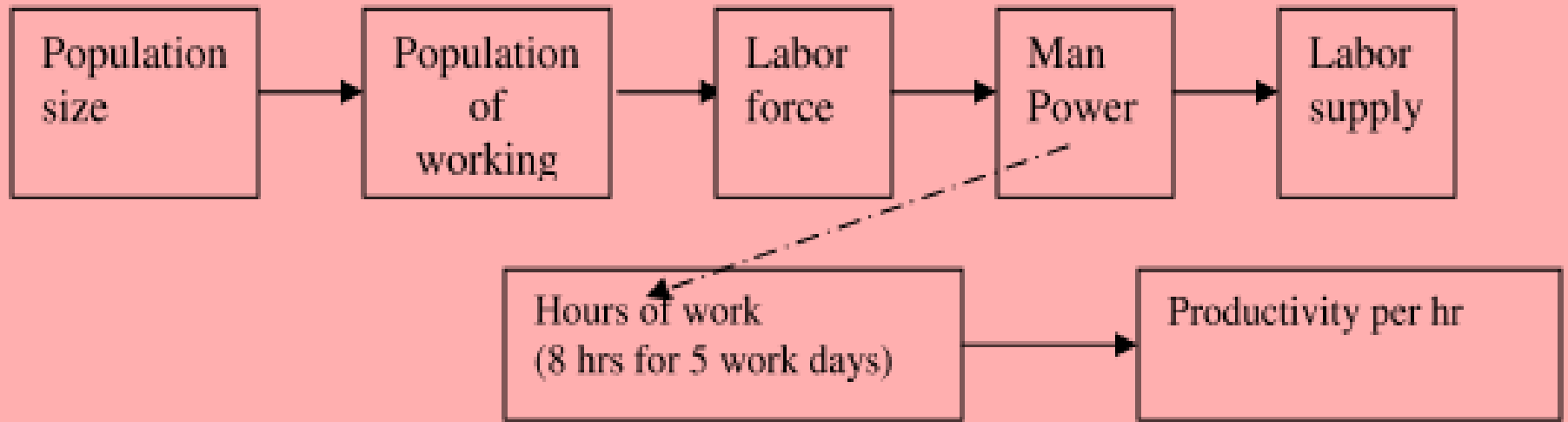
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- *But this classification has defective in the sense that it includes even currently inactive groups like disabled people and college student etc.*
- *3rd Labour force is an improvement in measuring labour supply over the crude head count approach.*
- *But the labour force does not bring clearly the relation (linkage) with output.*

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- *Even some think that the attributes of “productiveness” in labour supply is conveyed by the notion of skill, education, experience, etc. which are contained in the concept of manpower.*
- ***Manpower** contains the essential elements of labour supply such as education, skill, training, experience, etc. and hence, it is a better approximation of labour supply.*

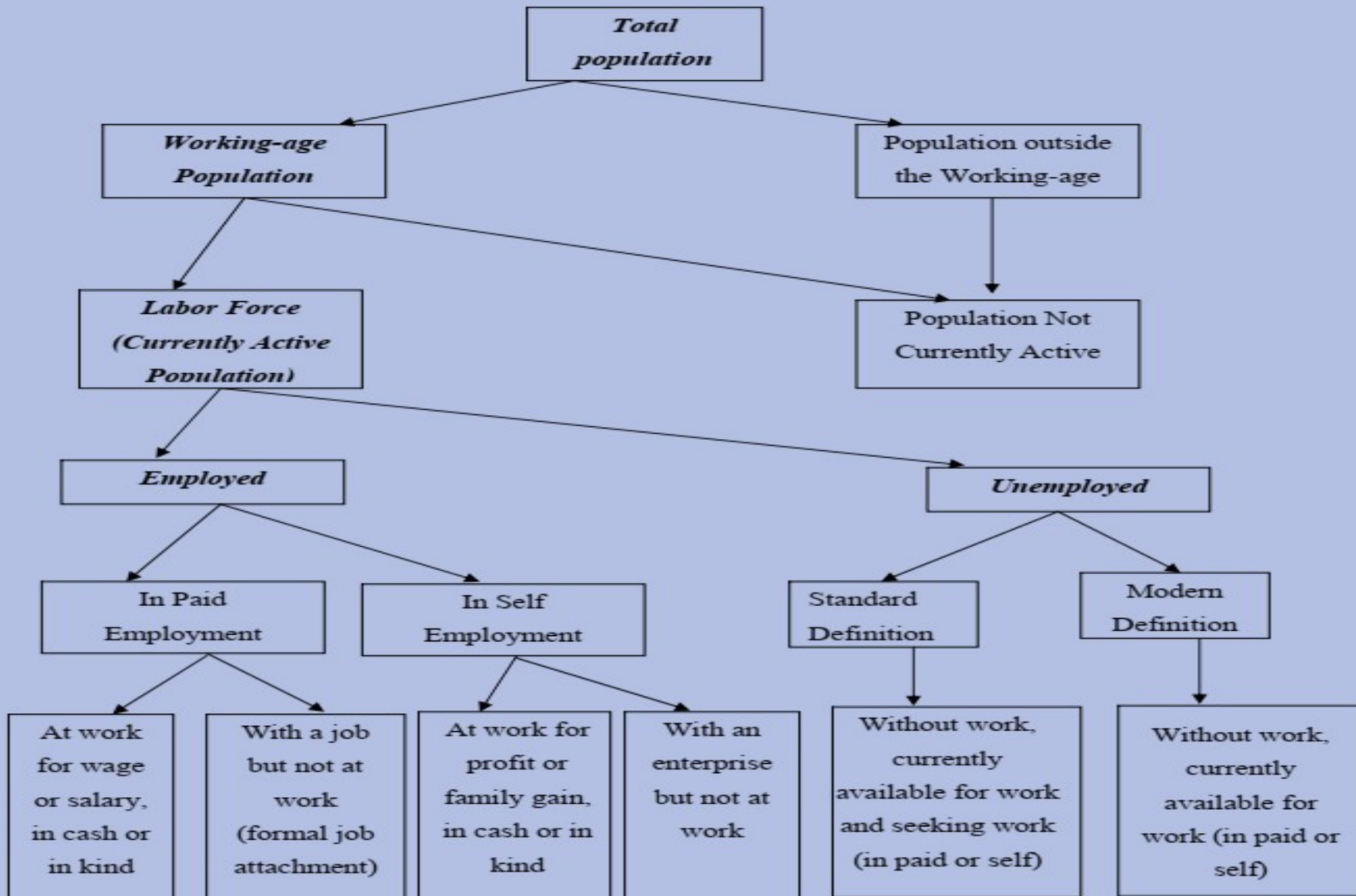
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Note that:

- *Whatever measure of labor supply is used, the reference period (whether a week, or a month, or a year ...) is an important dimension that should be disclosed.*
- *Whatever measure of labor supply is used, the shorter the reference period, the more credible the statistic is.*

Labour Supply Framework



labor force estimation procedure

- *First:* the working age population is defined and measured. working age ranges from years of 14/ 15 to 65.
- *Second:* the economically active population of working age or labor force is defined and measured.
- *The labor force is grouped into two: employed and unemployed.*
- *Labor force = employed + unemployed working age population*
- *Third:* identification and measurement of the population that is economically inactive population.
- *Those “not in the labour force” constitutes economically inactive population.*
- *The not economically active labour force include- full time students, home makers, retired or old age pensioners, and inmates of institutions such as prisons*

Difficulties in Estimating Labour Supply

- *The estimates provided by the approach commonly adopted (the labor force approach), like the concept and measurement of labor supply, cause many problems*
- *The concept of labour supply encompasses notions which go beyond the number of people in the labour force like hours of work, effort on the job,*
- *The reference period also affects the estimated labour supply. For instance, census of short reference period conducted during the slack seasons underestimate labour supply while it overestimates if it is conducted in peak seasons.*

- *The measurement of labor supply is difficult when the frequency of economic status change is high; i.e., It overestimates or underestimates labor supply if large proportions of the population are in and out of the labor force count too frequently.*
- *It is difficult to identify and measure comprehensively the level of economic activities. This problem is more serious in the least/less developed countries where a large proportion of production is non-marketable.*

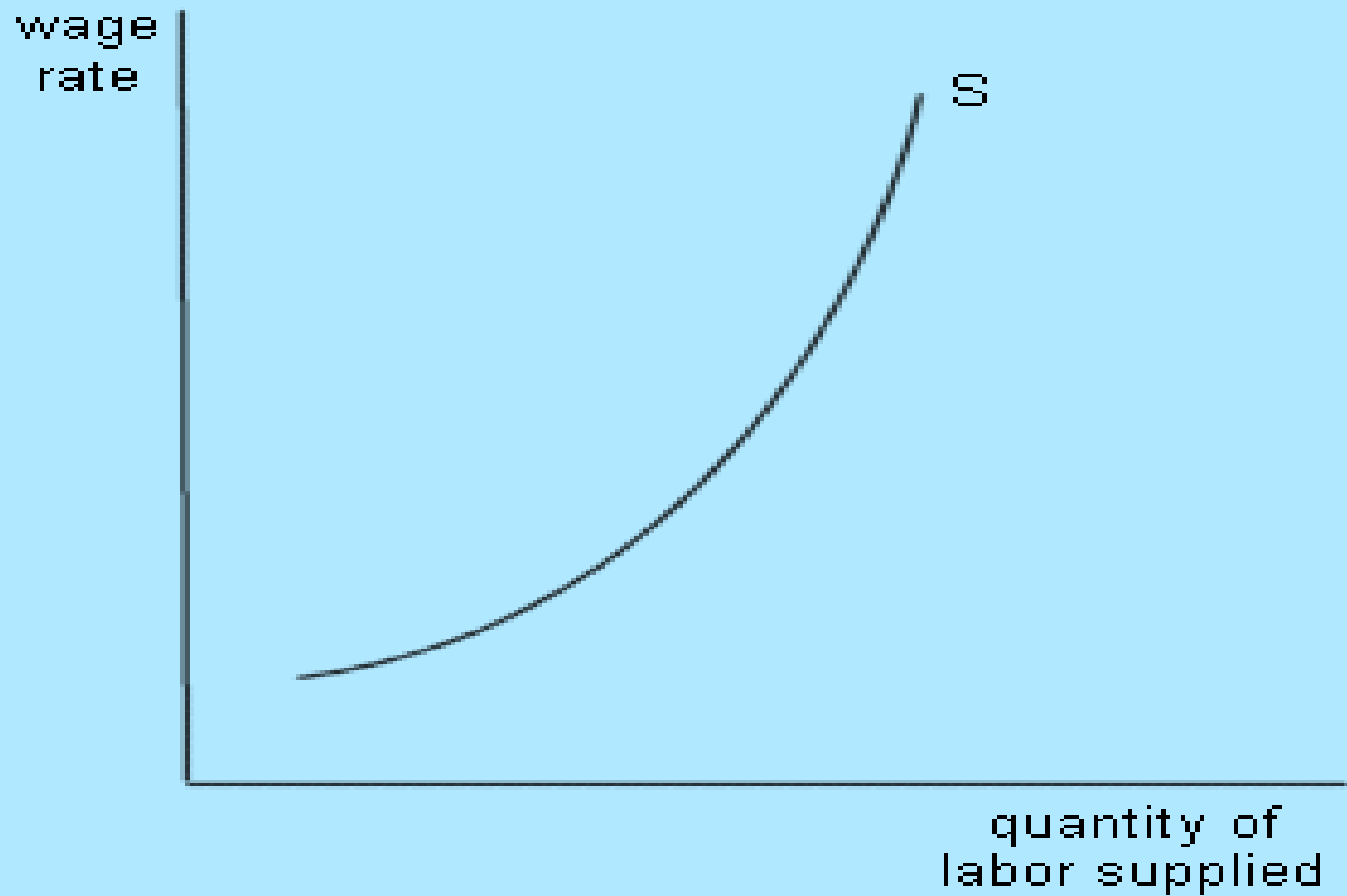
The aggregate of labor services depends on:

1. **The size and demographic** compositions of the population, which in turn depends on births, deaths, and net immigrations;
2. **The labor force participation rate:** i.e., the percentage of working age population which is actually working or seeking work. Though supply of labor depends primarily on the size of population of the country, we must also remember that not all people are available for work.
3. **The number of hours worked per week or year:-**Workers can give continuous and sustained work only if the working hours are reasonable. Unreasonable long working hours means fatigue and inefficient work.
4. **The quality of the labor force**
5. **Sex rate:** The ratio of males to females in the working age group
6. **Age of retirement:** Workers retire at a certain age.
7. **Social attitude:-** In certain occupations, sex and caste are taken into consideration. This restricts the mobility and availability of labor.
8. **Wage level:-** Labor supply is related to the wage level in a country. High wage level contracts the supply of labour and lower wage level forces the worker to work more due to economic compulsions.
9. **Potential desire to work**
10. **Training period:-** If labor is educated and trained, then the period of education and training will be an important factor in influencing supply of labor.

The Market supply curve

- *The market labor supply curve is expected to be upward sloping because an increase in the wage in a particular labor market will:*
 - ✓ *Cause some workers in this market to work additional hours,*
 - ✓ *Induce some workers to shift from other labor markets to this relatively good alternative employment, and*
 - ✓ *Will cause some individuals who are not currently in the labor force to enter this market.*

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Shifts in market labor supply curve

- *A change in salary will lead to a movement along labour demand or labour supply curves, but it will not shift those curves.*
 - *However, other events like:- changes in non labour income ,changing wages in other markets, changes in worker tastes and preferences, Number of worker, Required education etc.*
- ✓ *those outlined above will cause either the demand or the supply of labour to shift, and thus will move the labour market to a new equilibrium salary and quantity*

Activity Rate (Participation Rate)

- *Activity rate* measures the extent of (actual and potential) involvement in economic activities
- The *participation* in the production and distribution of goods and services.
- The activity status of an individual is determined by what the individual is doing during the reference period.
- Only if he is engaged in the production and distribution of economic goods & services will he be counted as economically active or currently active population or the labor force

Cont....

$$\text{Activity Rate} = \frac{\text{labor force}}{\text{working age population}} \times 100$$

Labor force = employed + unemployed

= actual labor force

Factors Affecting Participation Rates

- ***Job attachment:*** *Evidences show that extent of job attachment is not evenly distributed in the population*
- ✓ *Primary = Participation rate tend to high.*
- ✓ *Secondary worker = Participation rate tend to low.*
- ***Business cycle***
- ✓ *At recession = Participation rate tend to increase and move counter – cyclically and hence tend to stabilize employment.*
- ***Other factors***
- ✓ *Low wage income,*
- ✓ *Low level of development of non-wage income sources*
- ✓ *The inadequacy of non-traditional social security systems*

2.2 The Short Run Theory of Labour Supply

- The short run labor supply is the period in which the *size of the population of working age* and the *skills they possess* are held constant
- The long run is the period over which these are allowed to *vary*.
- The objective of the labor supply model is to represent an individual's choice of *hours worked* given *her market opportunities* and the value she places on her *nonmarket time*.
- The hours-of-work aspect of labor supply has a variety of dimensions including hours per day, days per week, and weeks per year.

The Work-Leisure Decision Model

- *An individual having a fixed amount of time available and must decide how that time should be allocated among work (labor market activity) and leisure (non- labor market activity).*

$$T = H + L, \quad H = T - L \quad \text{OR} \quad L = T - H$$

- ***Work** is time devoted to paying job*
- ***Leisure** is used here in a broad sense to include all kinds of activities for which the person does not get paid.*

Assumptions Underlying the Model

- ✓ *There are only two possible uses of time: labour and leisure.*
- ✓ *Each individual selects the combination of hours of work and leisure that maximizes his or her level of satisfaction (utility).*
- *Given these assumptions and the basic economic concepts of opportunity cost and choice,*
- ✓ *For individuals who are working, the opportunity cost of an additional hour of leisure time is the (market) wage rate.*
- ✓ *An individual will choose not to work if the value he/she places on leisure time exceeds the market wage.*

Cont..

- *Two sets of information are necessary to determine the optimal distribution of an individual's time between work and leisure.*
- ✓ *1st. Subjective (psychological) information concerning the individual's work-leisure preferences. This information is embodied in indifference curves.*
- ✓ *2nd. Objective (market) information, which is reflected in budget constraints.*

Utility Function

- *Shows the various combinations of real income and leisure time which will yield some specific level of utility or satisfaction to the individual.*

- ✓ $U = f(Y, L)$

- ✓ *U is an index.*

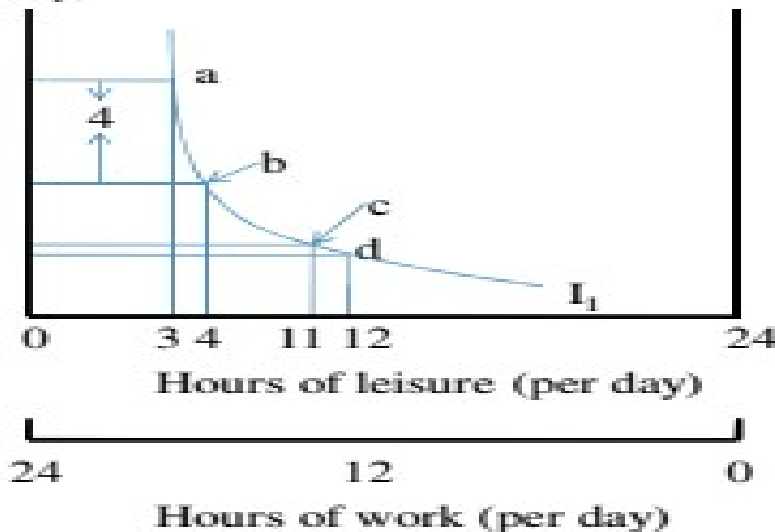
- ✓ *The higher is U, the happier is the person.*

.All rights rese

Indifference curves

- As applied to the work leisure decision, an indifference curve shows the various combinations of real income and leisure time which will yield some specific level of utility or satisfaction to the individual.*

Income (per day)



Properties of Indifference

Negative slope

Convex to origin

Indifference map

Different W-L Preference

- *The movement along the indifference curve involves no change in total utility, the loss is exactly compensated by the gain.*

Properties of Indifference curve

- *Negative slope*

- ✓ *It is because both are source of utility*
- ✓ *When an individual get more of one of them (leisure), some of the other (real income) must be surrendered to maintain the same level of utility*

- *Convex to the origin*

- ✓ *The slope of the curve decrease when we go down on the curve*
- ✓ *Slope of indifference curve shows individuals subjective willingness to substitute between leisure and income*

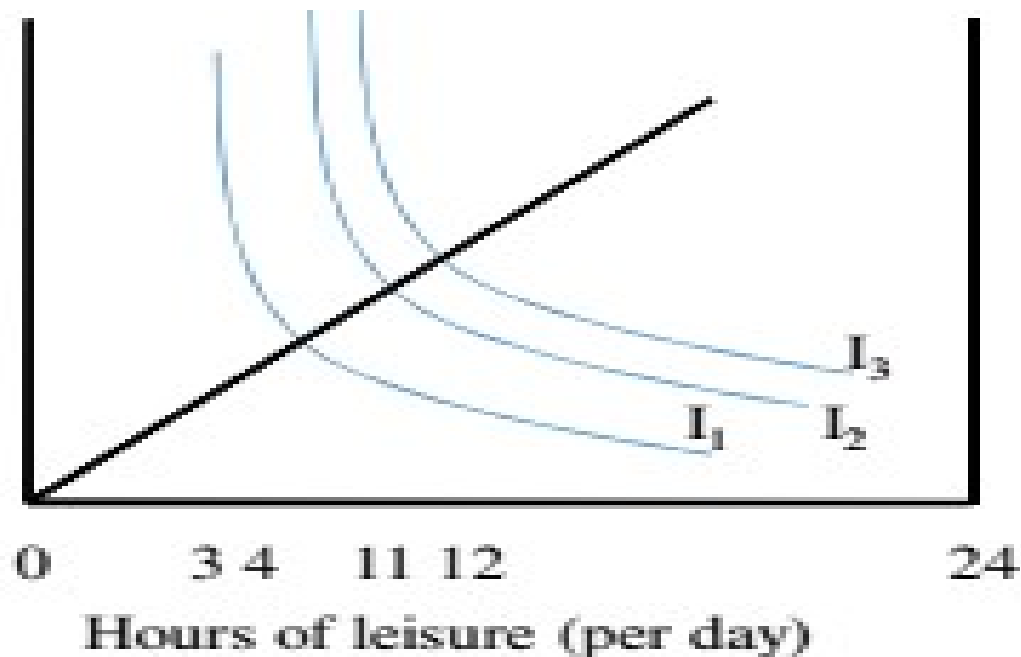
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- *The convexity reflects the notion that an individual becomes increasingly reluctant to give income as it becomes increasingly scarce. Compare ab with cd on our W - L Indifference curve.*
- *Technically speaking the slope of W - L indifference curve is measured by the marginal rate of substitution of leisure for income ($MRS_{L,Y}$)*
- *$MRS_{L,Y}$ is the amount income one must give up to compensate for the gain of 1 more unit of (hour) of leisure.*
- *The basic point is as one moves down on the W - L indifference curve, $MRS_{L,Y}$ or the slope of the indifference curve declines.*

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■ *Indifference map*

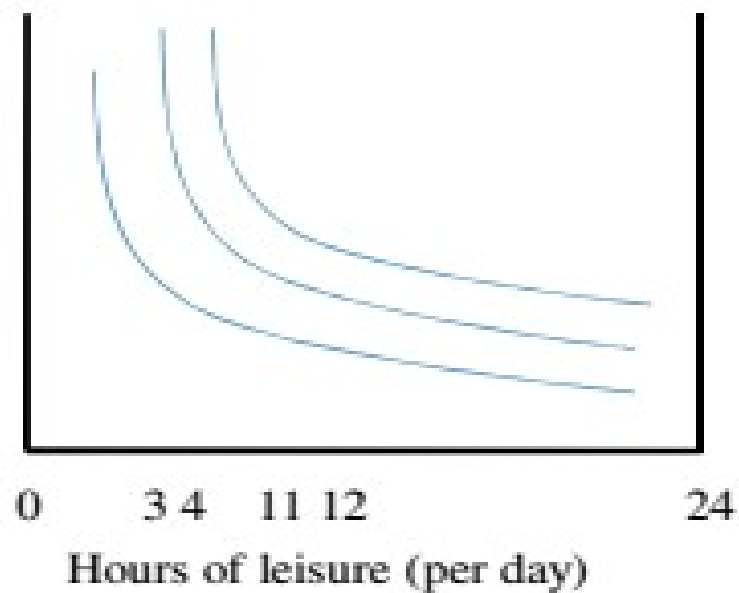
- ✓ *Curves further from the origin represent larger utility.*



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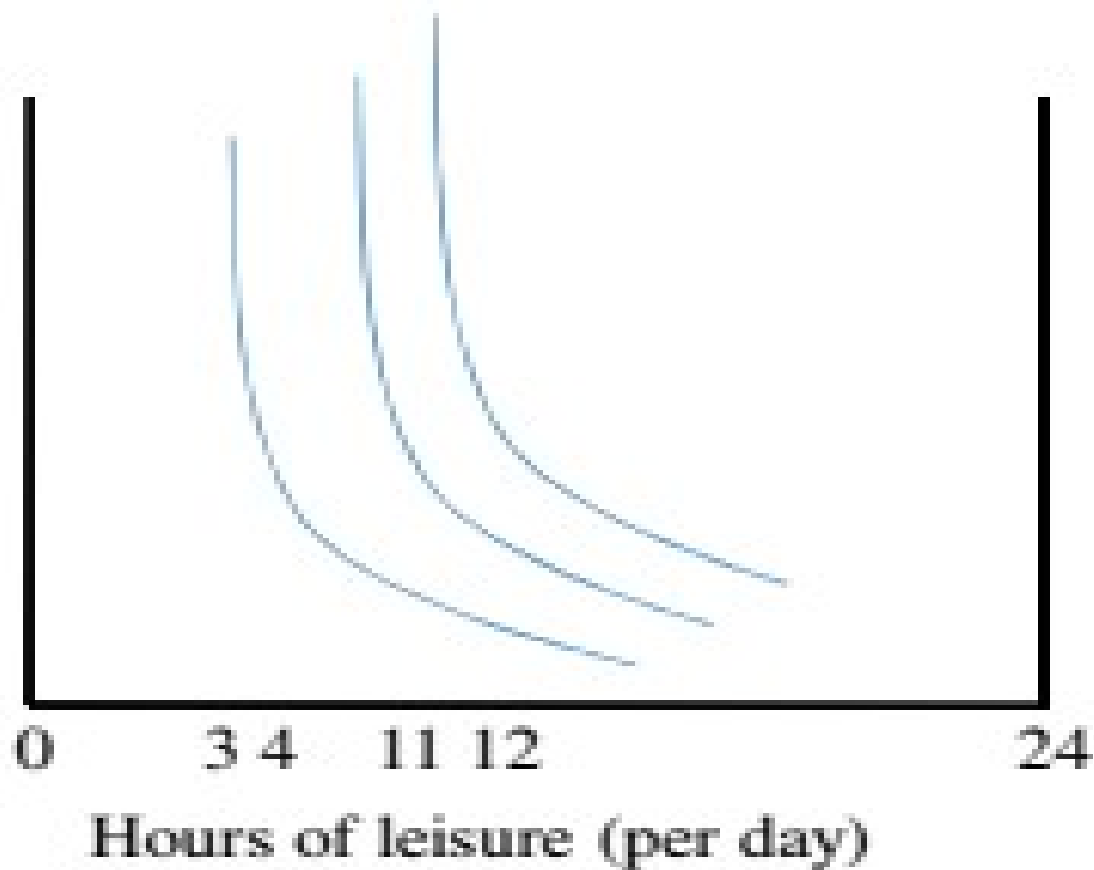
■ *Different work-Leisure preference*

- ✓ *Indifference curves for workaholic individual; who places a low value for leisure*



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- *Indifference curves for “leisure lover” individual; who puts high value for leisure*



Budget constraint

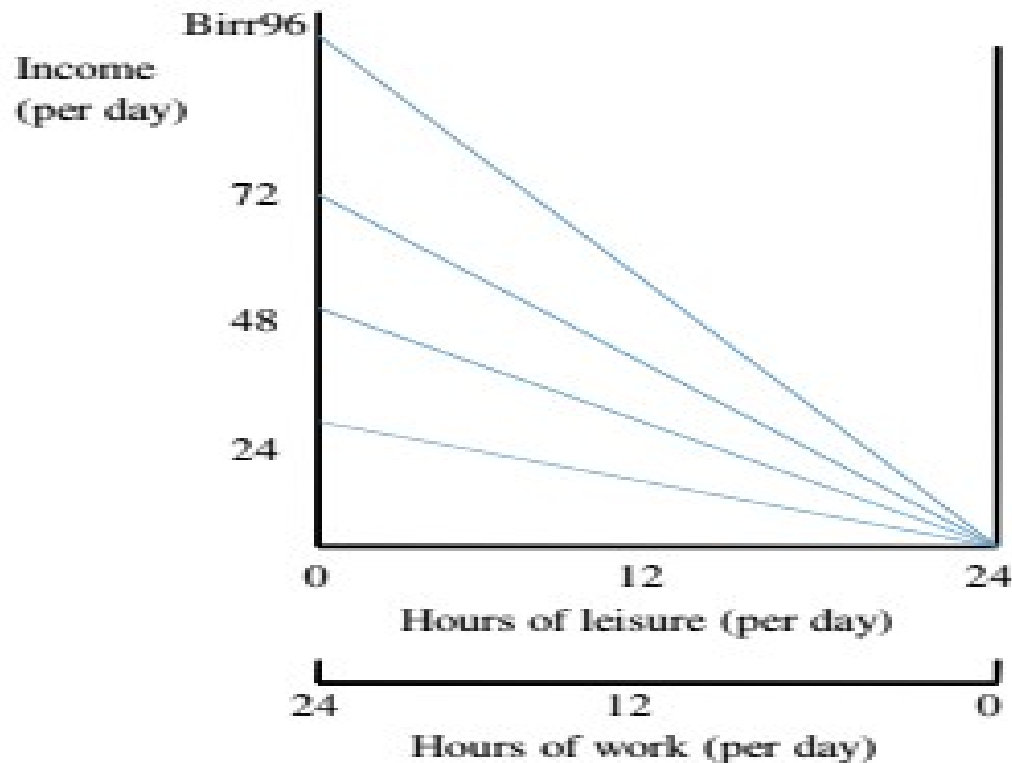
- *Indifference map shows unlimited number of indifference curves representing infinitely large level of utility.*
- *But, workers are constrained by budget (the amount of income) to reach at those indifference curves representing infinitely large level of utility.*
- *Thus, it is only possible to enjoy the level of satisfaction represented by those indifference curves which can be attained under the budget constraint.*

Budget constraint

- *It defines the worker's opportunity set, indicating all of the income–leisure combinations the worker can afford.*
- *The budget (wage) constraint line is the line which shows all the various combinations of income and leisure which a worker might realize or obtain, given the wage rate.*
- *The following assumptions are important:-*
 - ✓ *An individual's only source of money income is from work.*
 - ✓ *no accumulated savings to draw on, no borrowings*
 - ✓ *An individual is a wage rate taker.*
 - ✓ *Wage rate is fixed and an individual can't alter.*

Cont..

- The assumption that wage rate is given for the individual enable us to draw a linear budget constraint (strait line).*



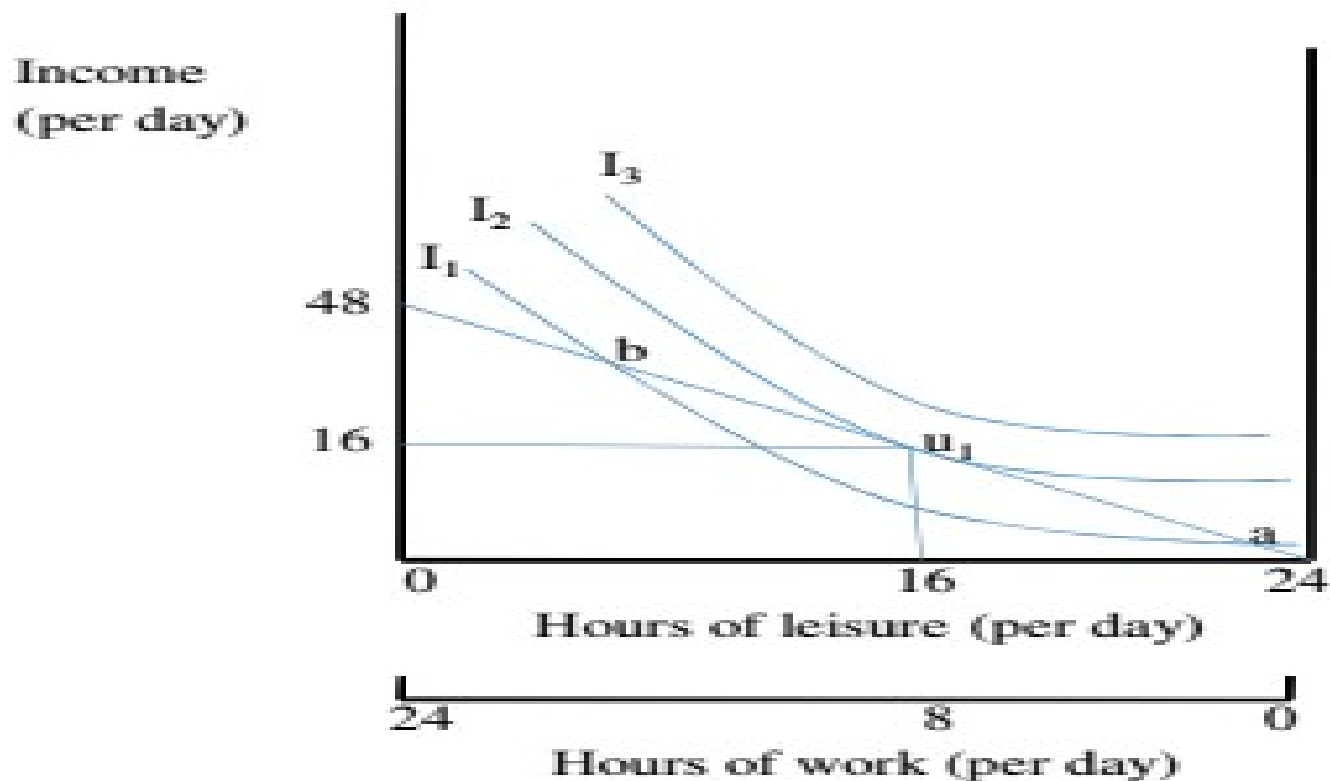
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- *The budget line intersecting the y-axis at birr 24 is drawn for a wage rate level of birr 1.*
- *The individual can get birr 24 if he or she allocates all the 24 hours for work.*
- *With an increase in wage, the budget line rotates to the right.*
- *The highest budget line intersecting the y-axis at birr 96 is drawn with a wage rate of birr 4.*

Utility maximization of workers

- *The utility maximizing position (the optimal distribution of time between work and leisure can be determined by bringing indifference curves and the budget line together on one plane.*
- *Recall that the further an indifference curve is from the origin, the greater the person's total utility.*
- *Therefore an individual will maximize total utility by attaining the highest possible indifference curve.*

Cont..



- ✓ Given birr 2 wage rate, one realizes the highest attainable level of utility at point u_1 on the utility maximizing figure above.
- ✓ At this point; the budget line is tangent to the indifference curve (I_2).
- ✓ The individual chooses to work 8 hours, earning a daily income of birr 16 and enjoying 16 hours of leisure.

Cont..

- *At the optimal (utility maximizing point) the individual and the market are in agreement as to the relative worth of leisure and real income.*
- *The individual's preference to substitute work and leisure embodied in the slope of the indifference curve is equal with the exchange rate of one leisure hour for work in the labour market (wage rate) embodied in the slope of the budget line.*
- *In other words; the optimal work-leisure distribution of time is attained when $MRS_{L,Y}$ is equal with the wage rate.*

Cont..

➤ *Why point a is not optimal point?*

- *At point a, the slope of the indifference curve is less than the slope of the budget line.*
- *The individual give less value for leisure than what the market is giving for an hour of leisure.*
- *Say; if the slope of the indifference curve is 1 and the slope of the budget line 2.*
- *This means; an hour of leisure worth only birr 1 for the individual but the individual can actually get birr 2 by giving up that hour of leisure which worth only birr 1.*
- *This makes the individual to feel under-employed and wants to work more and move up on the budget line from point a to point u1.*

Cont..

✓ *Why not point b?*

- *At point b, the slope of the indifference curve is larger than the slope of the budget line*
- *The individual give high value for leisure than what the market is giving for an hour of leisure.*
- *For illustration; if the slope of the indifference curve is 2 and the slope of the budget line is 1, an hour of leisure worth birr 2 for the individual but the individual can actually get only birr 1 by giving up that hour of leisure which worth birr 2.*
- *This makes the individual to feel over-employed and wants to work less and move down on the budget line from point a to point u1*

Cont..

For workers who stands to maximize utility:-

- **Objective function** :-Utility= $Y^a L^b$

- **Subjective function**:-

- ✓ $H + L = T, \quad H = T - L$

- ✓ $Y = WH$,total earning

- ✓ $Y = W (T - L)$

- ✓ $Y = WT - WL$

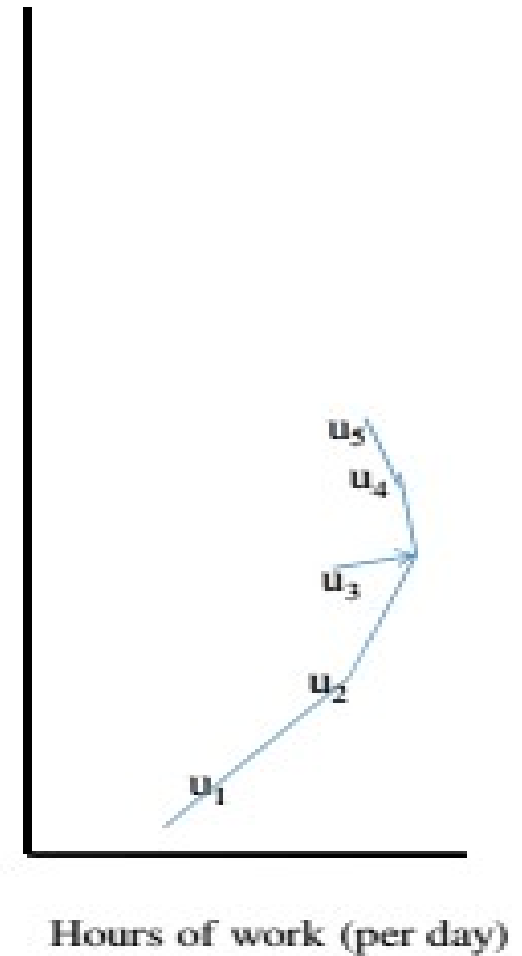
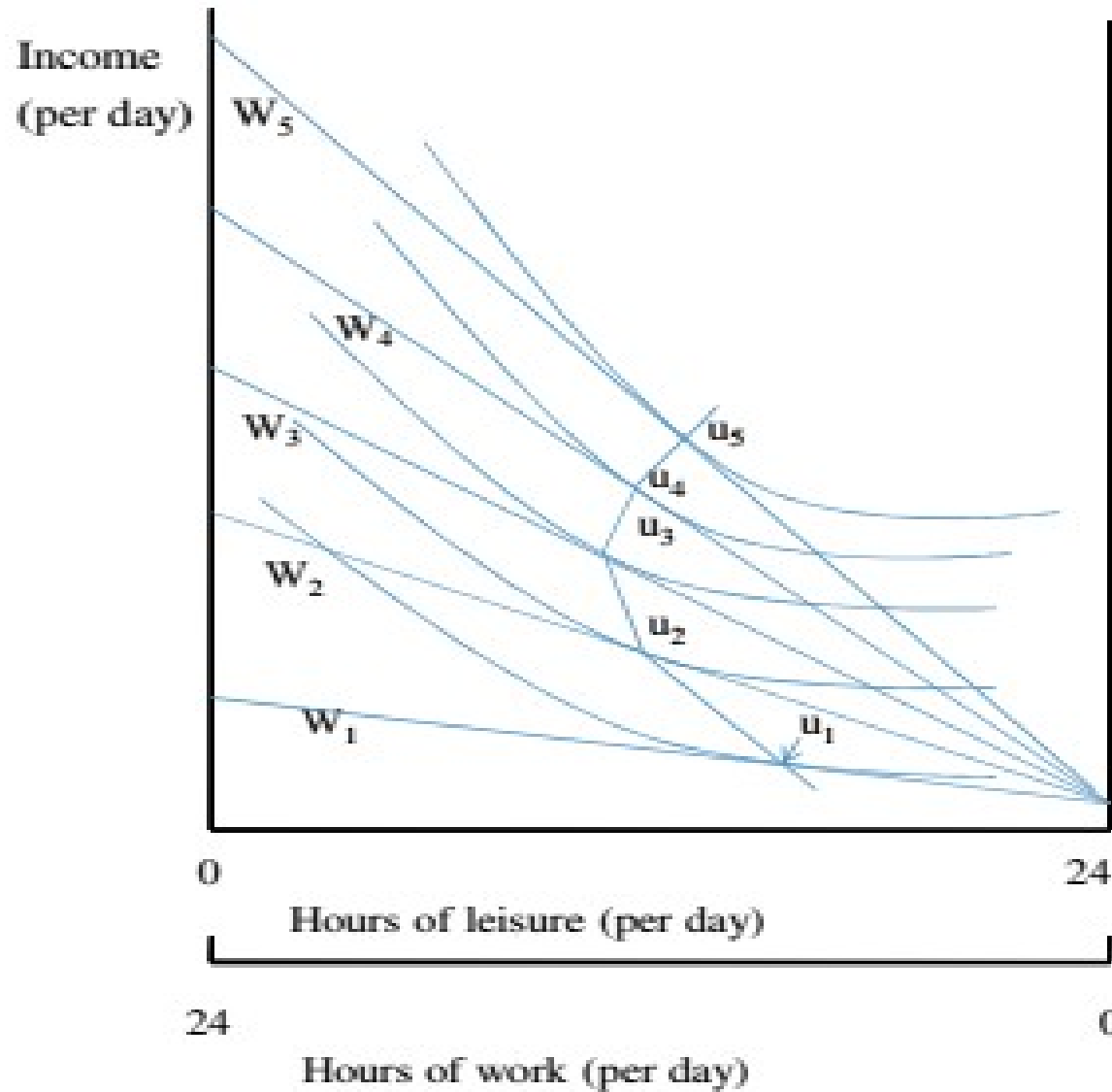
- Suppose $a=b=0.5$ $w = 10$ $T = 16$

- Then know labourer can determine the utility maximize level of L and H using Lagrangian method.

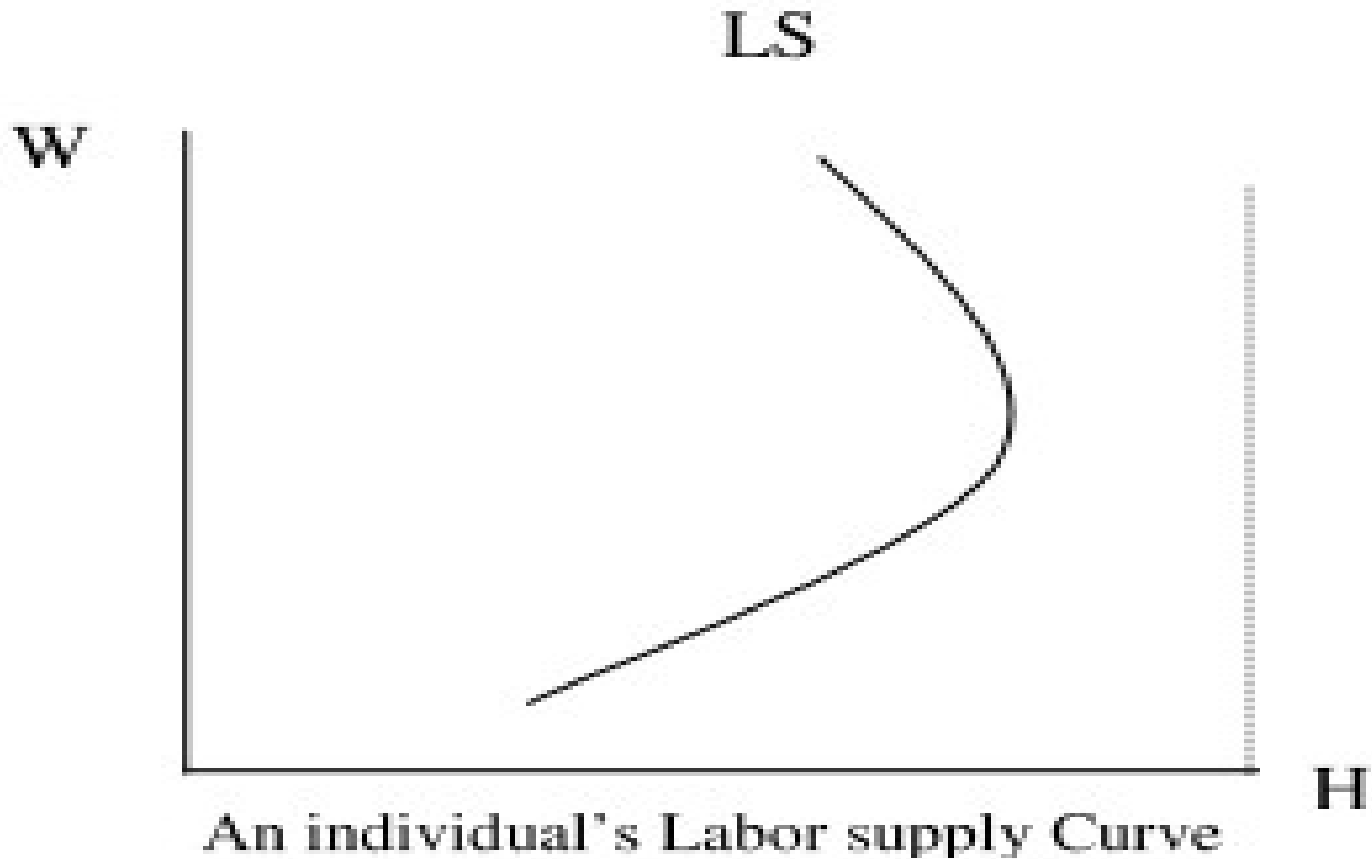
Wage rate changes; Income and substitution effects

- *Will an individual choose to work more or fewer hours as the wage rate changes?*
- ✓ *At lower levels of wage rate, wage rate increment makes an individual to work more.*
- ✓ *But after some level, an increase in wage rate makes an individual to work less.*
- ✓ *This is because after some level of wage increment, individuals get large amount income by doing few hours and value leisure more than income.*
- ✓ *Thus, when they see the wage increasing they feel it is better to work less hours.*

Cont..



The Backward Bending Labor supply curve



Why is it backward bending?

- ✓ *Income and substitution effects of the wage rate increment can explain this.*
- *Substitution effect*
 - *As the wage rate rises , the opportunity cost of leisure time rises.*
 - *In response to this higher wage rate, the individual consume less leisure time*
 - *spend more time at work (even at the same level of satisfaction)*
 - *Substitute work for leisure.*
 - *This is the substitution effect resulting from a higher wage.*

Cont..

✓ *Income effect*

- *The wage rate increment has also another component of effect; income effect*
- *When the wage increases, the income that the individual can get from his work increases.*
- *When income of a given individual increases, he/ she demand more of normal goods*
- *If we assume leisure to be normal good, the individual will buy more of leisure.*
- *But, how does the individual buy more of leisure? The answer is by working less.*
- *This is what we call income effect of wage rate increment.*

Net effect of income and substitution effect

1) An increase in wages

- Substitution effect: Leisure becomes more expensive—its price increased, so people buy less leisure, substituting work for leisure. Hours of work increase.
- Income effect: An increase in wages causes income to rise and people demand more leisure. Hours of work decrease.

The overall all effect of an increase in wages on labor supply is ambiguous. The substitution effect and income effect are moving in opposite directions.

- If the substitution effect dominates, hours of work increase from an increase in wages.
 - Labor supply curve slopes upward.
- If the income effect dominates, hours of work fall from an increase in wages.
 - Labor supply curve slopes downward
- If the substitution effect exactly offsets the income effect, hours of work remain unchanged.

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2) A decrease in wages

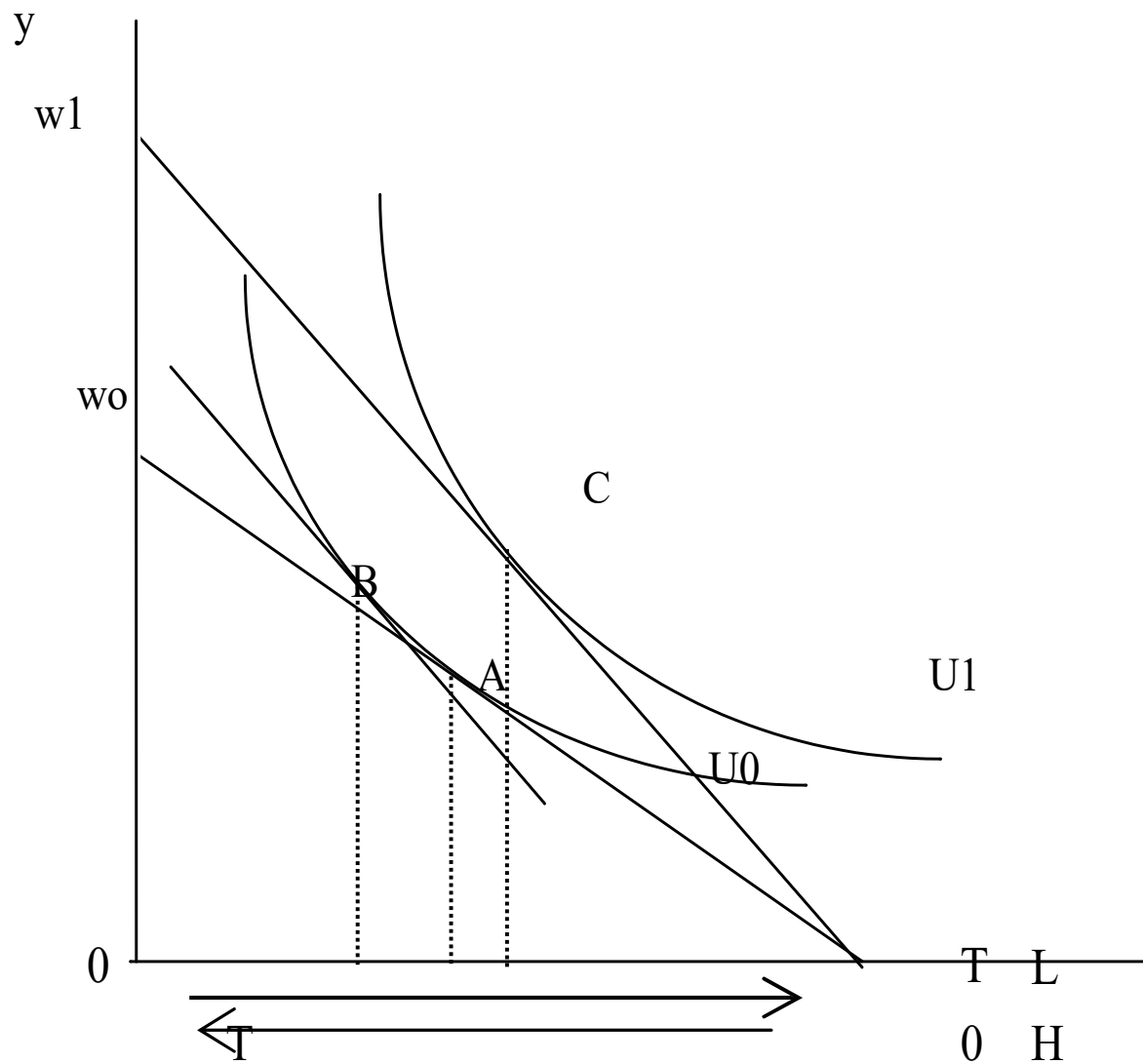
- Substitution effect: Leisure becomes less expensive—its price decreased, so people buy more leisure, substituting leisure for work. Hours of work decrease.
- Income effect: A decrease in wages causes income to fall and people demand less leisure. Hours of work increase.

The overall all effect of a decrease in wages on labor supply is ambiguous. The substitution effect and income effect are moving in opposite directions.

- If the substitution effect dominates, hours of work decrease from a wage decrease.
 - Labor supply curve slopes upward.
- If the income effect dominates, hours of work increase from a wage decrease.
 - Labor supply curve slopes downward
- If the substitution effect exactly offsets the income effect, hours of work remain unchanged.

Labor supply curve is vertical

The Substitution and Income Effects of a Wage rise



Reservation wage

- *Reservation wage* refers to the least amount of money a person would consider working for in a particular position or type of employment.
- In labour economics, the reservation wage is *the lowest wage rate at which a worker would be willing to accept a particular type of job.*
- A job offer involving the same type of work and the same working conditions, but at a lower wage rate, would be rejected by the worker.
- The absolute value of the slope of the indifference curve at the point corresponding to zero hours of work is the *individual's reservation wage* .

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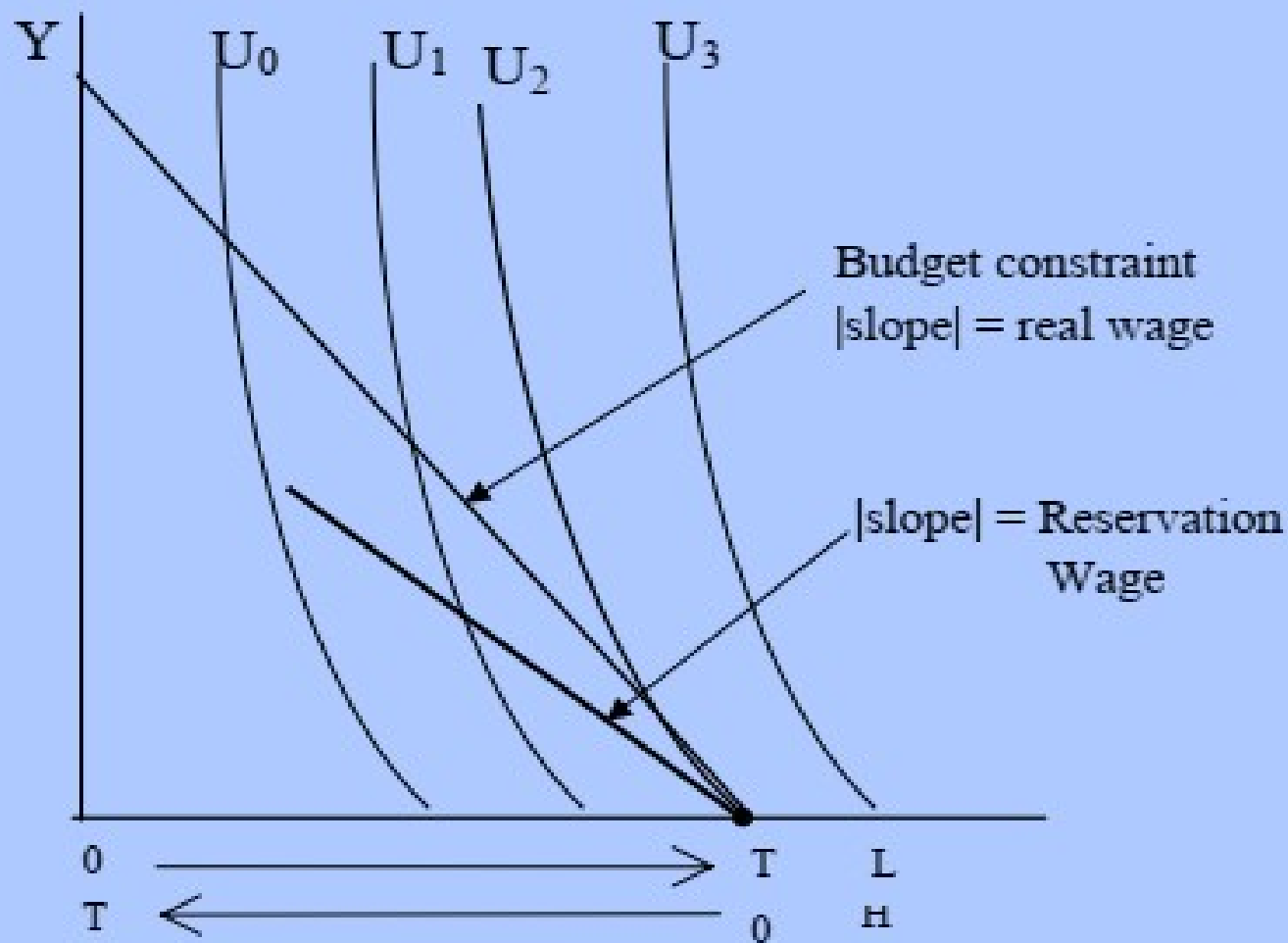


Figure 2.11: Reservation Wage Less Than Market Wage, Positive Quantity of Labor Supplied

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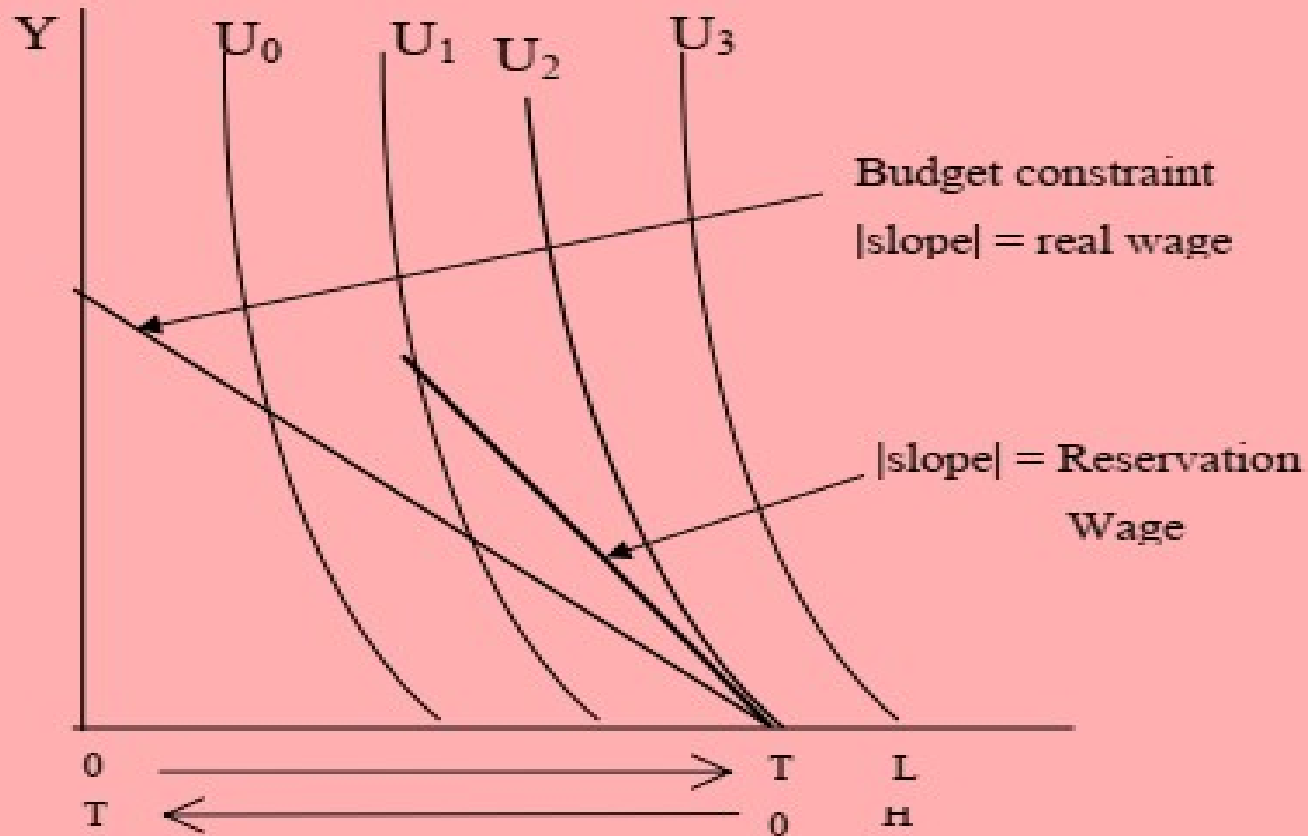


Figure 2.12: Reservation Wage Greater Than Market Wage, Zero Quantity of Labor Supplied

Effect of Payment System on Labor Supply

- *Firms manipulate the level and structure of the wages they offer workers in order to induce them to supply the desired quantity and quality of labor.*
- *For example, firms frequently require employees to work longer than normal hours and induce them to do so by **paying a higher hourly rate for all hours worked beyond normal hours.***
- *Such a premium for overtime has the effect of introducing a kink into the budget constraint facing the employee.*

Cont....

- *Again firms need to ensure that hourly paid workers deliver work of the appropriate quality.*
- *One mechanism for achieving this is the offer of an hourly wage above the market clearing level (**efficiency wage theory**).*
- *Another way is to monitor workers performance and sack those who do not perform to the required standard.*
- *An alternative to all these is to pay workers by results; i.e., to **relate pay directly to output**.*
- *In general, we have two basic pay systems.*
 - ***Time rate payment***
 - ***Payment by results,***

Time rate payment

- *In the method of time rate payment each hour of labor supplied was paid the same wage rate.*
- ✓ *Payment is related directly to the workers supply of time.*
- ✓ *Such a method of payment is appropriate where each hour of labor supplied is equally productive and the firm is confident that the numbers of hours contracted for will be delivered.*

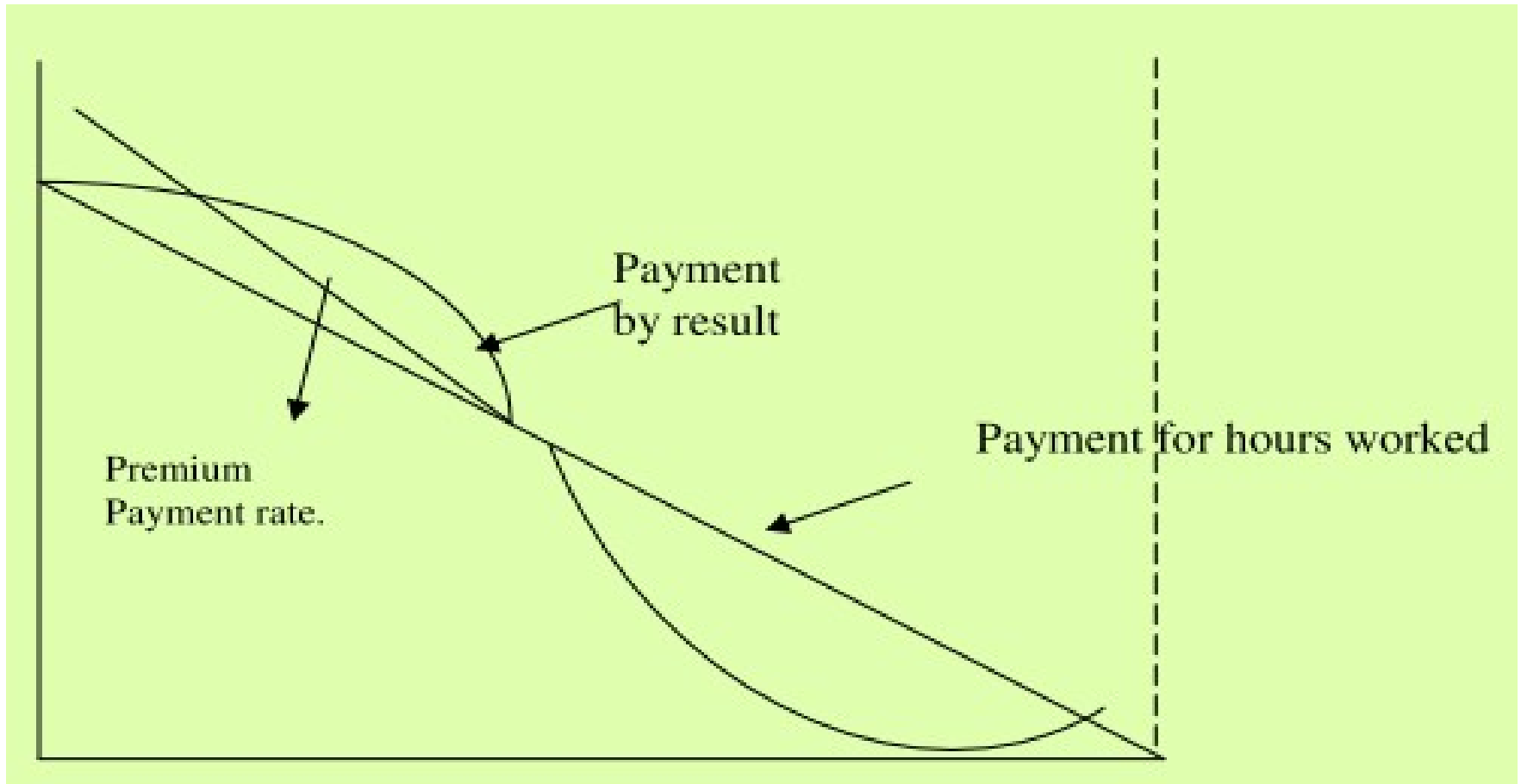
Payment by results,

- ✓ *However, in many jobs these conditions will not hold.*
- ✓ *The amount of output produced during the first hours of work each day, while employees are “warming up”, may be less than that produced during the middle hours of the day.*
- ✓ *Again if workers work longer hours, output may increase more slowly towards the end of the day as workers suffer from fatigue and exhaustion.*
- ✓ *Indeed, even with a much shorter working period, output may increase at a slower rate toward the end of the day as workers ‘get ready to go home’ or “ship off” early on Friday afternoon*

Cont....

- *It seems probable therefore that output will initially rise at an increasing rate but that later the rate at which output increases will decline.*
- *This implies that MP of workers is expected to rise at the beginning and then fall later.*
- *It is this change in the productivity of workers over time that brought the concept of payment by result schemes.*
- *Note that to reflect the time rates, the wage rate was drawn as a straight line while premium for over time has the effect of introducing a kink into the budget constraint facing the employee.*
- *On the other hand, although it depends on the productivity of the worker, the wage rate in the case of payment by result is a kind of twisted curve*

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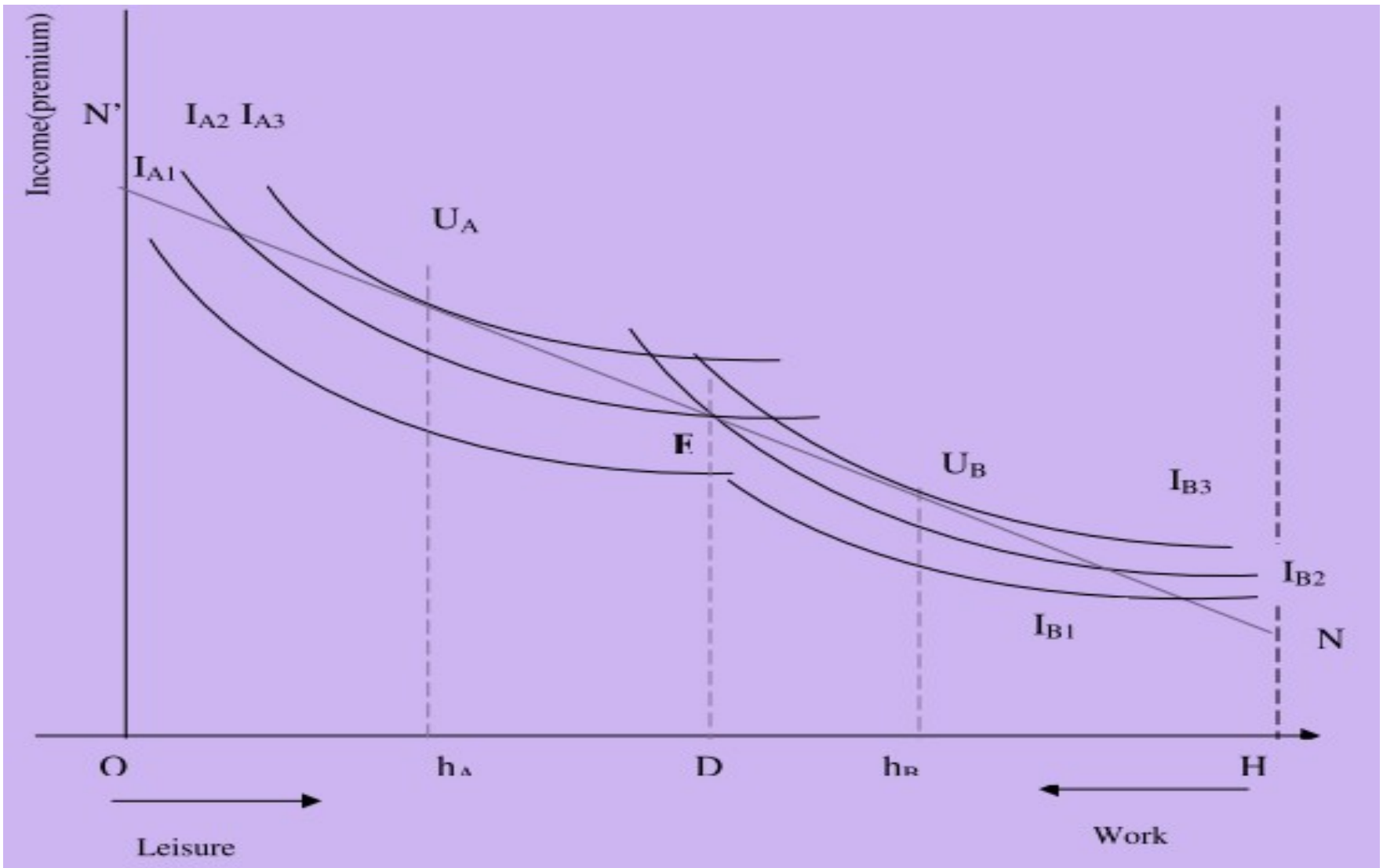


- *The consequence of a change in the slope of the budget constraint is to induce both an income and a substitution effect and it depends on the nature of the individuals preferences as to which effect dominates.*

The Effect of a Standard Work Day on Labor Supply

- *Our discussion thus far has implicitly assumed that workers can individually determine the numbers of hours they work.*
- *This is typically not the case.*
- *It is common practice in many countries to set legal limitations for hours of work. E.g. in Ethiopia normal hours of work are not greater than 8 hrs a day or 48 hrs a week .*
- *The imposition of such laws restricts the income – leisure choice of individuals.*
- *Some may find this 8 hr work day a bit long and feel over employed, while other individuals in similar circumstances find themselves under employed.*
- *To verify these possibilities, consider the following conditions. For two person A and B with their corresponding indifference curves, let D be the standard work hours and both of them (A& B) has to do D hrs a day or not at all*

The Effect of a Standard Work Day on Labor Supply



Cont....

- *Person B's optimal position is at U_B , where he prefers to work only h_B hours per day.*
- *But it is not a relevant choice.*
- *The relevant choice is between working the standard work day at D or being non participant at N.*
- *In this instance, it is preferable to work the standard work day because it entails a higher indifference curve I_{B2} as opposed to I_{B1} .*
- *At point D, Mr. B will feel "overemployed" (working more than desired hours) because at that level the worker values leisure more highly at the margin than does the market.*
- *(At D, slope of I_{B2} is greater than the slope of the budget line $\Rightarrow MRS_{LY}$ exceeds the wage rate).*

Cont....

- *To overcome the feeling of overemployment, workers engage in one of the following actions:*
 - ✓ *Absenteeism*
 - ✓ *Reading journals or news at work deteriorating the quality of work*
 - ✓ *Leaving work before time*
 - ✓ *Non punctuality at work*

Cont....

- *For person A, on the other hand, his optimal position is at UA where he prefers to work h_A hrs per day.*
- *But if the standard work day rule is applied he would be located at E where he will work only D hrs per day.*
- *Being at E put him on the lower indifference curve IA_2 than IA_3 (his optimal choice) and at F, his personal valuation of the price of leisure (the slope of the indifference curve is less than the market prices (the slope of the budget line)).*
- *Since this person is limited to work only D hours instead of h_A hours, he would feel “under employed”.*

Cont....

- *To overcome the feelings of under employment, workers can take one or more of the following actions:*
 - *moonlighting (taking two or more work)*
 - *extra hours of work (in the same place) = overtime work*
 - *part time work*

- *The elasticity of labor supply E_s is defined as the proportional change in the quantity of labor supplied divided by the proportional change in the wage rate.*
- *More specifically, wage elasticity of labor supply is defined as.*
$$E_s = \frac{\text{Percentage Change in Quantity of Labor Supply}}{\text{Percentage Change in Wage Rate}}$$
- *The elasticity will depend on the relative strengths of the income and the substitution effects generated by a wage rate change*

Cont..

Labour Supply in the Long Run

- *The long run is the period in which the size of the population of working age and the skills they possess are allowed to vary.*
- ✓ *Now we are concerned with the long run determinants of the size of the country's population.*
- ✓ *The principal determinants of the country's population are two: first, demographic factors such as birth and death rates; and second, migration.*

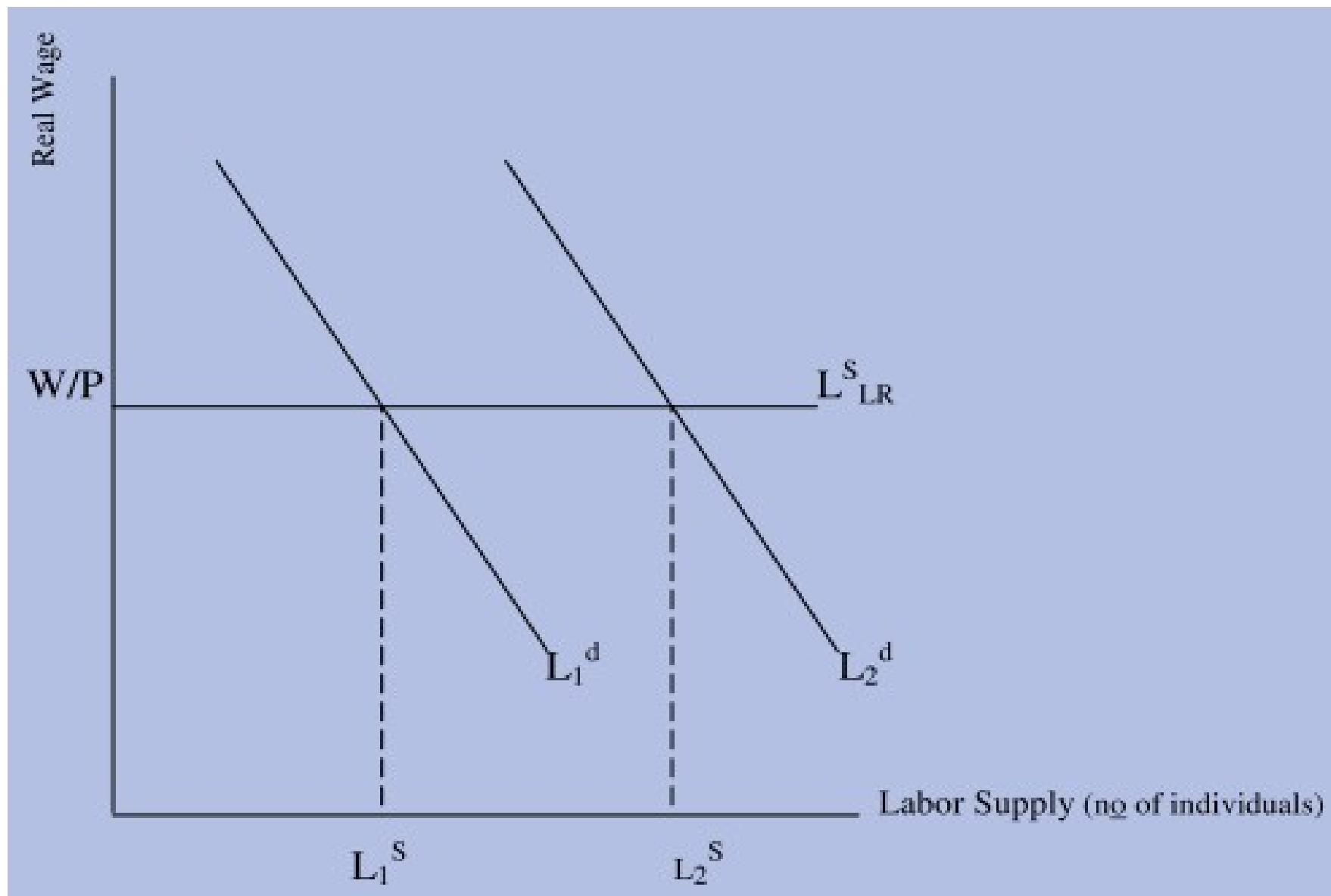
The Economic Determinants of Population Changes

- *Economic analysis of population change dated back to the Malthus' "Essay on Population" which was published in 1798.*
- *Thomas R. Malthus argued that population was determined by the response of both **fertility and mortality** to **change in income**.*
- *Although Malthus admitted to **one voluntary check** on the number of births, which was the **age at which people married**, the essential check on population was the **subsistence level of income**.*
- *Population, he predicted, would grow quickly if **wages** grew because the average person would marry earlier and for this reason have more children and deaths would fall.*

Cont..

- *He assume, In this analysis, the subsistence real wage is assumed constant at W/P .*
- *Now suppose that aggregate labour demand is given by $L1d$.*
- *This is just adequate to support a working population of $L1s$ and their dependences.*
- *Any increase in the number of dependents would reduce per capita incomes below subsistence level and result in a rise in the death rates.*
- *On the other hand, any increase in the aggregate labour demand to $L2d$ would be accompanied by a rise in the real wage which would encourage individuals to marry earlier and have more children.*

Cont..



Cont..

- *As the population grows, the labour supply increases and wage rates fall back toward W/P .*
- *Thus, in the long run labour supply is **perfectly elastic** as indicated by labour supply curve $LsLR$.*
- *One implication of this is that in the long run real wages are entirely supply determined.*

The Effect of an Increase in Non Labor Income on Labor Force Participation

- *Increase in non labour income, pure income effect (expansion of the opportunity set) impact on labour supply depends on whether leisure is*
 - ✓ *Normal good $\rightarrow$ labour supply decreases*
 - ✓ *An inferior good $\rightarrow$ labour supply increases*
- *Seems reasonable to think that leisure is a normal good (and this is confirmed by the empirical application discussed later in the lecture)*

Cont..

- *Initially we assume that there is zero non-labour income and the income budget line is JZ.*
- *For the given wage rate, the individual maximizes utility at point A and supplies L_a hours of labor; the individual participates in the labor force.*
- *Now suppose that the individual has a new source of non labor income (such as a lottery prize or an inheritance).*

Cont...

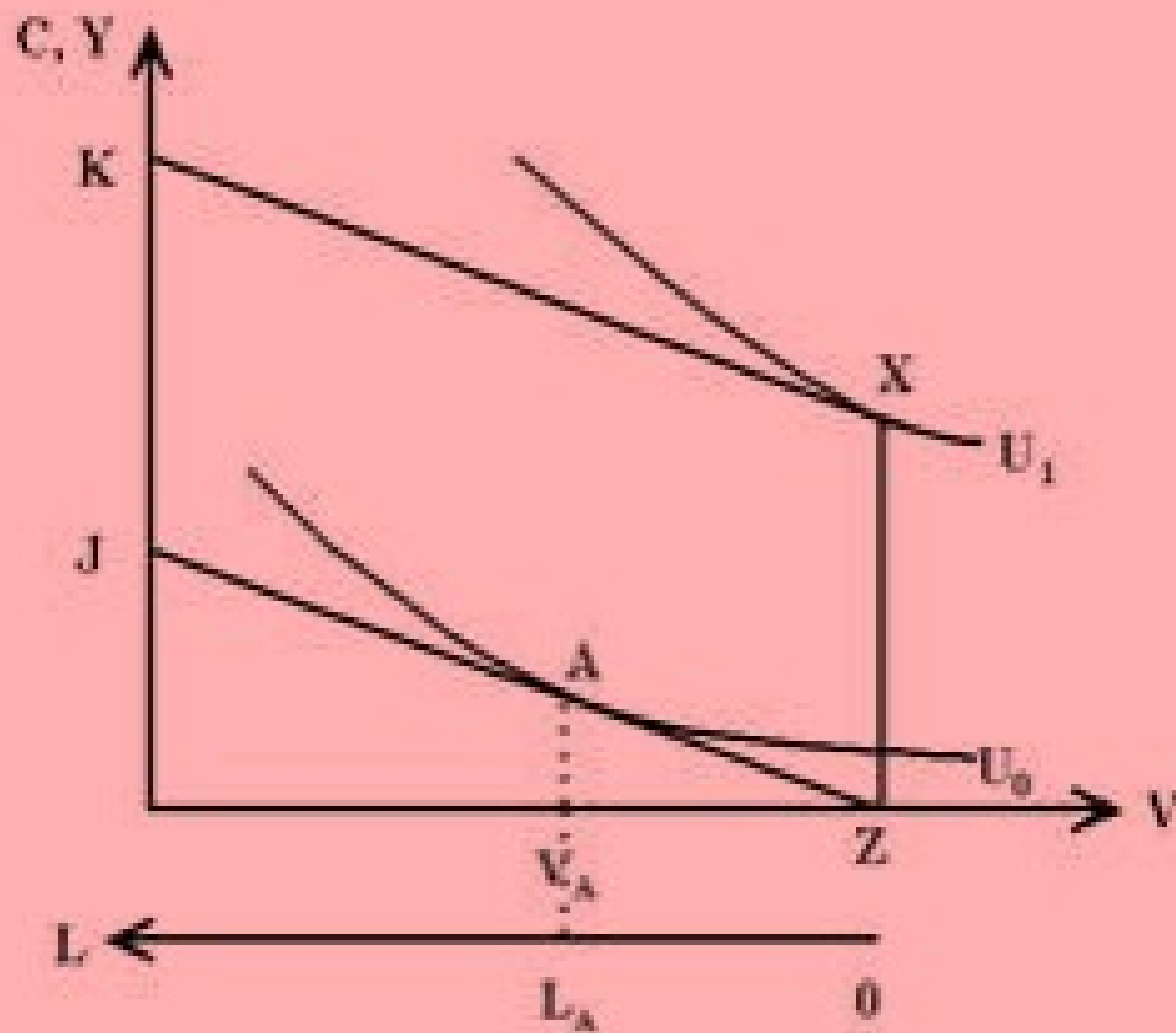


Figure 2-7

Chapter Three

Labour Demand

Introduction

- *Labour demand* is defined as the amount of labour that employers seek to hire during a given time period at a particular wage rate.
- The demand for labour is an economics principle *derived from the demand for a firm's output*.
- That is, if demand for a firm's output *increases*, the firm will demand *more labour*, thus *hiring more staff*.
- *In the short run*, it is assumed that labor is the only variable input (i.e., the amount used of the other inputs is fixed and cannot be changed).
- *In the long run*, labor is combined with variable factors

Basic concepts

- **MR**



$$MR = \frac{\Delta TR}{\Delta Q}$$

- **MPL**



$$MP = \frac{\Delta Q}{\Delta L}$$

- **MRPL**:- The extra income that a firm gets from the sale of the output produced by labor . **MRPL = MPL * MR**

$$MRP = \frac{\Delta TR}{\Delta L} = \frac{\Delta TR}{\Delta Q} \times \frac{\Delta Q}{\Delta L}$$

- **VMPL** :- This indicates the \$ increase in revenue generated by an additional worker, holding capital constant.

$$VMPL = P * MPL$$

- **VAPL** :- Value of Average Product (VAP) of labour is the dollar value of output per worker: **VAPL = P * APL**

Discussion area under this chapter

- *section 1. The demand of a firm for labor in the short run*
 - ✓ *Perfect competition in both the product and labour markets*
 - ✓ *Monopolistic power in the product market*
- *Section two: the demand of a firm for labour in the long run*
 - ✓ *Perfect competition in both the product and labor markets*
 - ✓ *Monopolistic power in the product market*
- *Section three: the market demand for labour and elasticity's of labour demand*
 - ✓ *The market demand for labour*
 - ✓ *Elasticity's of labour demand*
 - ✓ *Elasticity of factor substitution*

Section 1. The demand of a firm for labor in the short run

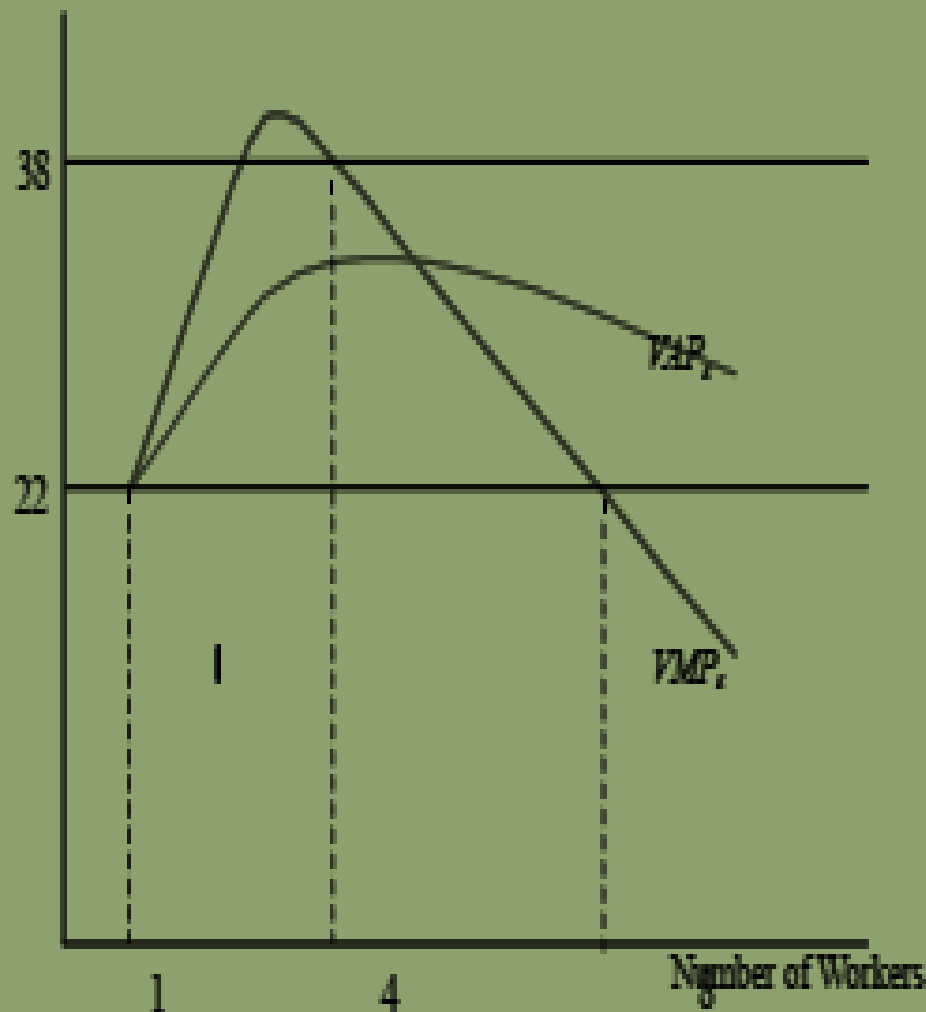
Section 1.1. Perfect competition in both the product and labour markets

- *When the firm is a perfect competitor in the **product market**, it is a price-taker and thus its marginal revenue is equal to the commodity price.*
- *In this case, consequently, the marginal revenue product of labor (MRP_L) equals the value of marginal product of labor ($VMPL = MPPL * P$).*
- *The extra cost of hiring an input or the **marginal expenditure (ME)** is equal to the price of the input (wage rate = w) if the firm is a perfect competitor in the labor market.*
- *Perfect competition in the **labor market** means that the firm demanding the input is too small, by itself, to affect the price of the input.*

Cont..

- In other words, *each firm can hire any amount of the input (labor services) at the given market wage rate.*
- Thus, a profit-maximizing firm should hire an additional unit of labor as long as the MRP_L exceeds the marginal expenditure on labor or wage rate (w).
- Profit is maximized at a point where $MRP_L (= VMP_L \text{ in this case}) = ME_L (= w \text{ in this case})$.

Firms hiring decision in the short run



A profit-maximizing firm hires workers up to the point where the wage rate equals the value of marginal product of labour. If the wage is \$22, the firm hires eight workers.

Cont..

Mathematically, $\max_L \pi = R - C$ where $R = P_X \cdot X$ and $C = \bar{w}L + F$; F = fixed cost.

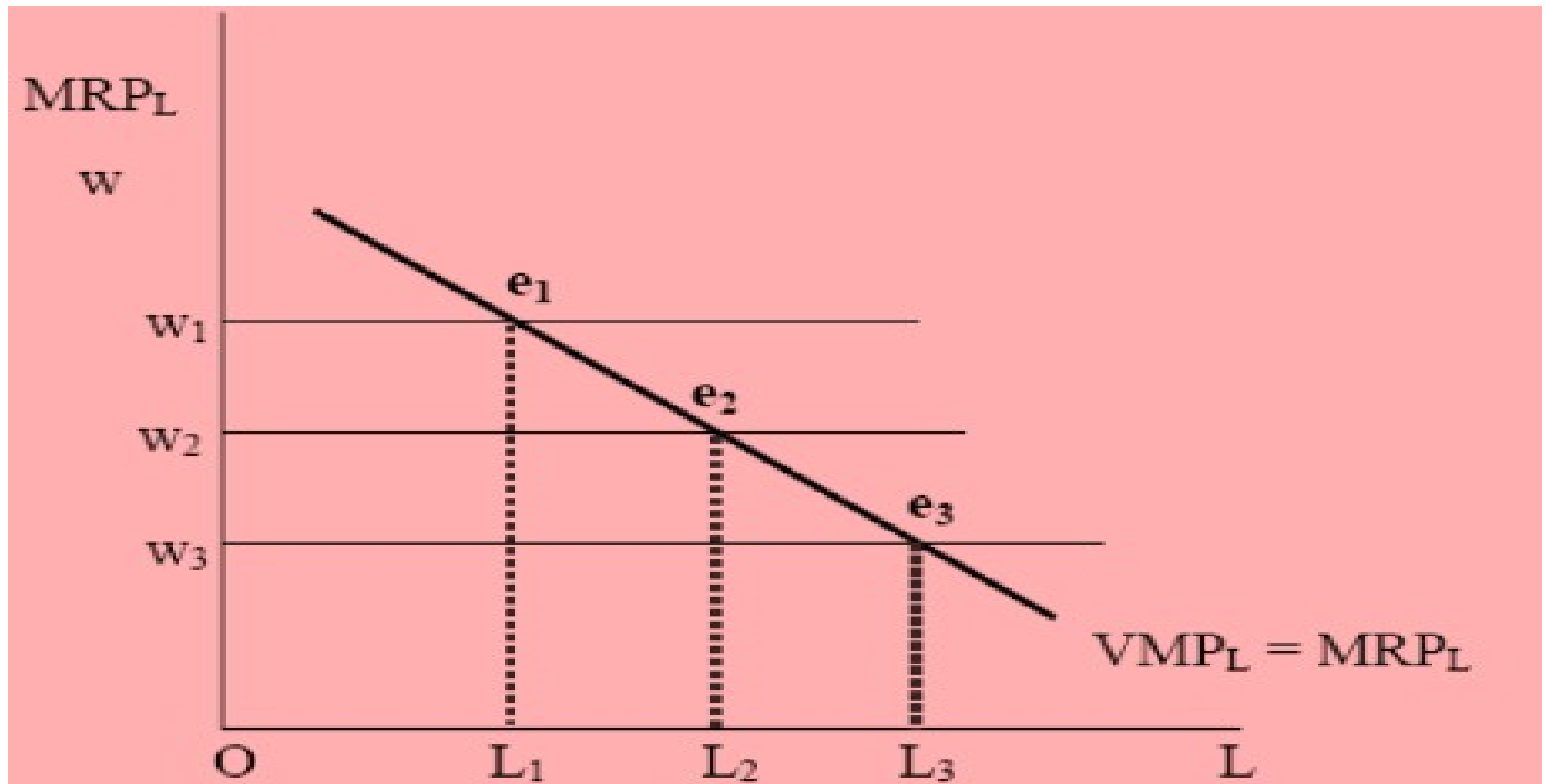
$$\Rightarrow \frac{d\pi}{dL} = \frac{d(\bar{P}_X \cdot X - \bar{w} \cdot L - F)}{dL} = \bar{P}_X \frac{dX}{dL} - \bar{w} \frac{dL}{dL} = 0.$$

$$\Rightarrow \bar{P}_X \cdot MPP_L = \bar{w}.$$

$$\Rightarrow \boxed{VMP_L = \bar{w}.$$

- *From your microeconomics, you may recall that a rational firm operates in the second stage of production where the MPP_L is declining but positive.*
- *Multiplying this, declining but positive, MPP_L by a fixed output price gives a downward sloping MRP_L curve as in Figure below.*

Cont..



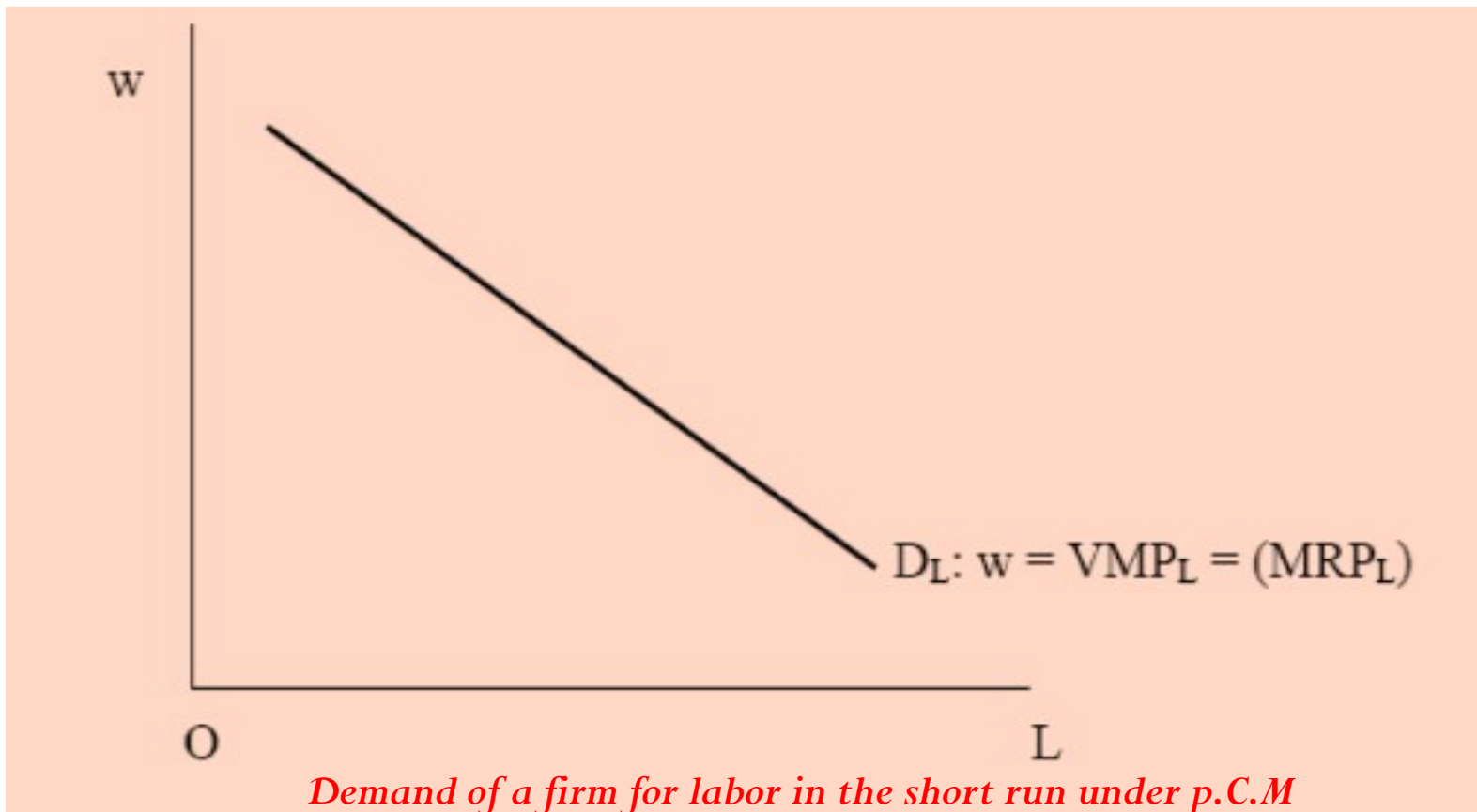
Equilibrium of a Firm in Perfectly Competitive Product and Labor Markets

Cont..

- *Given this MRP_L and the equilibrium condition $MRP_L = VMP_L = w$, the firm hires L_1 units of labor if the wage rate is w_1 .*
- *Similarly, L_2 units of labor will be hired at w_2 and L_3 at w_3 .*
- *At e_1 , $VMP_L = w_1$.*
- *The firm's profit is at the maximum for wage rate w_1 . To the left of e_1 , $VMP_L > w_1$.*
- *The firm would increase its profit by hiring more labor.*
- *The opposite holds to the right of e_1 .*
- *That is, the firm would increase its profit by reducing the amount of labor it uses.*

Cont..

- *The graph that shows this relationship between the wage rate and the quantity demanded (hired) of labor is the demand curve (Figure 3.2).*
- *Thus, the demand for labor is the value of marginal product of labor*



Cont..

Section 1.2 Monopolistic power in the product market

- Assumptions:

- ✓ The firm uses a *single variable factor – labor – whose market is perfect: the wage rate is given and the supply of labor to the individual firm is perfectly elastic.*
- ✓ The firm has monopolistic power in the output market.
- ✓ This implies that the demand for the product of the firm is downward sloping and the *marginal revenue curve lies below the demand curve* ($MR < P$) at all levels of output. why?

Cont.

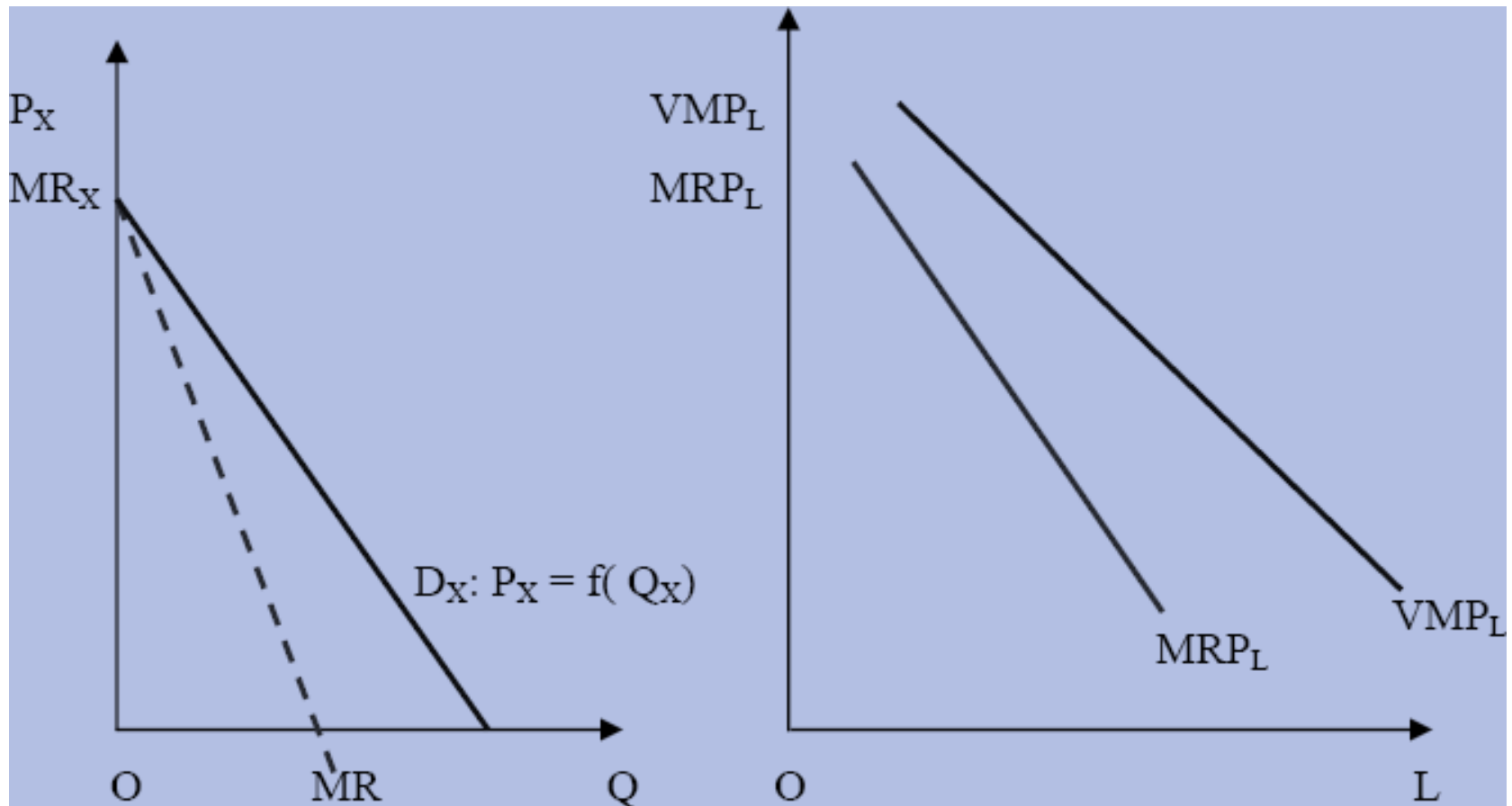
- ✓ *For a monopoly there is a **price effect**. ... So the marginal revenue on its additional unit sold is lower than the price, because it gets less revenue for previous units as well (it has to reduce price to the same amount for all units).*
- ✓ *Because it must reduce their price in order to sell more of their products and because the demand curve is a downward slope*

$$\triangleright MR_X < P_X$$

$$\triangleright MR_X \cdot MPP_L < P_X \cdot MPP_L$$

$$\triangleright MRP_L < VMP_L$$

Cont..



MRP_L less than VMP_L under Imperfectly Competitive Product Market

Cont..

- *The firm maximizes its profit with respect to the units of labor it employs.*
- *Given demand function $P_X = f_1(Q)$ and production function $Q_X = f_2(L, K \text{ is fixed})$,*

$$\pi = TR - TC$$

$$\Rightarrow \pi = P_X \cdot Q_X - (\bar{w} \cdot L + F)$$

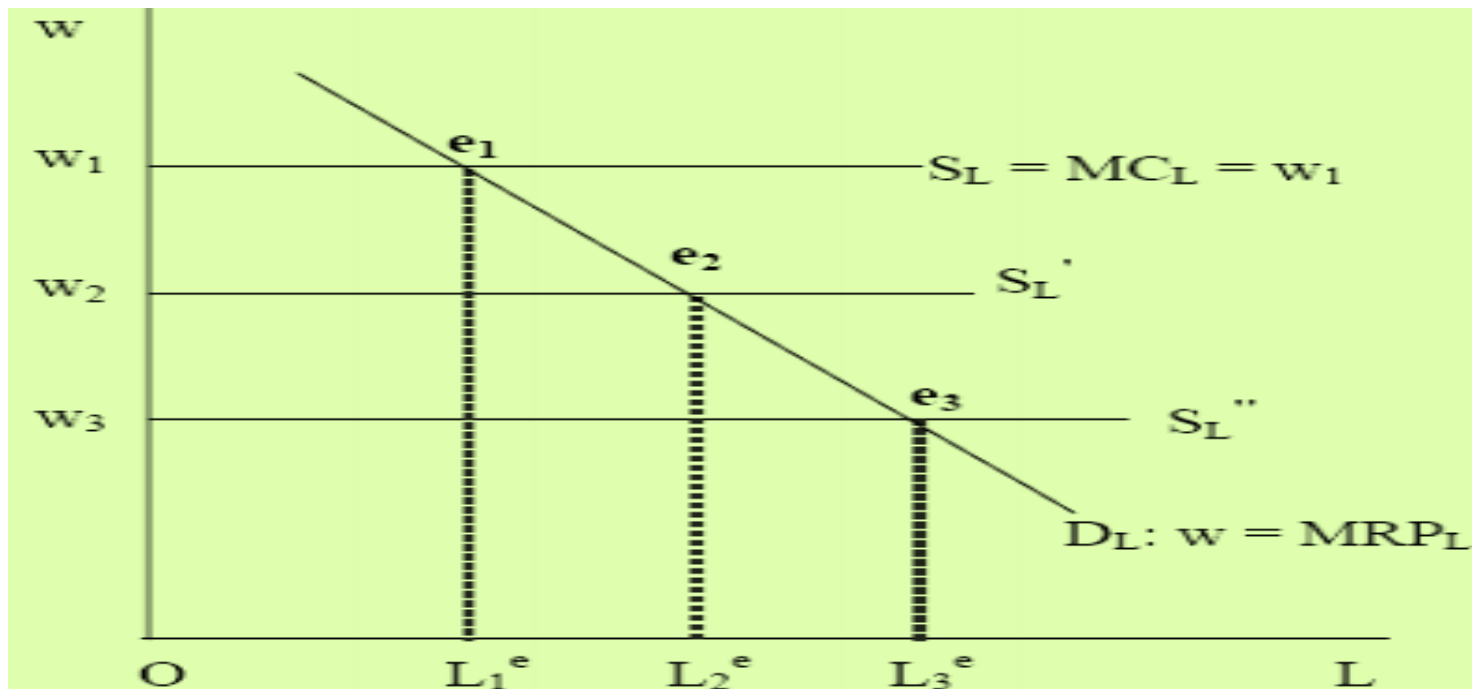
$$\max_{\underline{L}} \pi: \quad \text{F.O.C.} \frac{d\pi}{dL} = 0. \Rightarrow \frac{d(TR)}{dL} - \frac{d(TC)}{dL} = \frac{d(P_X Q_X)}{dL} - \frac{d(\bar{w} \cdot L + F)}{dL} = 0$$

$$\Rightarrow MRP_L - \bar{w} = 0$$

$$\Rightarrow \boxed{MRP_L = \bar{w}}$$

Cont..

- So the firm maximizes its profit when it employs labor in such a way that $MRP_L = w$ (i.e., l_1^e units of labor at wage rate w_1 , l_2^e at w_2 , and l_3^e at w_3 in the figure below).
- Joining the equilibrium points e_1 , e_2 and e_3 (at different wage rates) gives MRP_L as a demand curve that relates wage rate to labor employment.



Equilibrium of a Firm and the Demand of the Firm for Labor in monopolistic market

Example

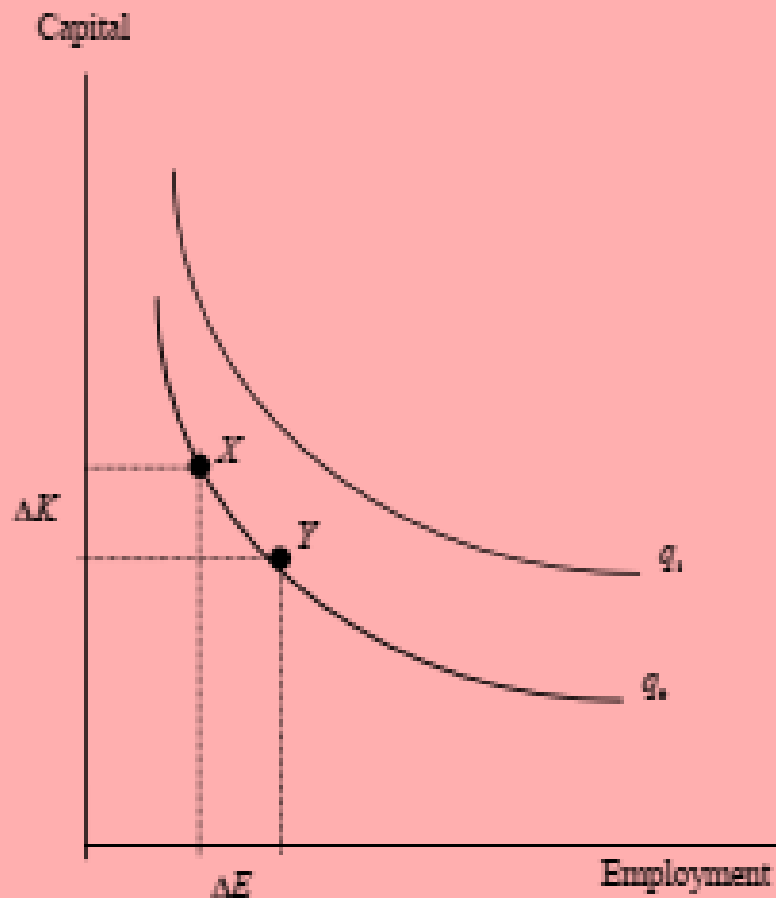
- *Suppose perfect competition going price is 2 and wage is 50 birr, then how many workers should we use ?*

No of worker	Out put
0	0
1	35
2	75
3	102
4	125
5	130

Section 2. The demand of a firm for labor in the long run

- *In the long run the firm can adjust all of the factors of production.*
- *In the long run, the profit-maximizing firm can vary the inputs of both labour (l) and capital (k)*
- *there are diminishing marginal returns to adding more units of k (holding l constant) and to adding more units of l (holding k constant)*
- *An **isoquant-isocost diagram** can be used :*
 - ✓ *to determine the optimal combination of L and K and*
 - ✓ *to derive the demand for labour when K is a variable input into the production process .*

Isoquant Curves



All capital-labour combinations that lie along a single isoquant produce the same level of output. The input combinations at points X and Y produce q_0 units of output. Input combinations that lie on higher isoquants produce more output.

Convexity implies diminishing marginal rate of technical substitution.

Properties of Isoquant Curves

- *The shape of the isoquant reflects the technological possibilities for substituting L and K in the production process*
- *Isoquants are convex to the origin*
- *The slope of the isoquant is equal to MPL/MPK , where MPL is the marginal product of labour and MPK is the marginal product of capital*
- *Given diminishing returns to hiring each factor, the slope of the isoquant exhibits a diminishing marginal rate of technical substitution between L and k*
- *larger and larger amounts of L must be substituted for each unit of K as the amount of K used in the production process decreases*

Cont....

- *Each isoquant depicts the various combinations of L and K which can be used to produce a particular level of output, say q_0 in figure 5-2*
- *Isoquants depicting larger quantities of output (such as Q_1 in figure 5-2) will be further from the origin (they require greater quantities of inputs)*

Isocost Lines

- *Assume that factor prices are given: 'W' is the wage paid to labour and 'r' is the cost(rental) of using capital*
- *For a given COST outlay, a firm can purchase different quantities of L and K according to the following equation:
 $COST = WL + rk,$*
- *An isocost line depicts the various quantities of L and K which can be purchased for a particular COST outlay*
- *Re-arranging the COST equation produces the following equation for an isocost line:
 $K = COST/r - (W/r)L$*
- *As shown in figure 5-2, the slope of the solid isocost line is $-(W/r)$, the horizontal intercept is $COST/W$ and the vertical intercept is $COST/r$*

Cont....

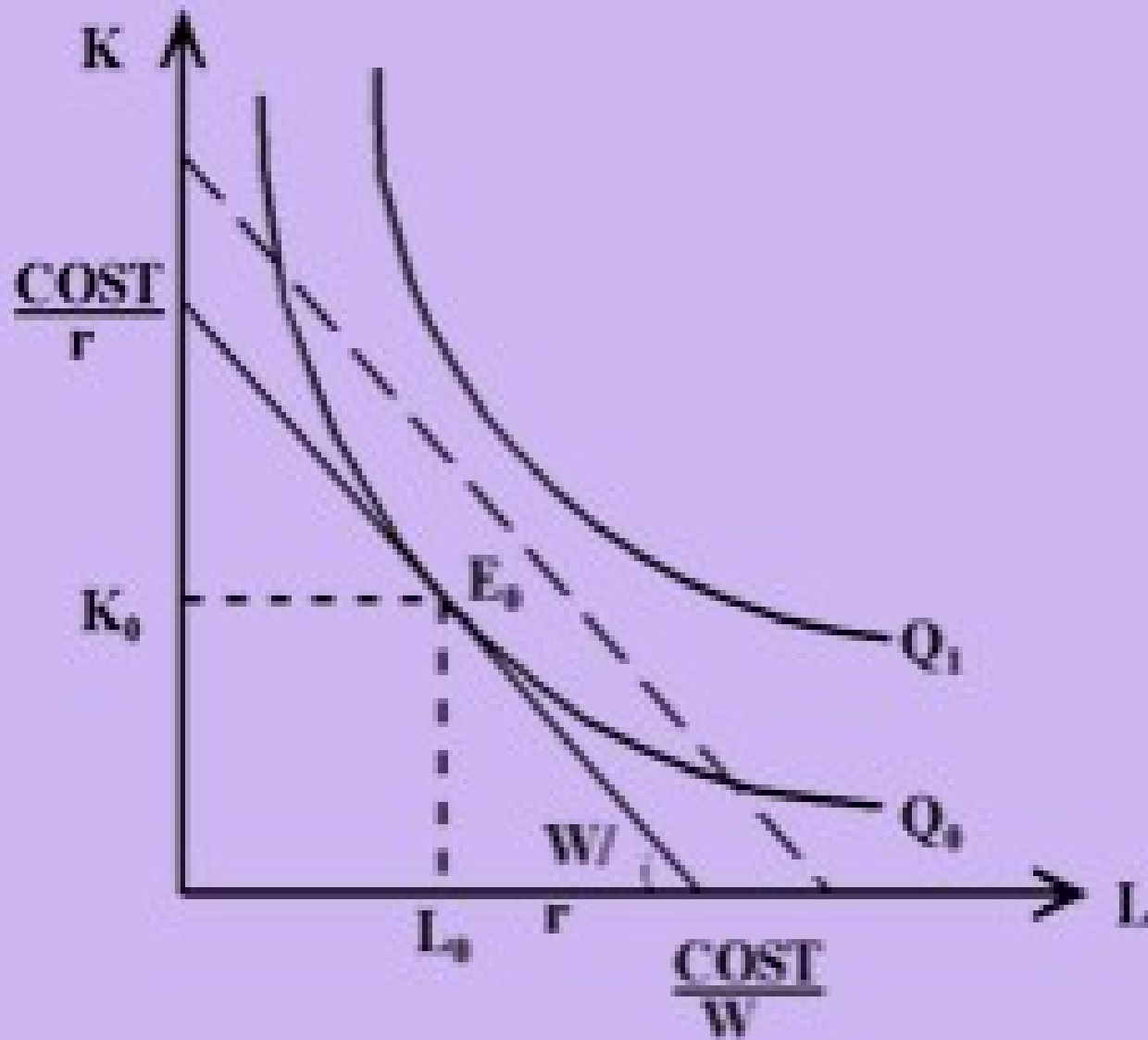


Figure 5-2

Cont.....

- *The greater the COST outlay, the further the isocost line will be from the origin (more of both L and K can be purchased)*
- *In figure 5-2, the dashed isocost line has a higher cost outlay than the solid isocost line; the dashed isocost line is parallel to the solid isocost line (both isocost lines have slope $-W/r$).*

The optimal quantities of l and k

- *A profit-maximizing firm will choose the least cost combination of L and K to produce a particular level of output, such as q_0*
- *The least cost factor input combination will be determined by the tangency of an isocost line with the q_0 isoquant*

Cont.....

- *The closer the isocost line to the origin, the smaller the cost.*
- *As shown in figure 5-2, the optimal tangency position is given by point e_o .*
- *Given factor prices (W, r) , the optimal combination of inputs to produce q_o is l_o and k_o .*
- *At this optimal tangency point, the slope (W/r) of the isocost line is equal to the MPL/MPK slope of the isoquant*
- *The ratio of factor marginal products is equal to relative factor prices; the firm's internal rate of factor substitution is equal to the rate at which the factors can be substituted in the market place*

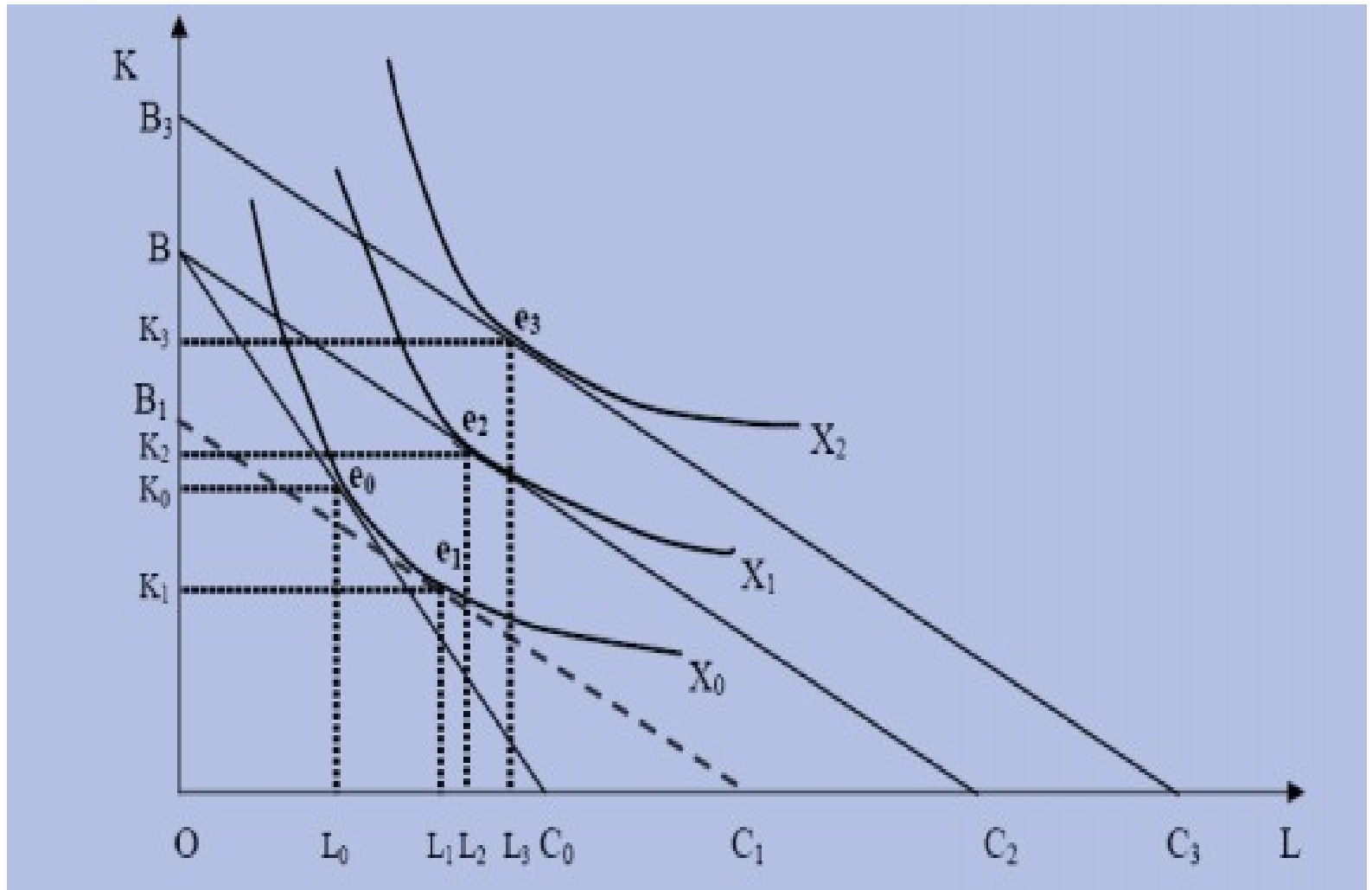
Section 2.1 perfect competition in both the product and labor markets

- *In the long run, when there are more than one variable factors of production, **the VMP_L is not the demand for labor.***
- *This is because various resources are used simultaneously in the production process so that a change in the price of labor (wage rate) leads to changes in the employment (use) of the other factors.*
- ***This in turn shifts the marginal (physical) product curve of labor (whose price is initially changed).***

Cont.

- *Let's assume that the price of labor (the wage rate) falls.*
- *Then this has three effects: a substitution effect, an output effect, and a profit-maximizing effect.*
- *Initially, the firm produces a profit – maximizing output x_0 with a combination of l_0 and k_0 , given factor prices w and r whose ratios are defined by the slope of the isocost line bc_0 (figure 3.5).*
- *When wage rate falls, the isocost bc_0 changes to bc_2 and this new isocost (bc_2) is tangent to the isoquant corresponding to output level x_1 at e_2 .*
- *K_2 units of capital and l_2 units of labor are used.*

Cont.



Substitution, Output and Profit Effects of a Fall in Wage Rate

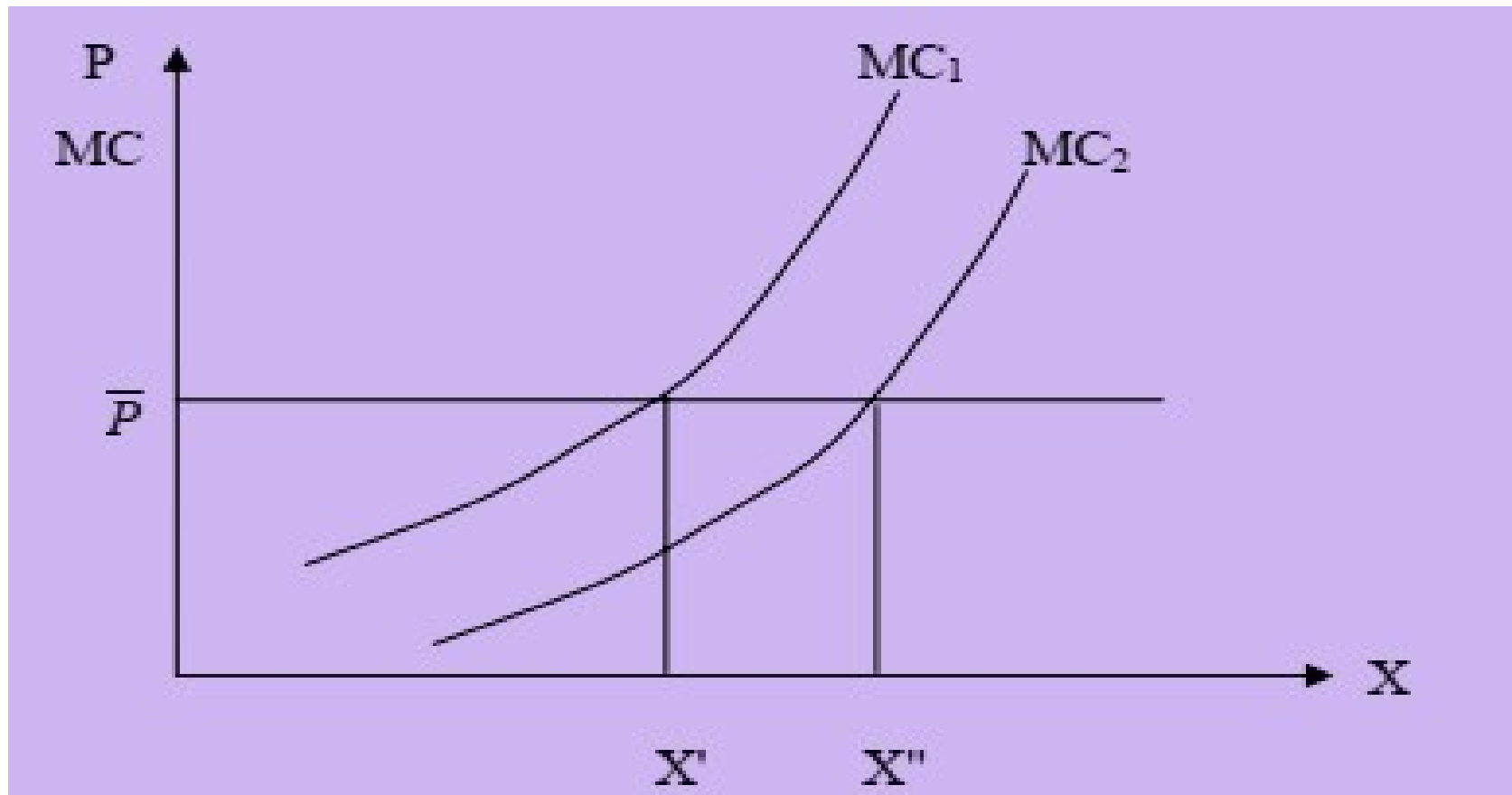
Cont.

- *This movement from e_0 to e_2 can be split into two: substitution effect and output effect.*
- *To see the substitution effect, we draw an isocost line (B_1C_1) parallel to the new isocost line (BC_2) but tangent the old isoquant (X_0).*
- *The movement from e_0 to e_1 is the substitution effect.*
- *This shows that the firm substitutes the cheaper labor for the relatively more expensive capital even if it were to produce the original level of output (X_0).*

Cont.

- But when wage rate falls, the firm can hire more of the two factors (L and K) with the **same expenditure**.
- Hence, the firm produces higher level of output with more labor and capital (L_2 and K_2) and, therefore, the movement from e_1 to e_2 is the **output effect**.
- Point e_2 is not the final equilibrium of the firm because keeping the total cost/expenditure constant doesn't maximize its profit.
- The fall in wage rate results in a shift in the marginal cost curve downward to the right (from MC_1 to MC_2 in Figure 3.6 below) and the profit maximizing output of the firm rises from X' to X'' .

Cont.



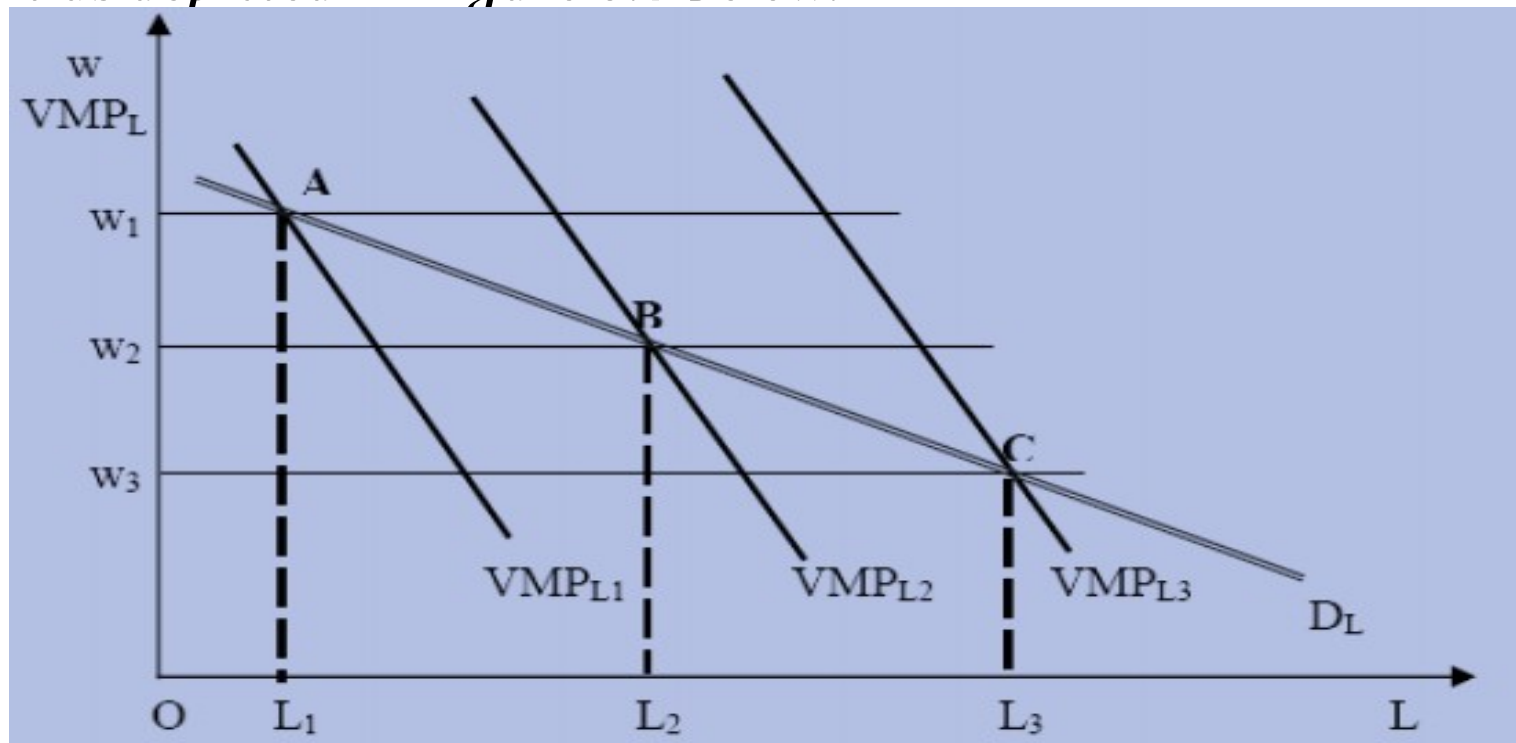
Fall in Wage Rate Reduces the Marginal Cost of Production of a Firm

Cont.

- *Thus, the isocost line BC_2 (in Figure 3.5) must shift upward parallel to itself.*
- *So, the final equilibrium is when isocost B_3C_3 is tangent to the highest possible isoquant (X_2) at point e_3 .*
- *The movement from e_2 to e_3 is the profit effect (or the profit-maximizing effect).*
- *The substitution effect of a decline in wage rate causes a decline in the marginal physical product of labor (as it increases the units of L and reduces that of K).*
- *The output and profit effects result in rise in the amounts of both labor and capital used.*
- *Both effects cause the MPP_L to shift upward and to the right (increase the MPP_L at a given level of employment).*

Cont.

- *The output and profit effects more than and it offset the substitution effect so that the final result of a fall in wage rate is an increase (and rightward shift of) the MPP_L (curve).*
- *Given the price of the final commodity P_x , the VMP_L also shifts to the right as depicted in Figure 3.7 below.*



Shifts in VMP_L Curve, Equilibrium of a Firm and Labor Demand

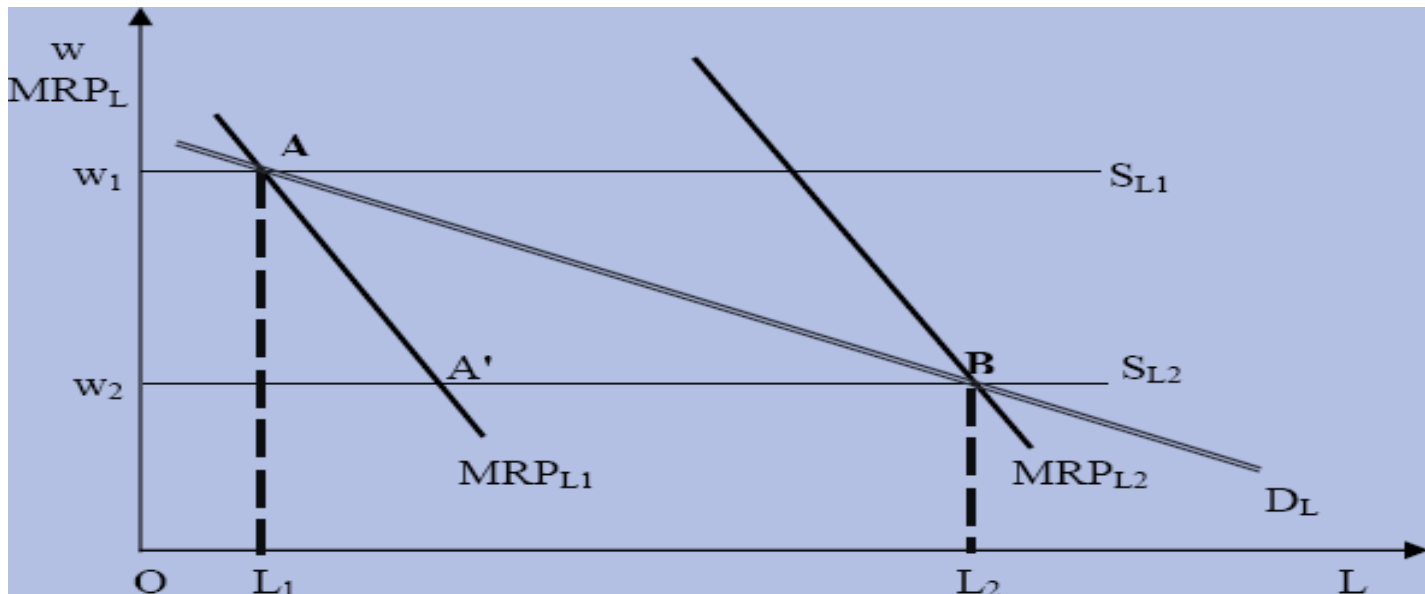
Cont.

- *At the initial wage rate w_1 , L_1 units of labor are employed (which is determined by the intersection of VMP_{L1} and the supply w_1).*
- *The new equilibrium demand for L (when wage rate falls to w_2) is at point B on VMP_{L2} .*
- *If w further declines to w_3 , the new equilibrium will be at point C .*
- *The locus of points A , B and C is the demand curve for labor by the firm when several variable factors are used.*
- *This is the long run demand for labor by a firm because all the factors used (assumed to be only labor and capital here for simplicity) are variable.*
- *In this case, the demand for labor is not the same as its VMP curve, but derived from changing (shifting) VMP curves.*

Section 2.2 Monopolistic Power in the Product Market

- With monopolistic power in the product market and more than one variable factor in the production process, the demand for labor is not MRP_L curve, but it is formed from points of shifting MRP_L curves.

Consider Figure 3.8 below.



Demand of the Firm for Labor

Cont.

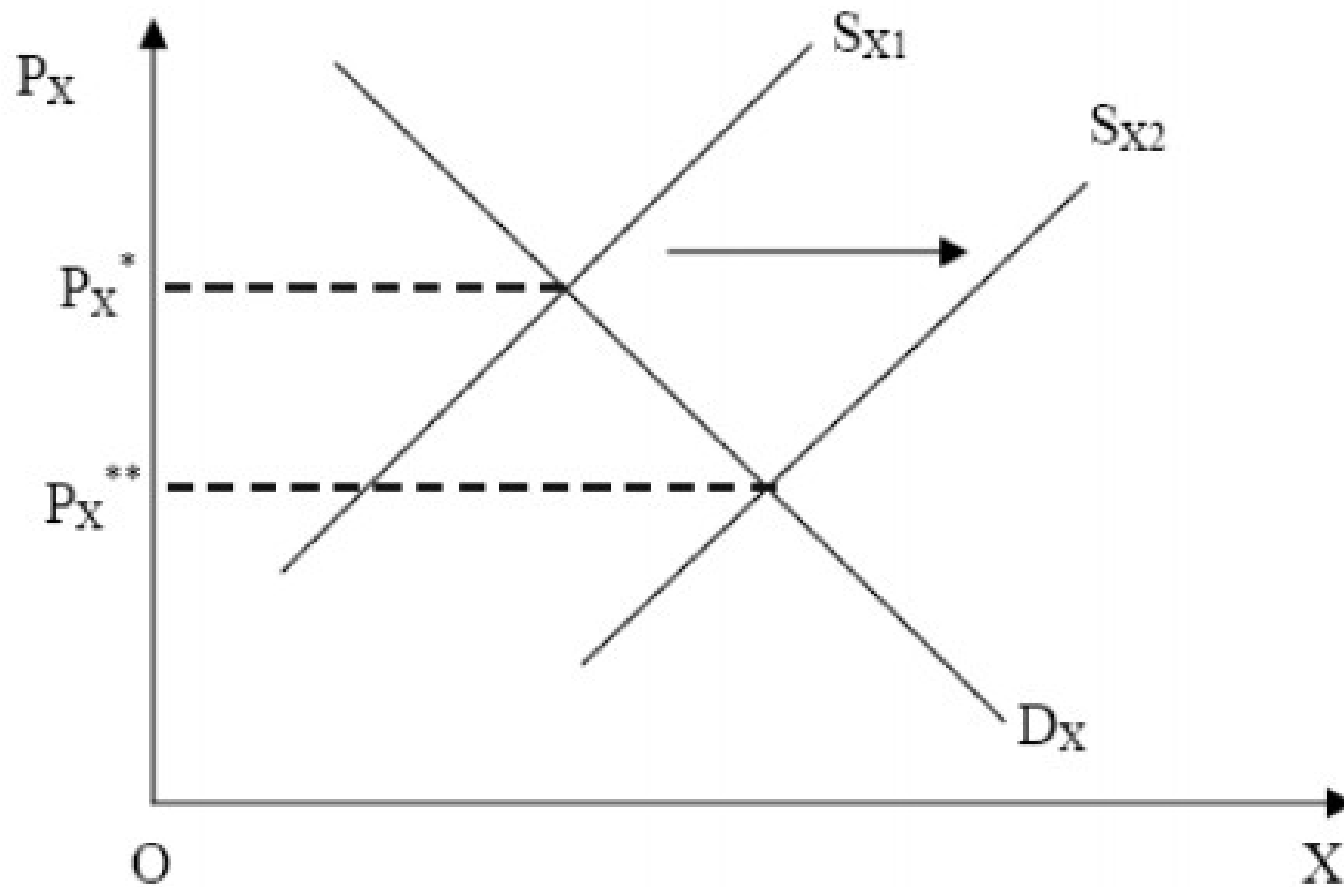
- *When the wage rate is w_1 , the equilibrium is at point A.*
- *If wage rate falls from w_1 to w_2 , the firm moves from A to A' along MRP_{L1} .*
- *The fall in wage rate has substitution, output and profit effects.*
- *The net result of these effects is a shift in MRP_L curve to the right which leads to equilibrium at B.*
- *So, line AB is the demand for labor in this case.*

Section 3: the market demand for labor and elasticity's of labor demand

Section 3.1 The Market Demand for Labor

- *The demand for labor in a particular market—called the market demand for labor—*
- *is the amount of labor that all the firms participating in that market will demand at different market wage levels.*
- *The market demand curve for labor is derived from the individual firms' demand curves for the input.*
- *But it is not the simple horizontal summation of the individual firms' demand curves.*
- *This is because when the price of labor falls, not only this firm but also other firms will employ more of this factor and other (complementary) inputs to expand production.*
- *Thus, the supply of the final commodity increases and its price falls*

Cont

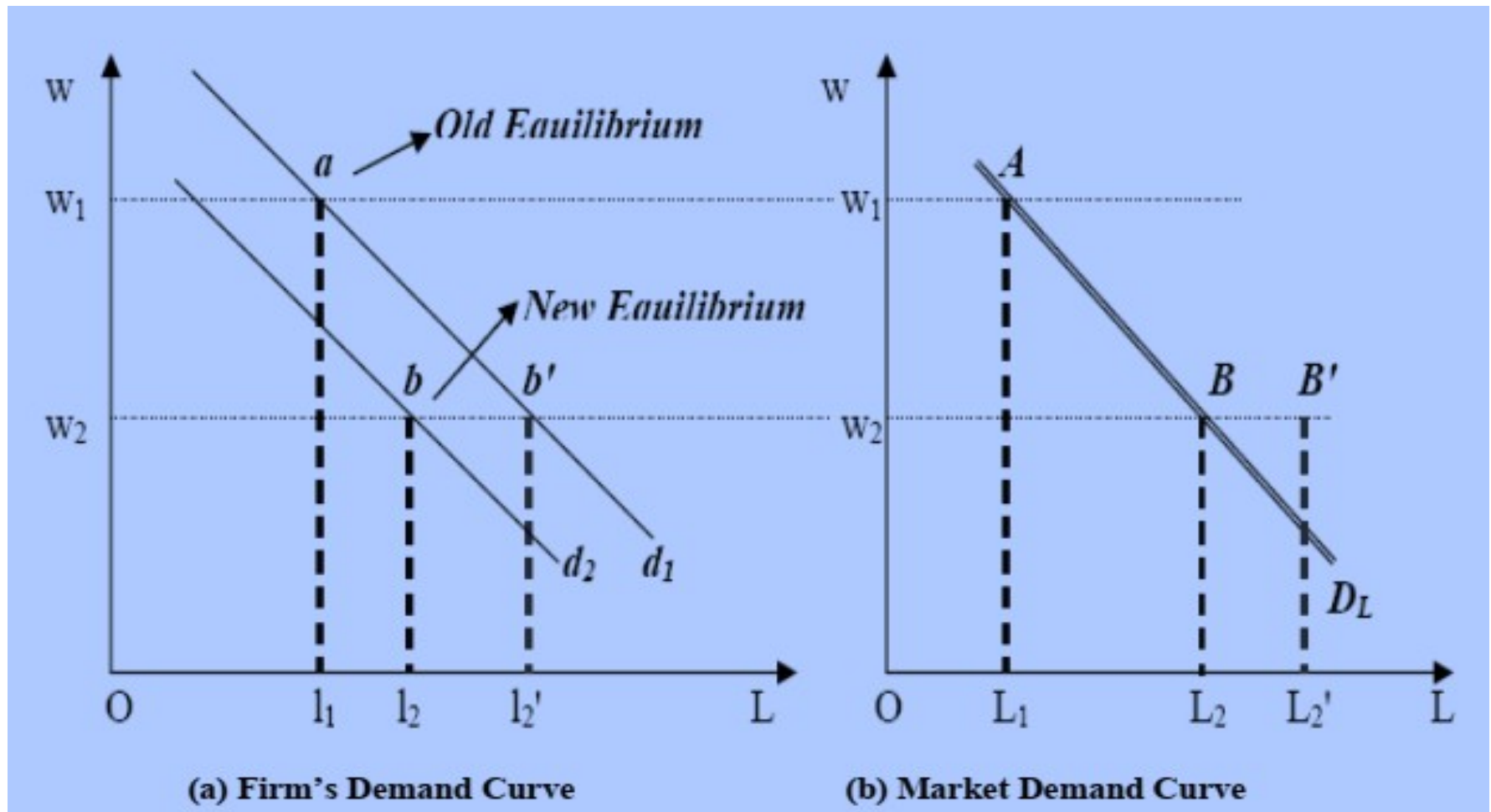


The Effect of a Fall in Input Price on Commodity Price

Cont.

- *If the wage rate falls and more of it is used, the supply of the commodity increases (shifts from S_{x1} to S_{x2}).*
- *This derives down the equilibrium price of the product from P_x^* to P_x^{**} .*
- *This in turn has a negative consequence on the demand for labor.*
- *Under perfectly competitive product and labor markets, since the $MRP_L = MP_L$ times MR (which is equal to the commodity price), the reduction in commodity price will cause each firm's MRP_L and demand curves for the input to shift down or to the left.*
- *The market demand curve for labor is then derived by the horizontal summation of the individual firms' demand curves for the input after the effect of reduction in the commodity price has been considered [See Figure 3.10 below].*
- *The market demand curve for a particular type of labor is the horizontal summation of the marginal revenue product of labor curves of every firm in the market for that type of labor after the price effect.*

Cont.



Deriving the Market Demand for Labor

Cont.

- *If the fall in commodity price were not taken into account, it would lead to an overestimation of the market demand for labor (which joins points A and B').*
- *The only difference when the product market is imperfectly competitive is that the individual demand curves are based on MRP_L (and not on VMP_L).*
- *If each firm is a pure monopolist (the only seller for its product), then the price of the final commodity is likely not to be affected, and in such cases the market demand curve is the simple horizontal summation of individual demand curves.*

In general, the demand for labor depends :-

- *The price of the input* – increase in wage rate reduces the quantity demanded of the inputs services, and vice versa. This represents a movement along a given demand curve.
- *The marginal physical product of the factor* – if the MPP_L increases, more of it will be demanded, and vice versa.
- *The price of the output (commodity)* – when the output the firm produces becomes cheaper, the firm will cut its production and thus demands less of the input used. The opposite happens when the commodity becomes more expensive.
- *The amount of other factors which are combined with the factor*
- *The price of other factors* – increase in prices of complementary inputs reduces the amount of the complementary inputs to be used with a given amount of labor, and thus reduces the productivity of (and demand for) labor.
- *Technological progress* – technological progress could increase or decrease the demand for labor depending on whether it increases or decreases the MPP_L .

- Section 3.2 The Elasticities of Labor Demand

- *The elasticity of labor demand is a measure of the sensitivity of labor demand to a change in one of its determinants.*
- *Below, we will see two elasticity's of labor demand: the own-wage elasticity and the cross-wage elasticity.*

- *Own-Wage Elasticity of Labor Demand*

- ✓ *The own-wage elasticity of labor demand is a measure of how sensitive is the demand for a particular category of labor to a change in the wage rate in that specific labor market.*
- ✓ *The own-wage elasticity of labor demand is defined as:*

$$\eta_{ii} = \frac{\text{percentage change in employment in market } i}{\text{percentage change in wage in market } i}$$

Cont.

- The i subscripts in the definition just refer to the specific labor markets used in the numerator and denominator of the expression above.
- It is expected that the elasticity of labor demand will vary substantially across labor markets.
- An inspection of the definition above should indicate that the own-wage elasticity of labor demand will always be negative as a result of the negative slope of labor demand curves.
- When labor demand is perfectly elastic ($/\eta_{ii} = -\infty$), a 1% increase in the wage will cause employment to fall by infinitely large.
- When labor demand is elastic ($/\eta_{ii} / > 1$), a 1% increase in the wage will cause employment to fall by more than 1%.
- If labor demand is inelastic ($/\eta_{ii} / < 1$), a 1% wage increase will cause employment to fall by less than 1%.
- Employment will fall by 1% when the wage rises by 1% if labor demand is unit elastic ($/\eta_{ii} / = 1$).
- When labor demand is perfectly inelastic ($/\eta_{ii} / = 0$), a 1% increase in the wage will cause no change employment.

Hicks-Marshall laws .

- *The Hicks-Marshall laws of derived demand state that own-wage elasticity of labor demand will be relatively high when:*
 - ✓ *The price elasticity of demand for the final product is relatively high,*
 - ✓ *It is relatively easy to substitute other factors for this category of labor,*
 - ✓ *The supply of other factors of production is relatively elastic, and*
 - ✓ *This category of labor accounts for a relatively large share of total costs.*

First Hicks-Marshall Law

- *Own-wage elasticity of demand is relatively high when the price elasticity of demand for the final product is relatively high.*
- *This works through the scale effect*
- *When the price elasticity of demand for the final product is relatively high, the increase in the price of the product occurring (because of higher wages and the implied higher marginal costs of production) results in a larger reduction in the quantity of output demanded.*
- *If output falls by more, then the firm will reduce its employment of labor (and all other inputs) by a larger amount.*
- *Therefore, a given change in the wage will result in a larger reduction in the quantity of labor demanded when the price elasticity of demand for the final product is relatively high.*
- *Since a wage change results in a larger reduction in employment when the price elasticity of demand for the final product is relatively high, we can see that labor demand is more elastic in this situation.*

Second Hicks-Marshall Law

- *Ease of Substitutability of Other Factors*
- *The own-wage elasticity of labor demand is relatively high when it is relatively easy to replace labor with other factors of production.*
- *If it is relatively easy to substitute other factors for this category of labor, a wage increase will result in a larger reduction in the quantity of labor demanded.*
- *This works through the substitution effect*

Third Hicks-Marshall Law

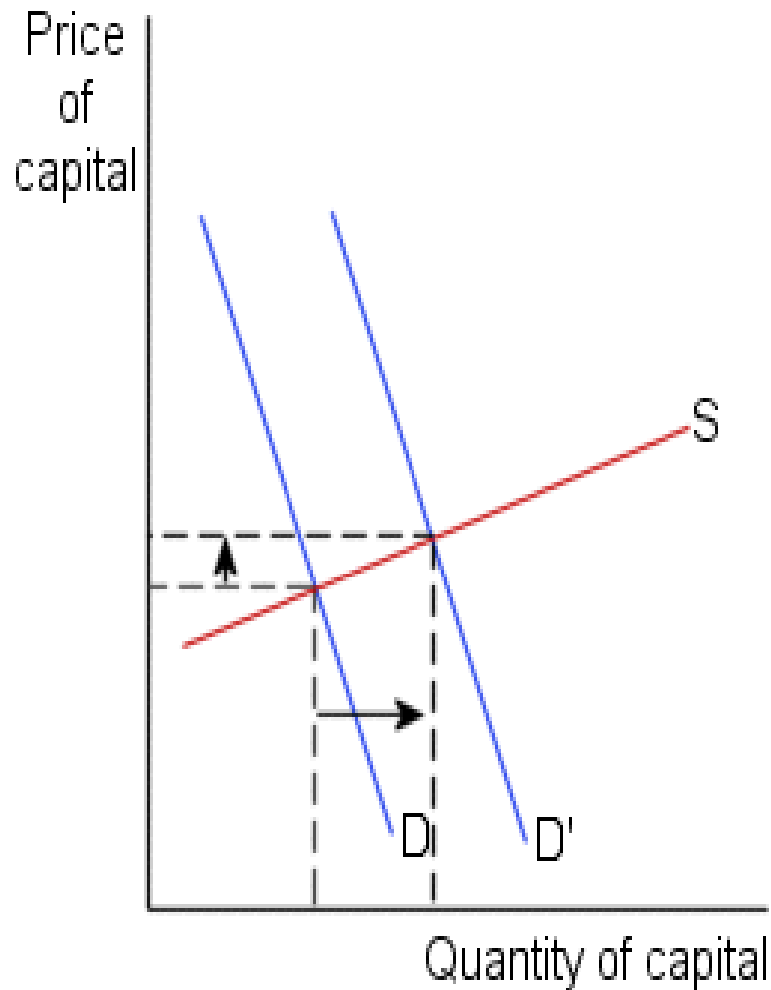
- *Elasticity of the Supply of Other Factors of Production*
- *This works through the substitution effect*

When the wage rate rises, firms will attempt to substitute other factors for labor.

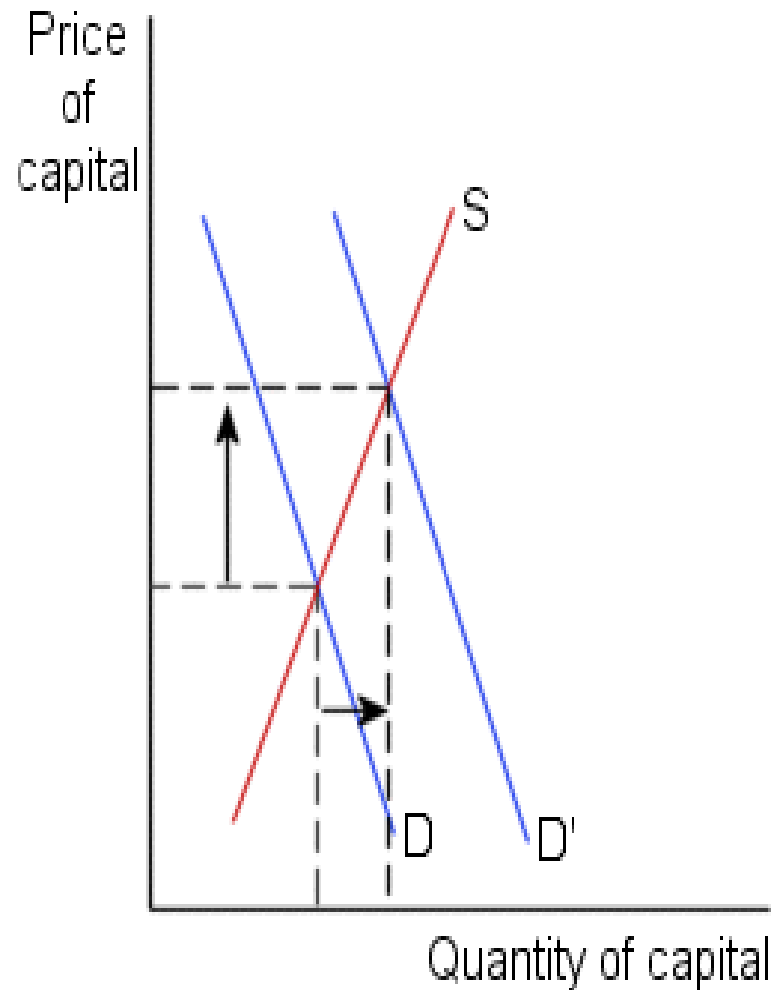
- *As they do so, the demand for these factors will increase.*
- *When the supply of capital is relatively elastic, this increase in demand results in a relatively large increase in the use of capital and a relatively small increase in the price of capital.*
- *When the supply of capital is relatively inelastic, however, the increase in the demand for capital drives up the price of capital by a relatively large amount but has a relatively small effect on the quantity of capital employed by the firm.*

Cont....

More elastic supply



Less elastic supply



Cont....

- *When this occurs, the increase in the price of capital limits the amount of additional capital that will be used as a substitute for labor.*
- *Thus, when the supply of other factors is relatively elastic, the substitution effect will be larger and labor use will fall by a larger amount. Since labor use falls by a larger amount in response to a wage increase when the supply of other factors is more elastic, own-wage elasticity will be relatively high*

Fourth Hicks-Marshall Law

- *The Share of Labor Costs in Total Costs*
- *When labor costs are a larger share of total costs, a wage increase will have a greater effect on the firm's costs and therefore a greater effect on the price of output.*
- *If output price rises by more, the scale effect (output effect + profit effect) will be larger and the reduction in labor use will be greater.*
- *Therefore, an increase in a given category of labor's share of total costs will result in a higher own-wage elasticity of demand for this type of labor.*
- *This works through the substitution effect*

Cross-Wage (Cross-Price) Elasticity of Labor Demand

- *The cross-wage elasticity of labor demand (also known more generally as the cross-price elasticity of demand) is a measure of the effect of the change in the price of one factor of production on the demand for another factor of production.*
- *It is defined as:*

$$\eta_{ij} = \frac{\text{percentage change in the quantity of resource } i \text{ demanded}}{\text{percentage change in the price of resource } j}$$

Cont.....

What is the relevance of the scale effect and the substitution effect in the labor market?

I

Workers and machines work together. Sometimes as complements, sometimes as substitutes.

There's a scale effect: When the price of capital goods or other inputs declines, your business produces output more cheaply so, at a given price, you sell a larger quantity.

- Production takes place on a larger scale, this requires more workers, and this increases the labor demand

Cont.....

There's a substitution effect: In the case of tasks that can be accomplished by workers or machines when the price of machines falls, the demand for labor that can be replaced by machines decreases.

I

The larger of the two effects depends on whether labor and capital are complements or substitutes;

Cont.....

Where labor and capital are complements: Scale effect dominates.

- Then a decrease in price of capital leads to a rightward shift of labor demand curve

Where labor and capital are substitutes: Substitution effect dominates

- Then a decrease in the price of capital leads to a leftward shift of the labor demand curve.

Cont.....

- *A Negative cross-price elasticity of demand between two inputs indicates that the two inputs are **gross complementary***
- *Negative cross-price elasticity occurs when the scale effect is larger than the substitution effect.*
- *A positive cross-price elasticity of demand between two inputs indicates that the two inputs are **gross substitutes**.*
- *This will occur only if the substitution effect outweighs the scale effect.*

Elasticity of Factor Substitution

- *How the relative price of factors affect the quantity of factors to be used in the production ?*
- *The size of this effect depends on the responsiveness of the change of the K/L ratio to the factor price changes.*
- *A measure of this responsiveness is the elasticity of factor substitution (σ).*

Cont.....

$$\begin{aligned}\sigma &= \frac{\text{percentage change in the } K/L \text{ ratio}}{\text{percentage change in the } MRTS_{L,K}} \\ &= \frac{\frac{d(K/L)}{(K/L)}}{\frac{d(MRTS_{L,K})}{(MRTS_{L,K})}} = \frac{d(K/L)}{dMRTS_{L,K}} \cdot \frac{MRTS_{L,K}}{(K/L)}\end{aligned}$$

In perfect input markets, the firm is in equilibrium when it chooses the input combination at which $MRTS_{L,K} = w/r$.

$$\Rightarrow \sigma = \frac{d(K/L)}{d(w/r)} \cdot \frac{(w/r)}{(K/L)}$$

Cont.....

- *The sign of σ is always non – negative because the K/L ratio and w/r ratio move in the same direction: $(w/r) \uparrow$ implies that labor is more expensive. implies that Capital is substituted for labor. implies that $(K/L) \uparrow$ implies that σ is non–negative.*
- *$\sigma = 1$ unitary substitutability*
- *$\sigma < 1$ inelastic substitutability*
- *$\sigma > 1$ elastic substitutability*

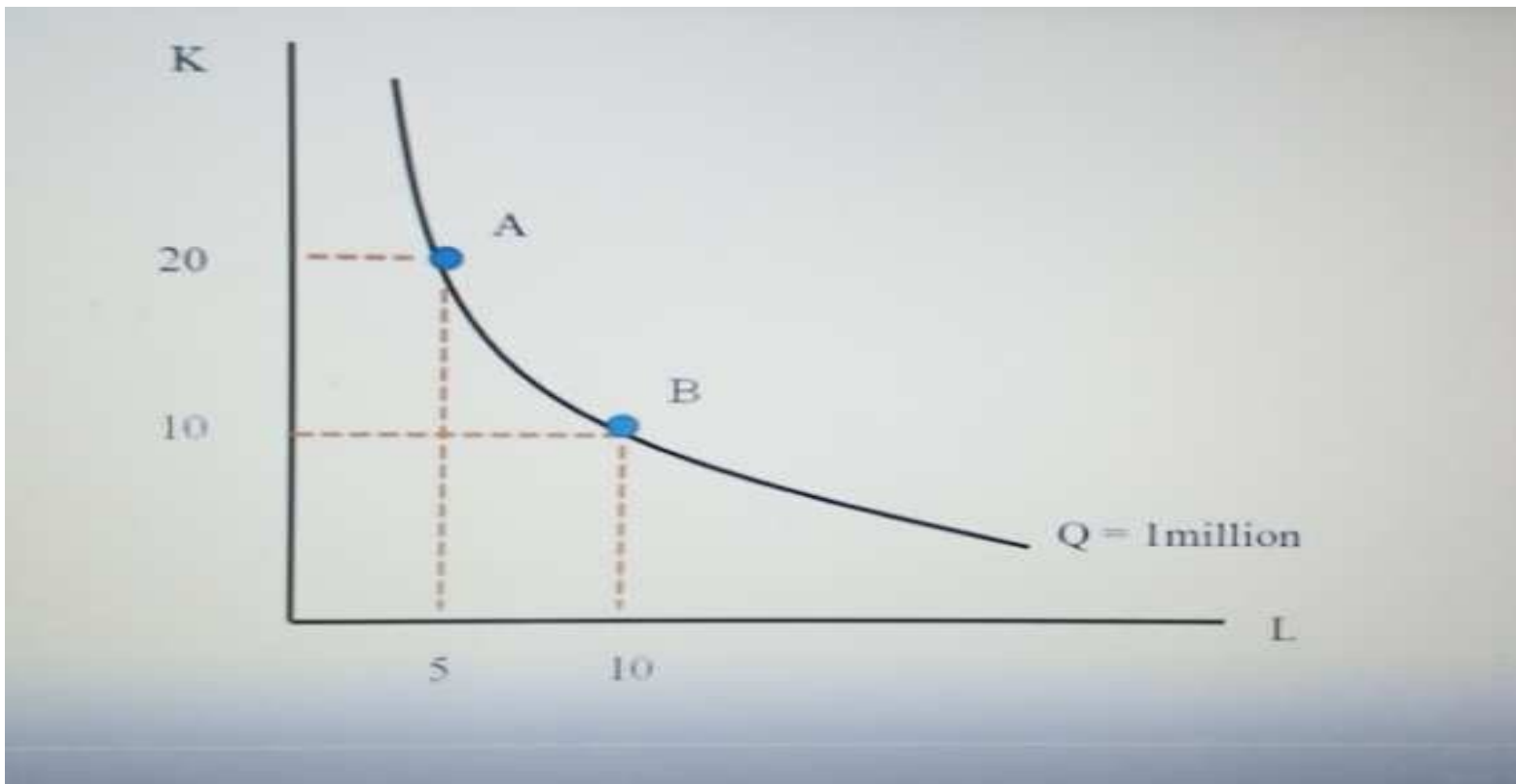
- *There is an important relationship between the values of σ and the distributive shares of factors.*
- ✓ *If $\sigma < 1$, a given percentage change in w/r ratio results in a smaller percentage change in K/L ratio, so that the relative share expression increases.*
- ✓ *Thus, if $\sigma < 1$, an increase in w/r ratio increases the distributive share of labor.*
- ✓ *If $\sigma > 1$, a given percentage change in w/r ratio results in a greater percentage change in K/L ratio, so that the relative share of labor decreases.*
- ✓ *If $\sigma = 1$, a given percentage change in w/r ratio results in an equal percentage change in K/L ratio, so that the relative share of labor remains unchanged*

Cont.....

- *In general, a change in the w/r ratio will cause the share of labor (relative to that of capital):*
- *to change in the same direction if $\sigma < 1$*
- *to change in the opposite direction if $\sigma > 1$*
- *The share of labor is not affected by a change in w/r ratio if $\sigma = 1$.*

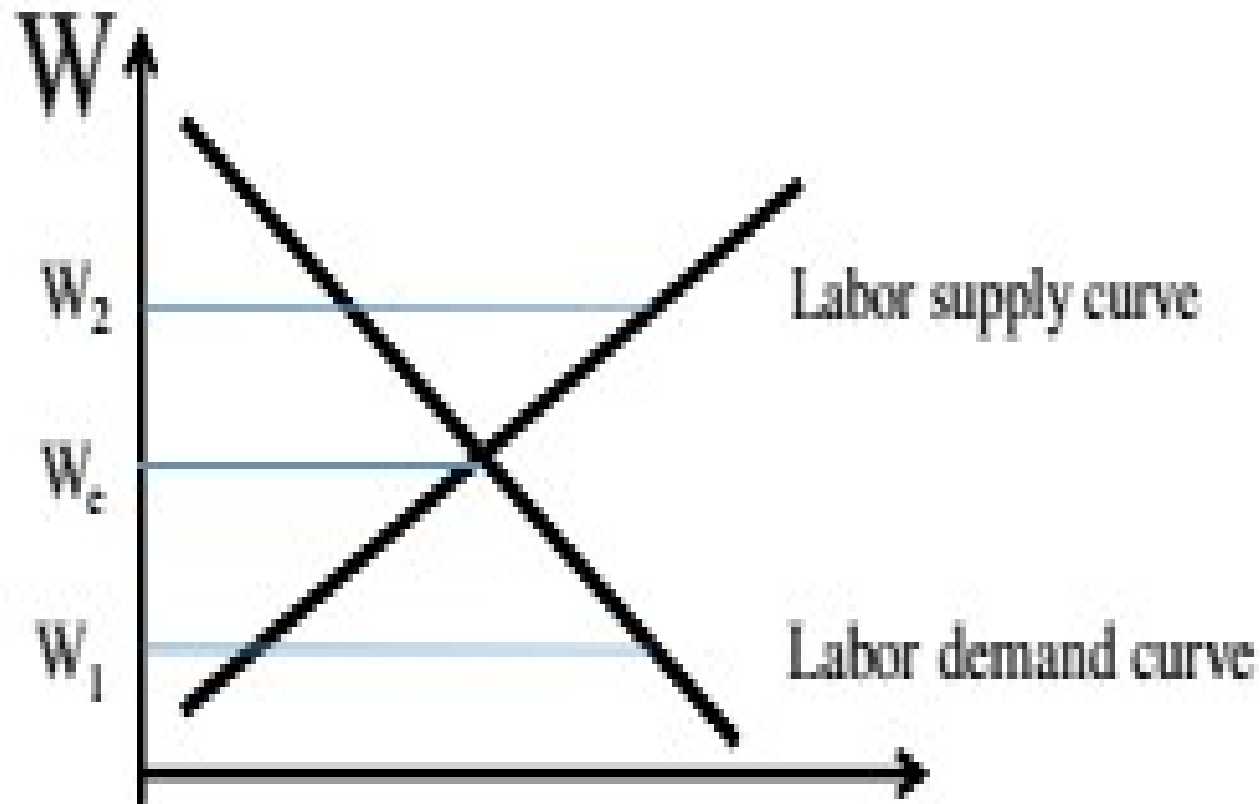
Example 1

✓ Find the Elasticity of Factor Substitution from point A to B



$$Q = L^{0.5} K^{0.25}$$

Labour Market equilibrium curve



Reading assignment

- *Disturbing the equilibrium*
- Try to see the effect of;
 - *Up ward shift of the L.supply curve*
 - Up ward shift of both L.DD and L.SS curves
 - *Dawn ward shift of L.SS curve*
- ✓ *dawn ward shift of the demand curve*
- ✓ *dawn ward shift of the L.DD and L.SS curves*
- ✓ *dawn ward shift of the L.DD and up ward shift of the L.SS curves*
- ✓ *Up ward shift of the L.DD curve and dawn ward shift of L.SS curve.*
- ✓ *On equilibrium wage and level of employmet*

Chapter Four

Wage Determination

Introduction

- *The wage rate is the price of labour per working hour, which could be measured in nominal and/or real terms:*
 - ✓ *Nominal wage* – what workers get paid per hour in current dollars.
 - ✓ *Real wages* or the real purchasing power of a worker's earnings – nominal wages divided by some measure of prices (usually the consumer price index – CPI).
 - ✓ *Earnings* refer to wages multiplied by the number of time units/hours worked.

Cont.....

- *Employee benefits can be either **payments in kind or deferred** .*
- *Examples of payments in kind are employer provided health care, health insurance, and paid vacation time.*
- *Examples of deferred payments are employer-financed retirement benefits – Social Security taxes – set aside money that enables employees to receive pensions later.*
- *Total compensation consists of earnings plus employee benefits.*

Cont.....

- *Income received by a family includes earnings, benefits, and unearned income, which included dividends or interest received on investment and government transfer payments.*
- *The wage rate that prevails in the labour market depends on **LD and LS**, regardless of whether labour unions and/or nonmarket factors are involved.*
- *The wage rate (W_e) at which LD equals LS is the market-clearing wage – that is, no labor surplus and/or no labor shortage.*

Cont.....

- For any wage ($W1$) lower than W_e : $LD > LS \rightarrow EDL$, and with adjustments from employers / demanders, wage rises to W_e .
- For any wage ($W2$) higher than W_e : $LD < LS \rightarrow ESL$, and with adjustments from workers / suppliers, wage falls to W_e .
- Changes in labour demand or changes in labour supply or the simultaneous changes in labour demand and supply will change the equilibrium wage (W_e) and employment (L).
- **Economic rents** can be defined as the difference between the wage workers are actually paid on a job and the workers' reservation wages.

Wage determination in perfectly competitive labour market

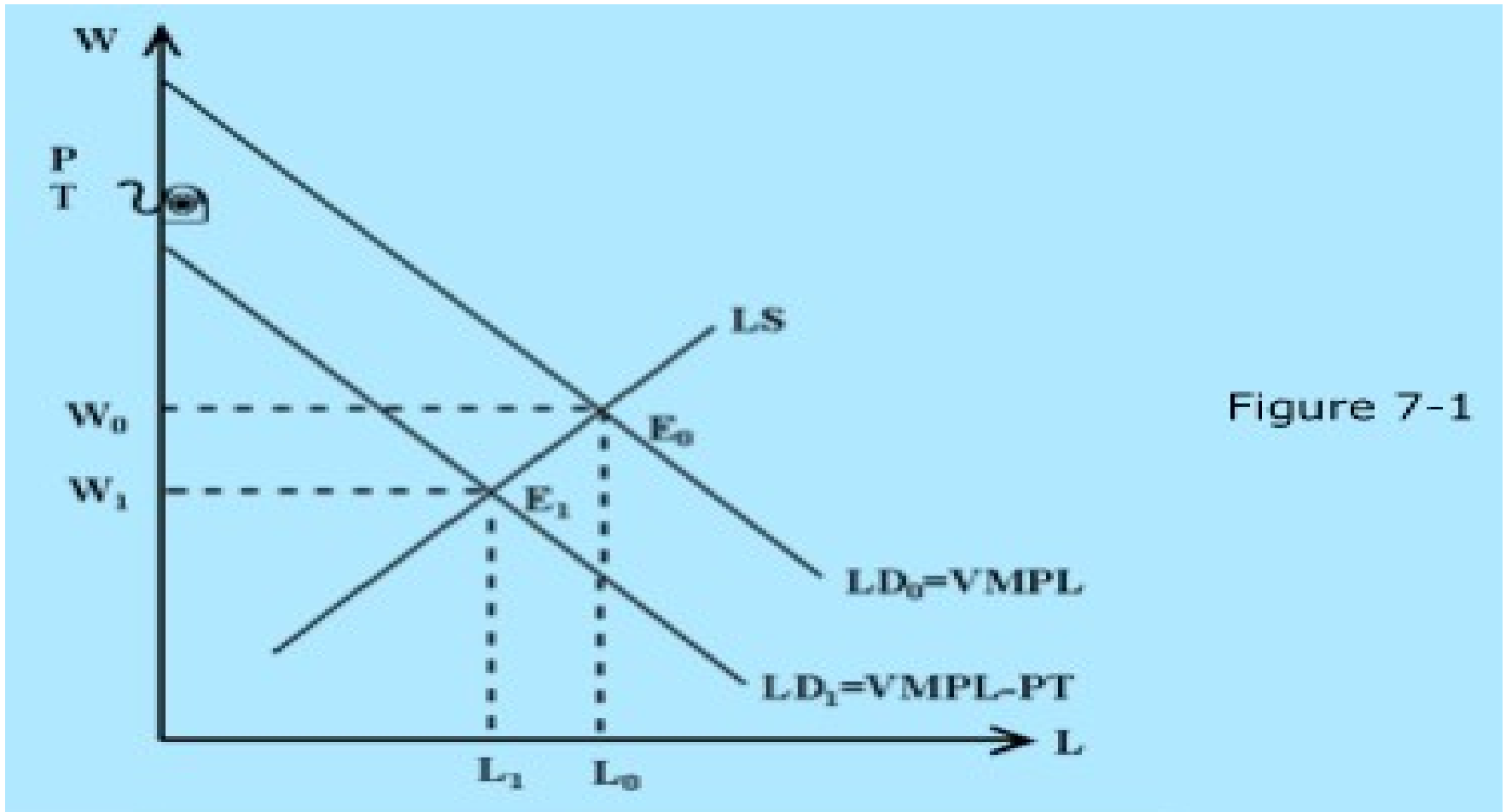
- *Wages and employment in a single labour market*
- *Assuming that firms can buy all the labour they wish at the going wage rate (firms are wage takers in the labour market),*
- *the market labour demand curve for perfectly competitive firms (firms that are price takers in the product market) is the horizontal summation of the labour demand curves of the individual firms*
- *As discussed in the previous two chapters, a perfectly competitive firm's labour demand curve is the value of the marginal product of labour ($VMPL = P * MPL$) schedule minus any (amortized) fixed costs.*

Cont.....

- *If the firm is a monopolist, the market demand curve corresponds to the monopolist's marginal revenue product of labour ($MRPL = MR * MPL$) curve minus any (amortized) fixed costs.*
- *The market labour supply curve is the horizontal summation of the individual labour supply curves (as derived in chapter 2)*
- *The intersection of the market labour demand curve and the market labour supply curve determines the equilibrium wage rate and the equilibrium employment level*

Cont.....

- Given the LD_0 and LS curves in figure 7-1, the competitive market equilibrium occurs at point E_0 ;
- the equilibrium wage rate w_0 'clears' the labour market .



- Assuming 'linear' curves, the labour demand and supply curves can be written as:
- $LD = a + bw + gx$, where b is a negative parameter and X represents exogenous (shift) variables such as product price, payroll taxes..
- $LS = c + fw + hz$, where Z represents exogenous (shift) variables such as consumer prices and income taxes
- Since economists have traditionally plotted the wage rate on the vertical axis of a labour demand and supply diagram, these two equations have to be re-arranged with W on the left hand side of the equation to represent the traditional labour demand and supply curves drawn in figure 7-1
- In figure 7-1 the vertical intercept of the labour demand curve would be $-a/b$ (recall that b is a negative parameter) and the vertical intercept of the labour supply curve would be $-c/f$

Cont.....

- *Reduced form equations for the equilibrium wage W and employment level L can be obtained by solving the structural equations for labour demand LD and labour supply LS*
- *Equating labour demand LD and labour supply LS produces the following equation*
- *$LDD = LSS$ implies that $a + bw + gx = c + fw + hz$,*
- *Which can be re-arranged to obtain the reduced form equation for W*
- *$W = [a + gx - c - hz] / [f - b]$*
- *Substituting the W reduced form equation into $L = LD = a + bw + gx$,*
- *$L = a + gx + b [a + gx - c - hz] / [f - b]$*
- *And re-arranging produces the following reduced form equation for L*
- *$L = [af - bc + gfx - bhz] / [f - b]$*

Cont.....

- *the W and L reduced form equations allow us to determine the effects on the equilibrium wage and employment level from a change in an exogenous factor (X or Z) .*
- *for example, if X increases by one unit, the equilibrium wage W will increase by $g/[f - b]$ and the equilibrium employment level L will increase by $gf/[f - b]$*
- *again recall that b is a negative parameter and thus the divisor $[f - b]$ is a positive number*

Shifts in labour market equilibrium

- *An increase in labour demand results in an increase in both the equilibrium wage and the equilibrium level of employment,*
- *A reduction in labour demand results in a decrease in both the equilibrium wage and the equilibrium level of employment,*
- *An increase in labour supply results in a lower equilibrium wage, but a higher equilibrium level of employment, and*
- *A reduction in labour supply results in a higher equilibrium wage, but a lower equilibrium level of employment.*
- *We now examine how equilibrium wages and employment levels in a competitive labour market are affected by the imposition of a payroll tax .*

The Labour Market Effects of Payroll Taxes

- *A **payroll tax** is a tax levied on employers with the amount of tax payable based on the size of the payroll (the wage rate multiplied by the quantity of labour employed).*
- *how it affects wage and employment level ?*
- *Employer responsibilities*
- *All employers are required by law to deduct **Payroll Taxes to the government** .*
- *Example Employers are required by law to deduct Canada Pension Plan (CPP) contributions and unemployment insurance (UI) premiums from most amounts they pay to their employees.*
- *Employers must remit these amounts to the Canada Revenue Agency (CRA) along with their share of CPP contributions and UI premiums.*

Cont.

- *UI premiums: in 2002 employers paid 3.08% of payroll costs for annual labour earnings up to \$39,000 .*
- *In 2002 the maximum annual UI premium paid by the employer for one employee is \$1,201.*
- *Cpp premiums: in 2002 employers paid 4.7% of payroll costs for annual labour earnings from \$3,500 to \$39,100 (the cap); in 2002 the maximum annual CPP premium paid by the employer for one employee is \$1,673.20.*
- *for employees earning less than the cap (\$39,000), in 2002 employers pay a combined UI and CPP premium of 7.78% of the employee's earnings; the maximum annual premium paid to the federal government by the employer for one employee is \$2,874.20.*

Cont.....

- *The firm must pay part of the value of the marginal product of labour (VMPL) to governments in the form of UI, CPP, and WC premiums.*
- *The demand for labour curve will be represented by the vmpl schedule minus the payroll tax (pt)*
- *A profit-maximizing firm will hire labour up to the point where $w = vmpl - pt$.*
- *For example, if the vmpl for the n th worker is \$110 and payroll taxes are 10%, a profit-maximizing firm will hire the n th worker as long as the wage is not greater than \$100 ($w = vmpl - pt$)*
- *The imposition of a payroll tax shifts the labour demand curve downwards by the amount of the payroll tax*

Cont.....

- *Figure 7-1 illustrates the labour market effects of the imposition of a payroll tax*
- *With no payroll tax, the labour demand curve is represented by the $l_{do} = VMPL$ line and the equilibrium position in the labour market is e_o ; the equilibrium wage rate is w_o and the equilibrium quantity of labour is l_o .*
- *Imposing a payroll tax shifts the labour demand curve down to $LD1 = VMPL - PT$.*
- *The vertical intercept of the labour demand curve shifts down by PT , the portion of the $VMPL$ which must be paid to the government (payroll taxes) the new equilibrium position in the labour market is $E1$; the equilibrium wage rate decreases to $w1$ and the equilibrium quantity of labour decreases to $L1$*

Cont.....

- *The imposition of a payroll tax (paid by the employer) reduces the demand for labour, the equilibrium wage paid to labour and the equilibrium quantity of labour .*
- *In the long run (when all factors are variable in the production process), the imposition of a payroll tax on labour will cause profit-maximizing firms to substitute other non-taxed factors of production (such as capital) for labour, further reducing the demand for labour*
- *The incidence of the payroll tax (who actually pays it) depends on the elasticity's (slopes) of the labour demand and supply curves*

Cont.....

- Given a downward-sloping labour demand curve, *employees will pay part of the payroll tax levied on firms in the form of lower wage rates (and less employment).*
- If the labour supply curve were perfectly inelastic (vertical in figure 7-1), then the equilibrium wage rate will fall by the full amount of the *employer-paid payroll tax* and there will be *no effect on employment levels* .
- while the employer remits the payroll tax to the government, the tax is shifted back on to the employee who implicitly pays the payroll tax (wages decline by the full amount of the payroll tax if the labour supply curve is *perfectly inelastic*)

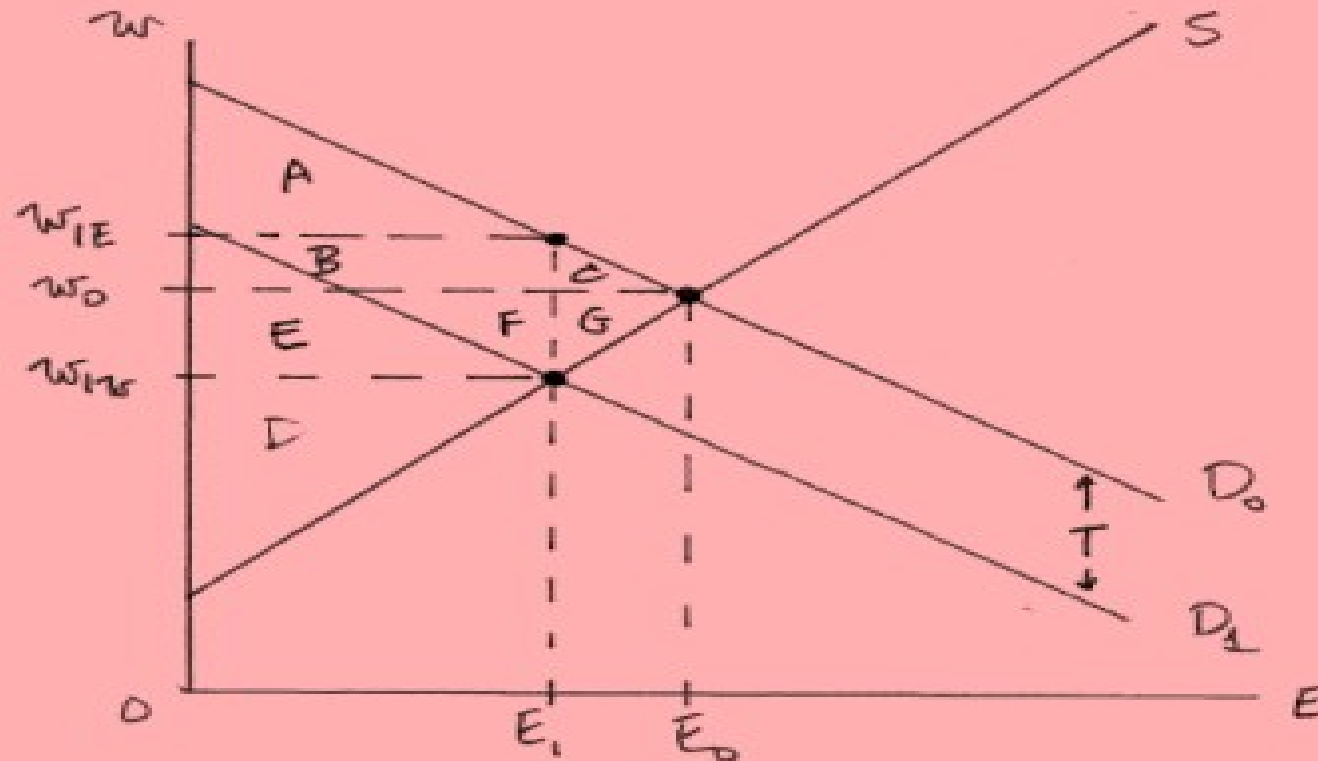
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- *From the reduced form equations above, a 1 unit increase in X from an increase in payroll taxes results in a $g/[f-b]$ decrease in wages and a $gf/[f-b]$ decrease in employment;*
- *The size of these wage and employment effects depends on **b , the slope of the LD curve, and f , the slope of the LS curve***
- *The growing body of empirical evidence seems to support the conclusion that in the long run ... the incidence of payroll taxes falls largely on workers, and that the misemployment effect is small."*

Cont.....

- *Workers are the big losers when the government imposes payroll taxes on employers;*
- *while the employer remits the payroll tax to the government, the payroll tax comes out of the pockets of the employees*
- *Besides any loss in employment, a payroll tax paid by employers reduces the wage rate received by employees and thus reduces employees' utility level (the employee's income budget-line in the income-leisure choice model rotates inwards given a fall in the wage rate).*
- *Since UI and CPP premiums are also paid by employees (both employees and employers pay UI and CPP premiums), workers suffer a double hit in utility .*
- *UI and CPP premiums lower 'net' take-home pay in two ways: there is a **lower 'gross' wage** (because the firm shifts employer-paid UI and CPP premiums back onto the employee in the form of lower wages) and the employee also pays UI and CPP premiums which further reduce 'net' pay (after tax deductions)*

Cont.....



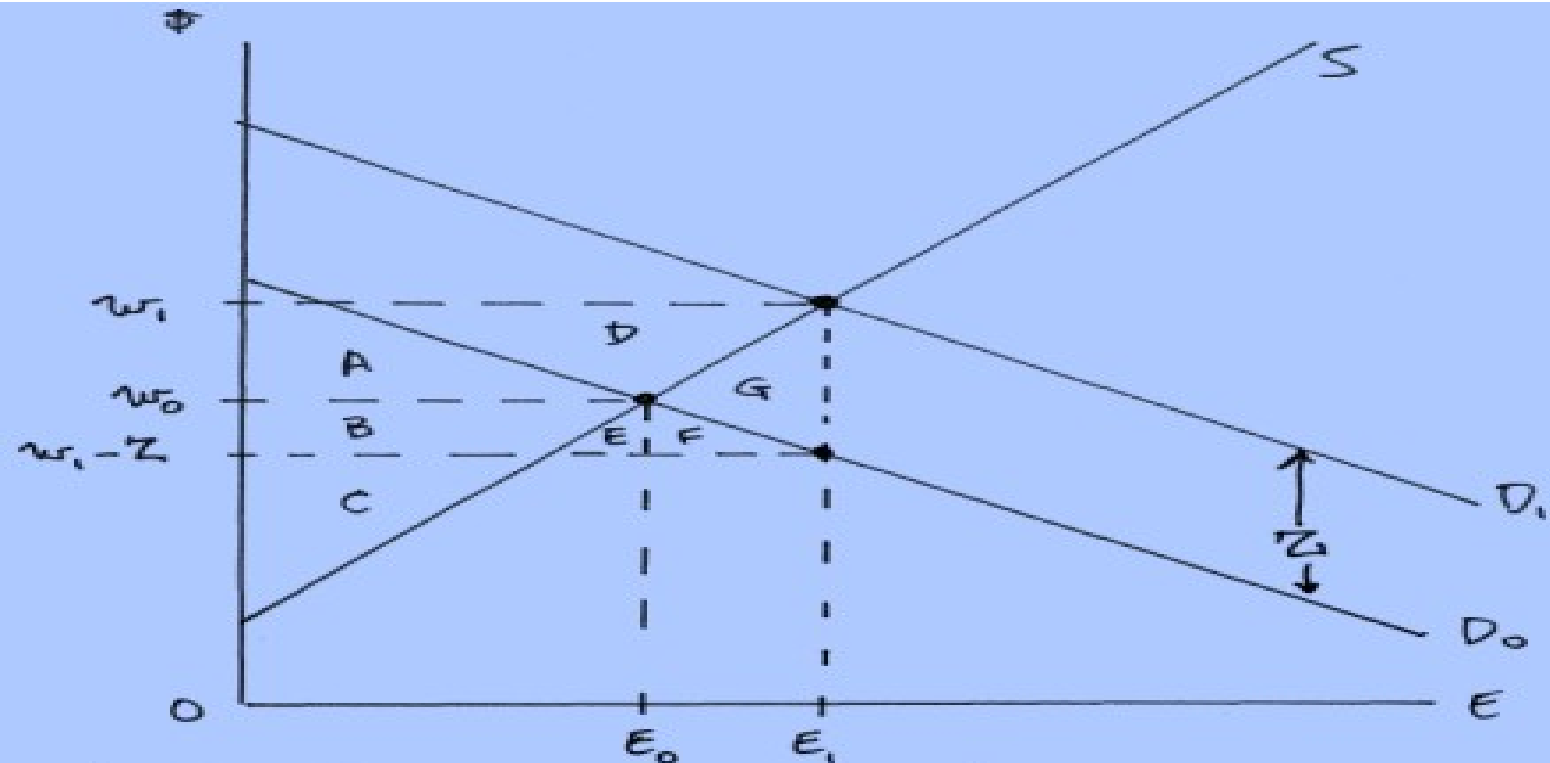
surplus	before	after	change
worker	$D+E+F+G$	D	$-(E+F+G)$
employer	$A+B+C$	A	$-(B+C)$
government	0	$B+E+F$	$B+E+F$
deadweight loss			$-(G+C)$

Cont.....

➤ *net impact of subsidy:*

- *workers and employers end up sharing the subsidy (workers get a higher wage, firms pay a lower wage)*
- *employment rises*
- *subsidy generates a deadweight loss (sum of employers' + workers' surplus falls) – intuitively, this is because labour is now paid more than the value of its marginal .*
- *product (VMPL): $w_{1w} > w_{1E} = VMPL$*

Cont.....



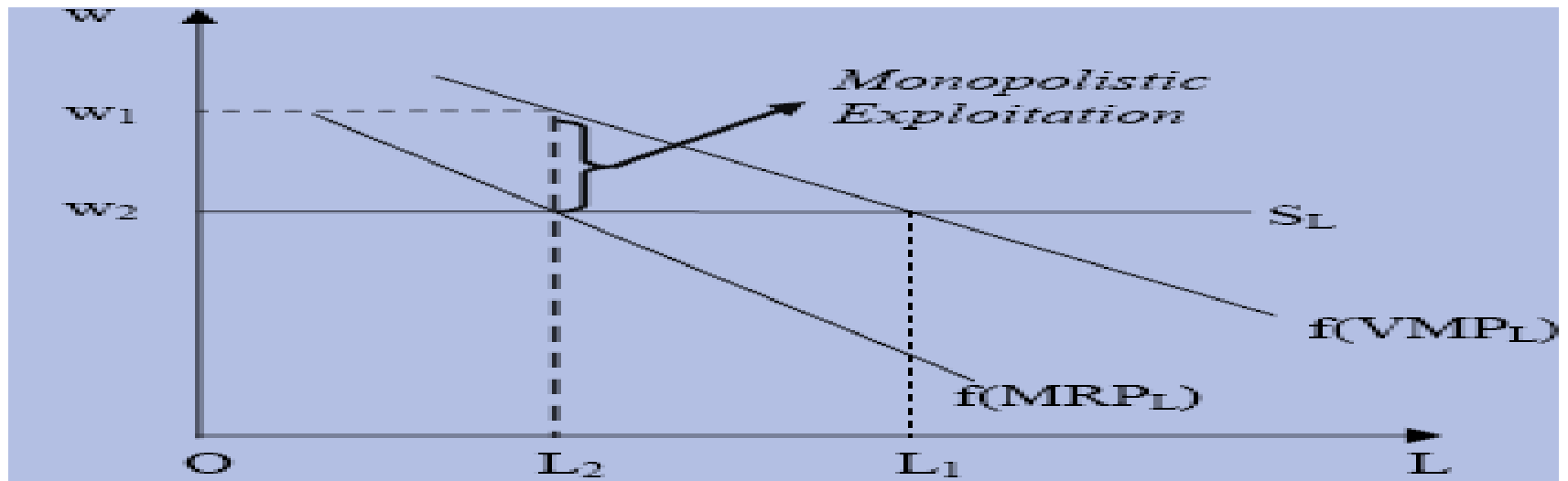
* subsidy = Z per worker $\times E_1$ workers
 = area of $(A + B + D + E + F + G)$

surplus	before	after	change
worker	$B+C$	$A+B+C+D$	$A+D$
employer	A	$A+B+E+F$	$B+E+F$
* government	0	$-(A+B+D+E+F+G)$	$-(A+B+D+E+F+G)$
deadweight loss			$-G$

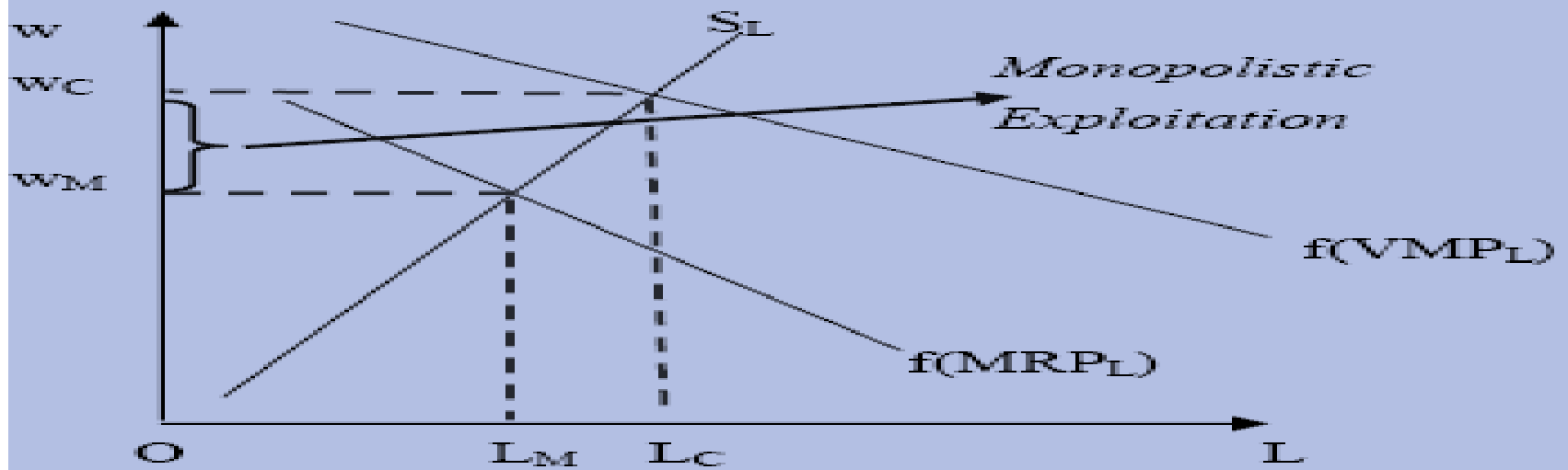
Monopoly in the product market

- *When firms have monopolistic power, labor is paid its MRP, which is smaller than the VMPL.*
- *This effect is called **monopolistic exploitation**.*
- *It represents the difference between the amount labour is paid under perfect competition and the amount the same factor (labor) is paid under the imperfection introduced to the product market.*
- *If firms under the two scenarios have to use the same amount of labor (L_2 in panel (a) of Figure 4.2) the firm in perfect competition pays a wage rate of w_1 while the other firm pays w_2 .*
- *$w_1 - w_2$ measures the level of monopolistic exploitation by the firm.*
- *The same concept is depicted in panel (b), but at the market level.*

Cont.....



(a) Firm



(b) Market

Monopolistic Exploitation (a) at Firm's Level, and (b) at the Market Level

The cobweb model

- *Our analysis of labor market equilibrium assumes that markets adjust instantaneously to shifts in either supply or demand curves, so that wages and employment change swiftly from the old equilibrium levels to the new.*
- *Many labor markets, however, do not adjust so quickly to shifts in the underlying supply and demand curves.*
- *There is some evidence, in fact, that markets for highly skilled workers, such as engineers and other specialized professionals, exhibit systematic periods of booms and busts which dispute the notion that labor markets attain competitive equilibrium quickly and cheaply.*
- *Consider, for example, the market for new engineering graduates.*
- *It has long been recognized that the market for newly minted engineers fluctuates regularly between periods of excess demand for labor and periods of excess supply.*

Cont.....

- *As a result, there is a cyclical trend in the entry wage of engineering graduates over time.*
- *In a series of provocative studies, Richard Freeman proposed a model that showed how these trends in the entry wage could be generated.*
- *Two key assumptions underlie the model:*
 - *It takes time to produce a new engineer and*
 - *Individuals decide whether or not to become engineers by looking at conditions in the engineering labor market at the time they enter school.*
- *Initially, labor market is in equilibrium where the supply curve S intersects the demand curve D , so that there are E_0 new engineering graduates and the entry wage is W_0*

Cont.....

- *Suppose there is a sudden increase in the demand for newly trained engineers.*
- *The demand curve for engineers shifts to D' , and firms would like to hire $E^*(E^* > E_0)$ new engineers at a wage of $W^*(W^* > W_0)$.*
- *Firms will find it extremely difficult to hire this desired number of new engineers.*
- *New engineers do not come out of thin air simply because firms want to hire them. It takes time to train new engineers.*
- *Because engineering schools are only producing E_0 engineers annually, the short-run supply curve is perfectly inelastic at E_0 workers.*
- *The combination of this inelastic supply curve and the demand shift increases the entry wage of engineers to W_1 ($W_1 > W^*$)*

Cont.....

- *While all this is happening in the engineering labor market, a new generation of high school and college students is deciding whether to enter the engineering profession.*
- *These students see a relatively high wage in the engineering market and, hence, have a large incentive to become engineers.*
- *In fact, at the current wage of W_1 , a total of E_1 persons will want to enroll in engineering schools.*
- *After a few years, therefore, E_1 new engineers enter the marketplace. At the time in which this cohort of engineers enters the market, the short-run supply of new engineers is again perfectly inelastic at E_1 workers*
- *Hence the current market situation is summarized by this inelastic supply curve and the demand curve D' (assuming that demand conditions have not changed any further).*
- *Equilibrium occurs at a wage of W_2 , which is substantially below the wage that the new engineers thought they were going to get ($W_2 < W^*$).*
- *In effect, high school and college graduates presumed that they would get a wage of W_1 dollars; therefore, there was an oversupply of engineers.*

Cont.....

- *But this is not the end of the story.*
- *Still another generation of high school and college students is trying to decide whether to become engineers.*
- *At the current low wage of W_2 the engineering profession does not look very attractive, and hence few persons will decide to attend engineering school.*
- *At a wage of W_2 only E_2 persons become engineers.*
- *When these students graduate and enter the labor market, the entry wage rises to W_3 ($W_3 > W_2$) because there was an undersupply of engineers.*
- *This high wage induces the next generation of students to oversupply to the marketplace, and so on.*
- *The analysis illustrates the cobweb that is created around the equilibrium point as the engineering labor market adjusts to the initial demand shock.*
- *The entry wage exhibits a systematic pattern of booms and busts as the market slowly drifts toward its long-run equilibrium wage w^* and employment E^* .*

The cobweb model curve



The initial equilibrium wage in the engineering market is w_0 . The demand for engineers shifts to D' , and the wage will eventually increase to w^* . Because new engineers are not produced instantaneously and because students might misjudge future opportunities in the market, a cobweb is created as the market adjusts to the increase in demand.

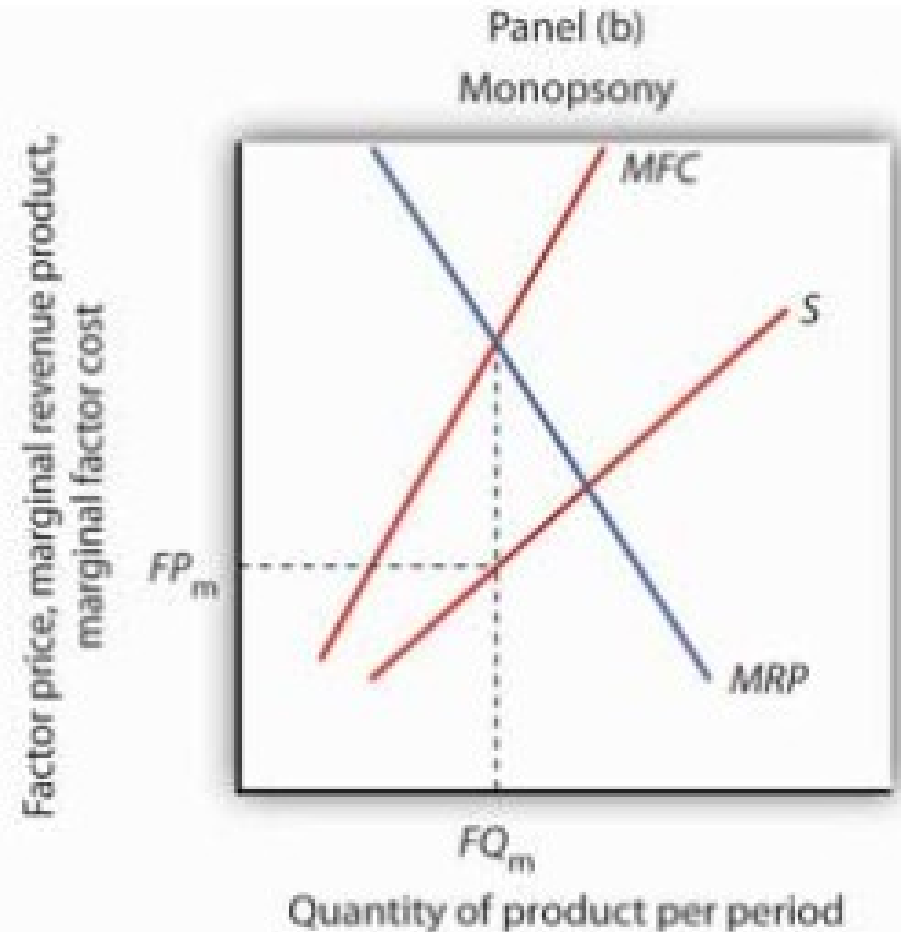
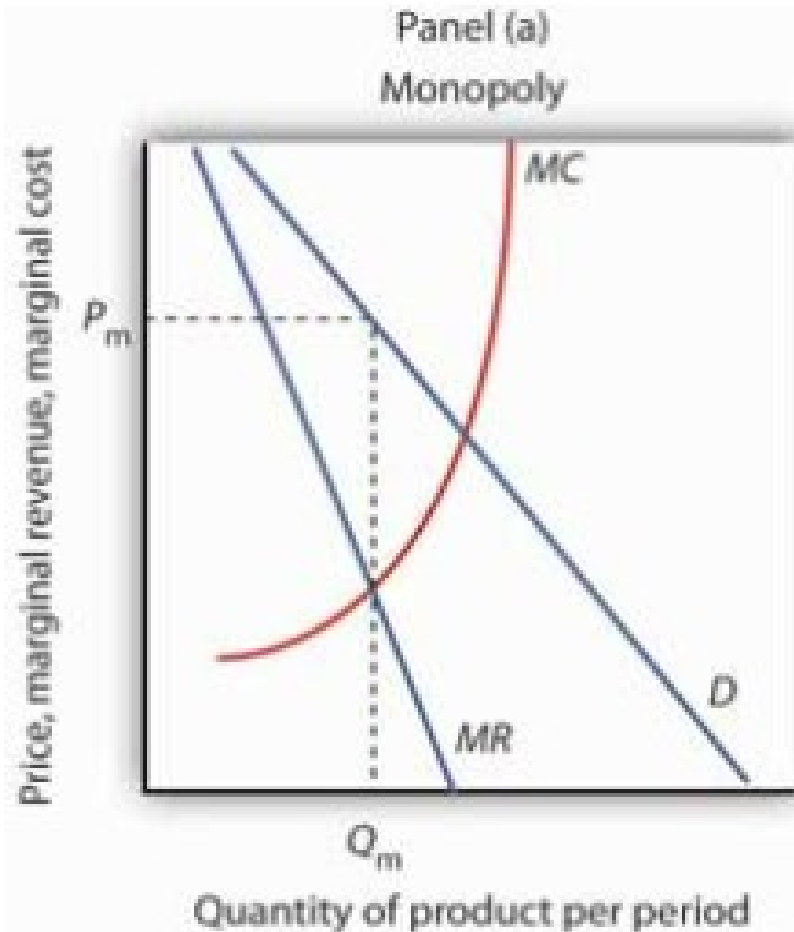
- ✓ *In any occupation, the higher the wage, the more people are willing to work in it, so the supply curve is upward sloping. On the other hand, the higher the wage, the fewer workers employers are willing to hire, so the demand curve is downward sloping.*

Monopsony

■ *Monopoly and Monopsony*

- ✓ *Both types of firms are price setters:*
- ✓ *The monopoly is a price setter in its product market; the monopsony is a price setter in its factor market.*
- ✓ *Both firms must change price to change quantity:*
- ✓ *The monopoly must lower its product price to sell an additional unit of output, and the monopsony must pay more to hire an additional unit of the factor.*
- ✓ *Because both types of firms must adjust prices to change quantities, the marginal consequences of their choices are not given by the prices they charge (for products) or pay (for factors).*
- ✓ *For a monopoly, marginal revenue is less than price; for a monopsony, marginal factor cost is greater than price.*

Monopoly and Monopsony



Monopsony

- *Is the sole employer of labour in a given profession.*
- *In monopsony (“one single buyer”) there is little or no competition among firms.*
- *so that the individual employer enjoys some, or even the full market power.*

Modern cases of monopsony:

- *A firm which is monopolist in the product market is likely to be monopsonist in the labour market if it is the only ‘buyer’ of a particular type of labour:*
- *E.g. An hospital hiring specialised nurses.*
- *Oligopoly: few employers*

Cont.....

- *As a consequence of its market power, the monopolistic firm gains positive profits.*
- *The monopsonistic employer faces directly the aggregate labour supply curve, which is upward sloping implying that he can choose the wage along that curve, even if the wage is reduced,.*
- *the firm does not lose all its workforce!*
- *a monopolist faces a downward-sloping product demand curve and sets the price (by restricting output) to maximize profits*
- *a monopolist equates marginal revenues (MR) to marginal costs*
- *Figure 7-3 depicts the monopsonist's upward-sloping labour supply (LS) curve*

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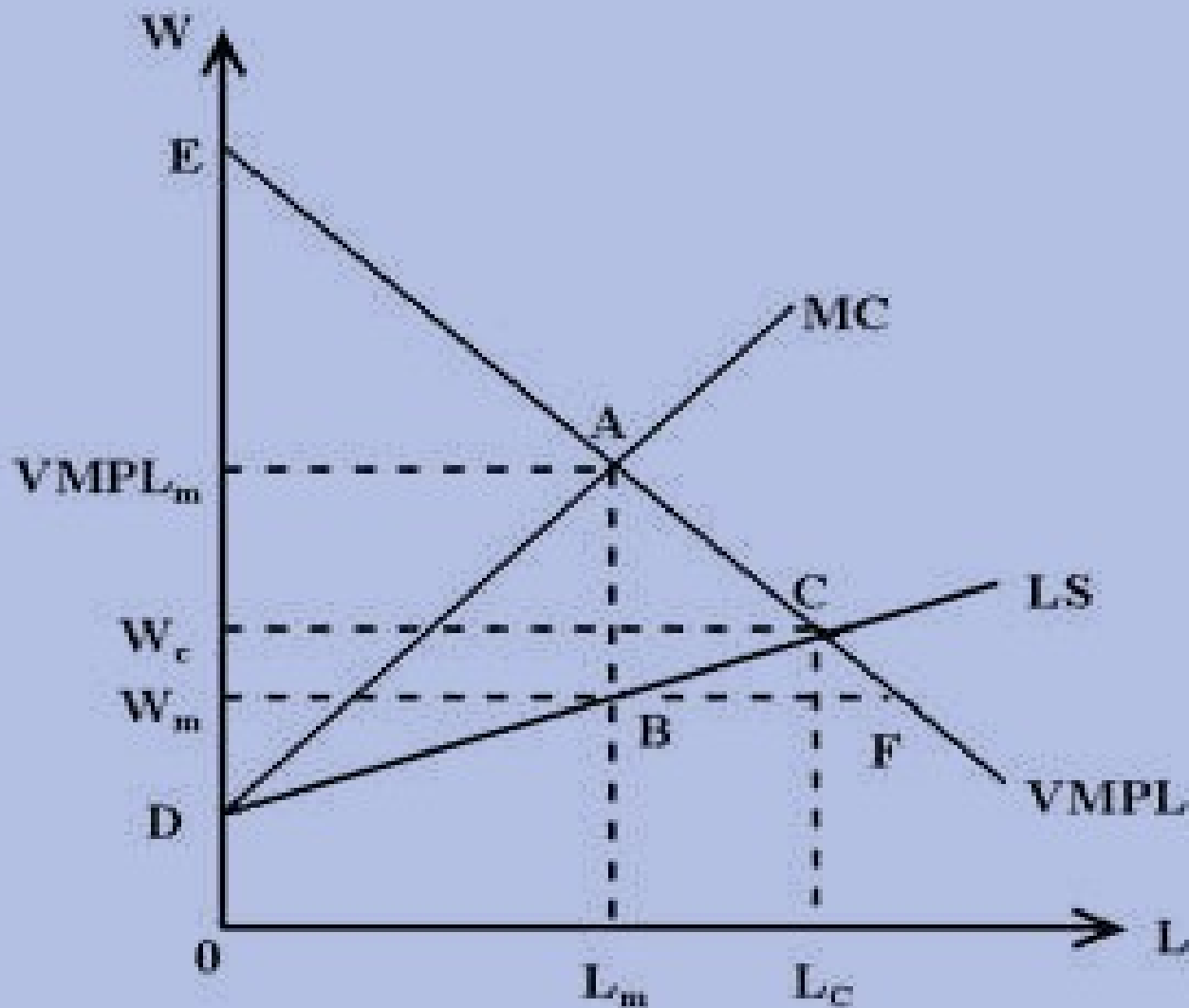


Figure 7-3

Cont.....

- *To attract and hire additional labour, the monopsonist must raise the wage (LS slopes up)*
- *If a monopsonist reduces the wage, some (but not all) workers will withdraw their labour services .*
- *If a perfectly competitive firm reduces the wage below the market-clearing level, all workers will leave the firm*
- *Given an upward-sloping labour supply (LS) curve, there is an upward-sloping marginal cost (MC) of labour curve*
- *To hire additional workers the monopsonist has to raise the wage rate; since all existing employees will also receive the new higher wage, the marginal cost of hiring an additional worker is greater than the wage paid to the additional worker*

Cont.....

- *For example, suppose that a firm has 10 employees being paid \$100 each; to hire an additional worker, the firm must raise the wage rate, say to \$105; the marginal cost of adding the 11th worker is \$155, i.E., \$105 for the new worker and an additional \$5 for each of the 10 workers already employed*
- *The slope of the MC curve is steeper than the slope of the LS curve*

$$\pi = Y(L) - w(L)L \quad \text{firm's profits (Y(L) is total product value)}$$

From profit maximisation we derive the following firm's equilibrium condition

$$\underbrace{y(L)}_{\text{marginal of hiring benefit}} = \underbrace{w + \frac{\partial w}{\partial L} L}_{\text{marginal of hiring cost}}$$

where $y(L)$ is the marginal product value

Cont.....

- *A profit-maximizing monopsonist will keep hiring workers up to the point where the marginal cost (MC) of hiring an additional worker is equal to the value of the marginal product of labour (VMPL), or equal to the marginal revenue product of labour (MRPL) if the monopsonist is also a monopolist.*
- *In figure 7-3, profit maximization occurs at point A, the intersection of the marginal cost (MC) of labour curve and the VMPL curve*
- *The monopsonist will hire l_m workers, pay each worker w_m , and make a profit equal to the profit triangle AED (the sum of $VMPL - MC$ for each worker).*
- *The last worker hired by the monopsonist has a $vmpl_m >$ the wage (w_m).*
- *At the profit-maximizing wage w_m , there are vacancies equal to the horizontal distance BF in figure 7-3; the profit-maximizing monopsonist does not fill these vacancies because that would require an increase in wages for all existing employees.*
- *Compared to perfect competition in the labour market (point C in figure 7-3), the monopsonist hires fewer workers ($l_m < l_c$) and pays each worker a lower wage ($w_m < w_c$).*

Cont.....

The degree of monopsonistic power

$\pi = Y(L) - w(L)L$ profits; we can derive the max profits condition by deriving π with respect to employment L .

$$\frac{\partial \pi}{\partial L} = 0 \rightarrow y(L_m) = \frac{\partial w}{\partial L} L + w_m \quad \text{where } y(L_m) \text{ is the marginal product value.}$$

By dividing and multiplying the right-hand side by w we obtain

$$y(L_m) = w_m (\varepsilon + 1) \quad \text{where } \varepsilon = \frac{\partial w}{\partial L} \frac{L}{w} \text{ is the inverse supply elasticity.}$$

Thus we can measure the monopsony power as $\frac{y(L_m) - w_m}{w_m} = \varepsilon$

The lower the supply elasticity (the higher ε), the higher the monopsony power, and the larger the gap between the marginal product and the wage paid to the workers.

On the contrary, if elasticity tends to infinity ($1/\varepsilon \rightarrow \infty$ that is $\varepsilon \rightarrow 0$), meaning that workers are able to move freely from an employer to another according to their utility, the monopsony power vanishes and the wage equates the marginal product.

Cont.....

■ Remedies

- **Minimum wage:** *contrary to what happens in a competitive market, in a monopsony a minimum wage may increase both efficiency and equity at the same time!*
- *Fostering competition and supporting workers' mobility (output markets liberalisation, antitrust laws, active labour market policies and training, low price transportation facilities, information dissemination)*
- *Collective bargaining institutions: laws and other 'industrial relations' arrangements supporting collective bargaining, so that the market power of the monopsonist is counterbalanced by workers' unions. (Collective bargaining is when a representative of the labor union sits down with the representative of the firm or the management to negotiate about wages and working conditions.)*
- *In all these cases the elasticity of labour supply increases and the monopsony power falls*

Cont.....

Monopsony: exercise 1

$w(L)=10+0.5L$ labour supply faced by the monopsonistic employer
 $mhc=10+L$ as $mhc=w+\partial w/\partial L \cdot L=(10+0.5L)+0.5L$

$y(L)=85-2L$ labour marginal productivity

Monopsony equilibrium:

$$\begin{aligned} 10+L &= 85-2L & L_m &= 75/3=25 & w_m &= 10+0.5 \cdot 25=22.5 & y(L_m) &= 85-2 \cdot 25=35 \end{aligned}$$

monopsony power $(y(L_m)-w_m)/w_m=(35-22.5)/22.5=0.55$

Perfect competition equilibrium:

$$\begin{aligned} 10+0.5L &= 85-2L & L^* &= 75/2.5=30 & w^* &= 10+0.5 \cdot 30=25 \end{aligned}$$

Lost surplus in monopsony $\frac{1}{2}(y(L_m)-w_m)(L^*-L_m)=1/2 \cdot 12.5 \cdot 5=31.25$

Total potential surplus (in perfect competition) $\frac{1}{2}(85-10)L^*=1/2 \cdot 75 \cdot 30=1125$

% Lost surplus in monopsony $31.25/1125 \approx 2.8\%$

Cont.....

Minimum wage in perfect competition

$\hat{w}=27$ above $w^*=25$

Supply	$27=10+0.5L$	$L=17/0.5=34$
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Demand	$27=85-2L$	$L=58/2=29$
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Unemployment	supply <i>minus</i> demand	$34-29=5$
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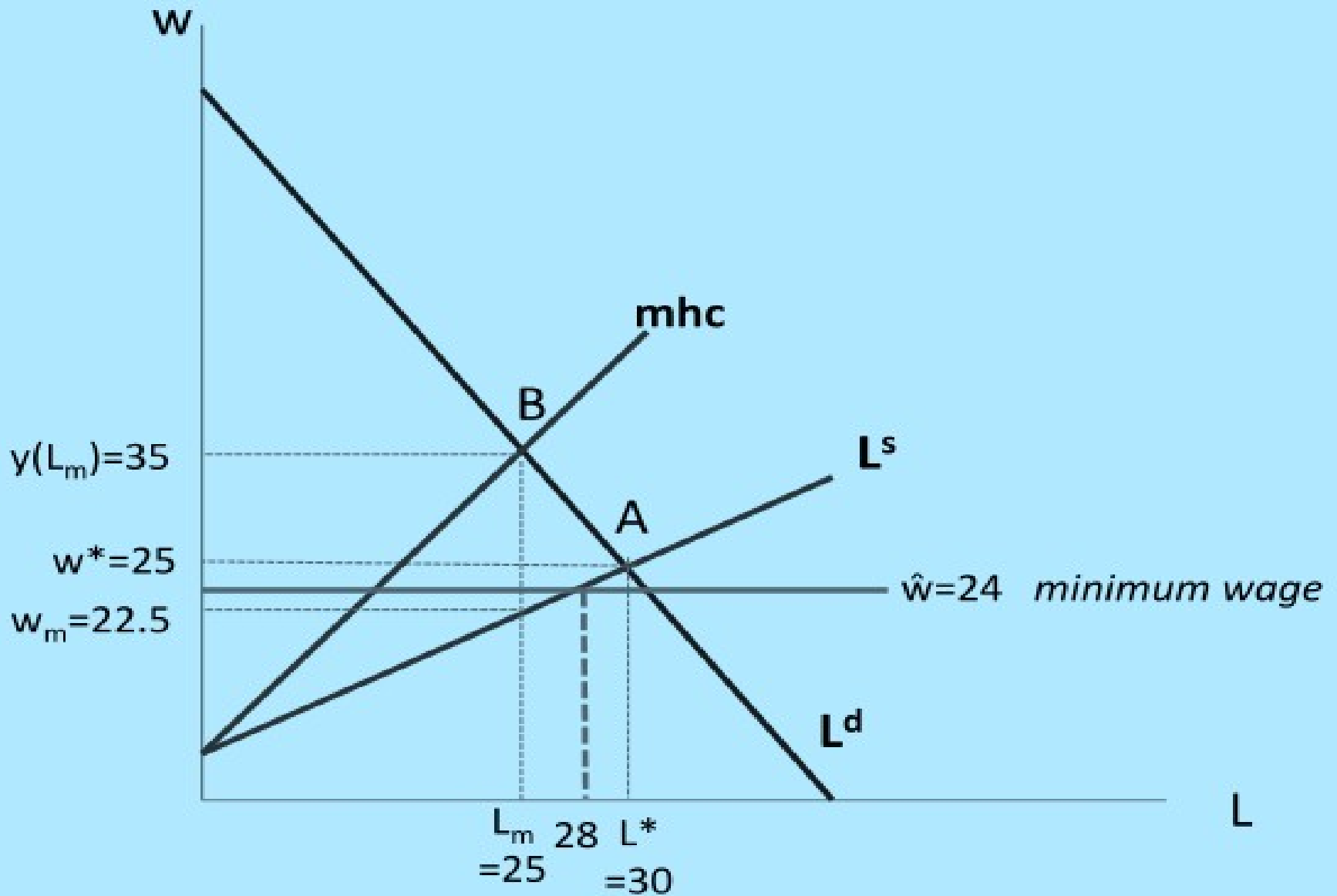
Unemployment rate	n. unemp/labour force	$5/34=14.7\%$
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Minimum wage in monopsony

$\hat{w}=24$ above $w_m=22.5$

Supply	$24=10+0.5L$	$L=14/0.5=28$
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Cont.....



Cont.....

Monopsony: exercise 2

$$w(L)=12+0.6L$$

labour supply faced by the monopsonistic employer

Total labour cost

$$w(L) \cdot L$$

Marginal hiring cost

$$\begin{aligned} mhc &= \partial(w(L) \cdot L) / \partial L = w + (\partial w / \partial L) \cdot L = \\ &= (12 + 0.6L) + 0.6L = 12 + 1.2L \end{aligned}$$

$$y(L)=68-0.4L$$

labour marginal productivity

Monopsony equilibrium:

$$mhc(L_m) = y(L_m)$$

$$12 + 1.2L = 68 - 0.4L \quad L_m = 56 / 1.6 = 35$$

$$w_m = 12 + 0.6 \cdot 35 = 33$$

$$y(L_m) = 68 - 0.4 \cdot 35 = 54$$

$$\text{monopsony power } (y(L_m) - w_m) / w_m = (54 - 33) / 33 = 0.63$$

Perfect competition equilibrium:

$$w(L^*) = y(L^*)$$

$$12 + 0.6L = 68 - 0.4L \quad L^* = 56 / 1 = 56$$

$$w^* = 12 + 0.6 \cdot 56 = 45.6$$

Cont.....

Can you show that at the monopsony equilibrium point the degree of monopsony power is equal to the inverse labour supply elasticity?

Inverse supply elasticity ε : $\frac{1}{\varepsilon} = \frac{\partial L/L}{\partial w/w} = \frac{\partial L}{\partial w} \frac{w}{L}$ then $\varepsilon = \frac{\partial w}{\partial L} \frac{L}{w}$

$\partial w/\partial L=0.6$, $L_m=35$, $w_m=33 \rightarrow \varepsilon=0.6 \cdot 35/33=0.63$ which corresponds to the value of monopsony power derived above

Which is the effect of a minimum wage equal to 36 on the monopsony equilibrium

$\hat{w}=36$ above $w_m=33$

Supply

$$36=12+0.6L$$

$$L=24/0.6=40$$

Alternative Pay Package

- *Piece Rates*

- ✓ *Piece rates are compensation paid in proportion to the number of units of output.*
- ✓ *Piece rates limit the amount of shirking.*

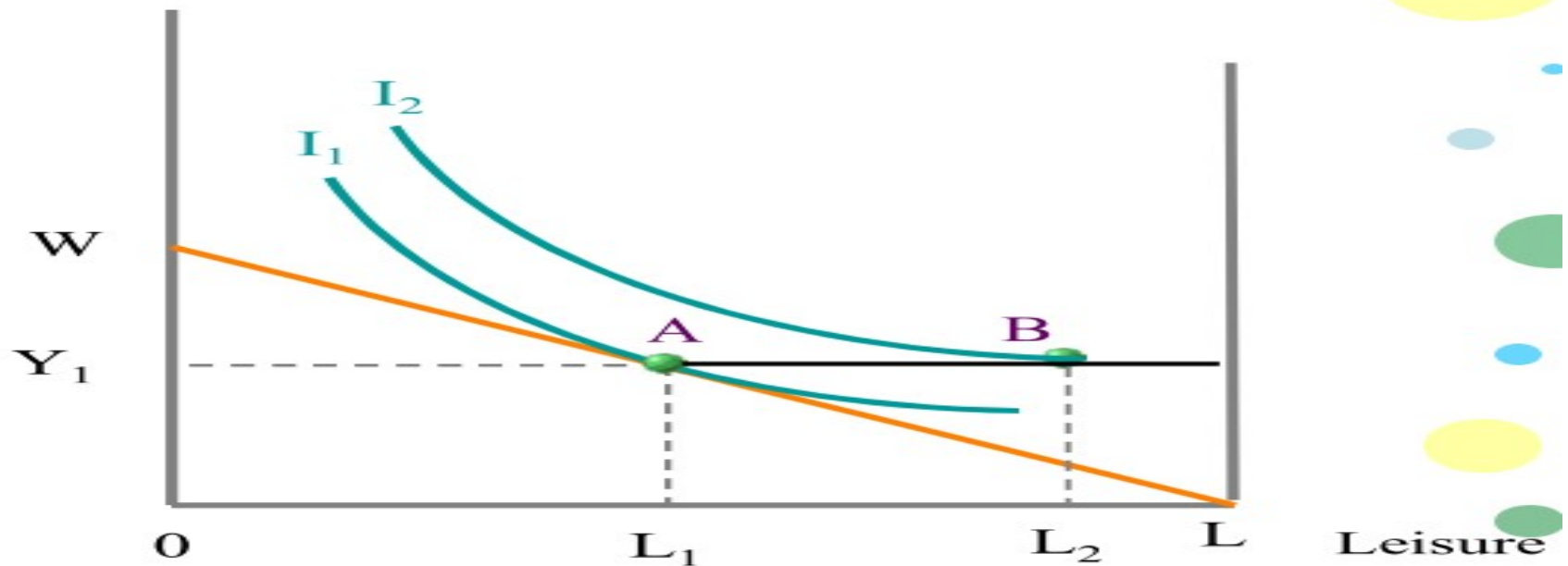
- *Drawbacks*

- ✓ *May be difficult to set rate as APL changes.*
- ✓ *They increase income variability and so firms will have to pay a premium.*
- ✓ *Difficult to use where team production is key: who made what & decreased cooperation*
- ✓ *Workers may decrease quality.*

Commission and Royalties

- *Unlike piece rates, commissions and royalties are compensation paid in proportion to the value of sales.*
- ✓ *These are efficient where work effort is difficult to observe but job performance tied to sales*
- Hourly Wage vs. Salary

Annual Income



Cont.....

- *If a worker is paid by the hour, the worker will choose point A with an annual income equal to Y_1 with L_1 hours of leisure*
- *An equivalent annual salary of Y_1 , the worker can get to a higher indifference curve I_2 by increasing hours of leisure to L_2 .*
- *The worker can get this higher level of utility by shirking.*
- *The firm can overcome this incentive problem by offering future raises or promotions to those who consume L_1 hours or less of leisure*

Bonuses

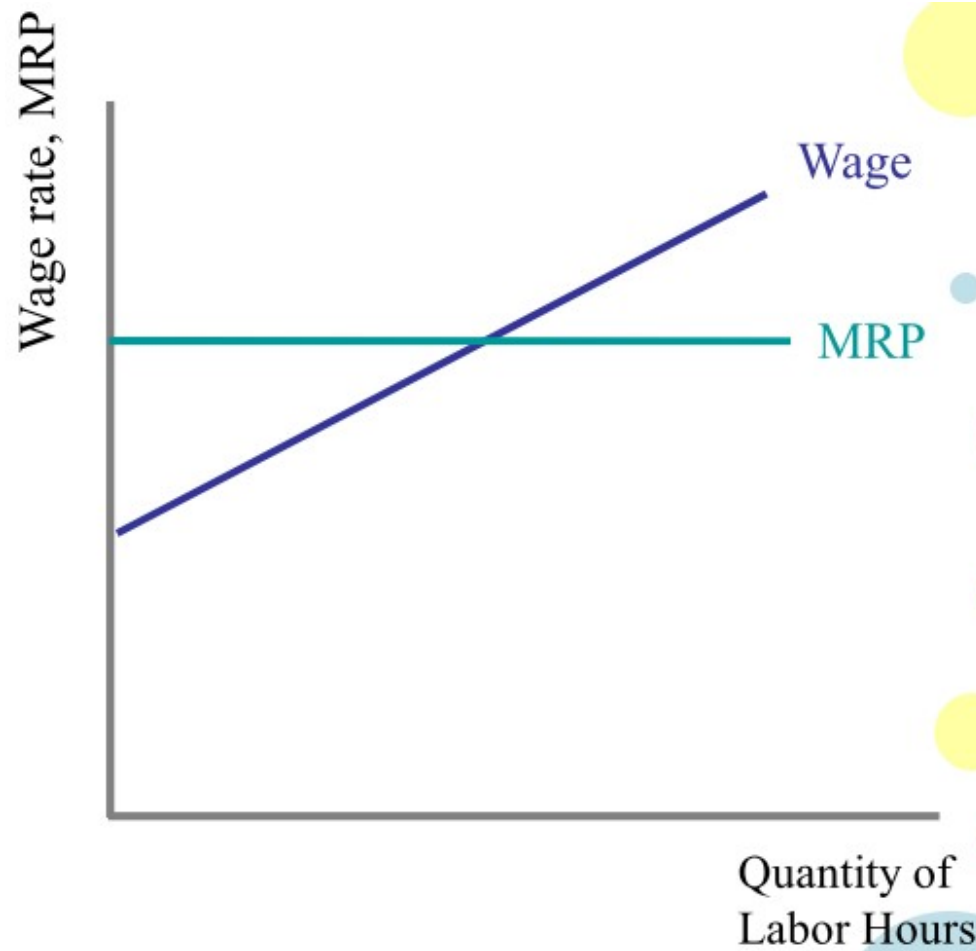
- *Bonuses are payments beyond the annual salary based on some factor such as personal or firm performance.*
- ✓ *Elicit extra work effort and are not permanent costs.*
- *Personal performance bonuses*
- ✓ *Based on evaluation by superiors or quantifiable measure.*
- ✓ *May have unintended effects.*
- *Workers schmooze superiors.*
- *Workers work to bonus items, nothing else*
- *Team performance bonuses*
- ✓ *Based on team performance.*
- ✓ *Leads to the free-rider problem.*
- *Workers have less incentive to work hard as the size of the group rises since their own effort matters less.*
- ✓ *Team performance bonuses work best when the size of the group is small*

Profit Sharing

- - Profit sharing is a pay system that allocates a portion of the firm's profits to its employees.
- - Supporters argue that profit sharing gives workers the incentive to work harder to increase firm profits.
- - Critics argue that it suffers from the free-rider problem.
- - Evidence indicates a modest positive effect on productivity.
- Tournament Pay
 - - Tournament pay plans base compensation on relative performance.
 - - A large prize exists for top performer, smaller prize for second place, and so on.
 - - Encourages all participants to exert more effort.
- The CEO position may be first place in a tournament.
 - CEO's are usually paid more than their personal MRP, but other executives increase their MRP in hopes of getting the top prize.

- Efficiency Wages
 - Firms may reduce shirking by monitoring the efforts of workers.
 - Monitoring workers is costly in some cases.
 - o Babysitters, security guards, managers
 - One solution is to pay an above-market wage.
 - An efficiency wage is one that minimizes an employer's wage cost per effective unit of labor employed.
 - o The marginal benefit of a higher wage equals the marginal cost of the higher wage

Deferred Pay Contracts

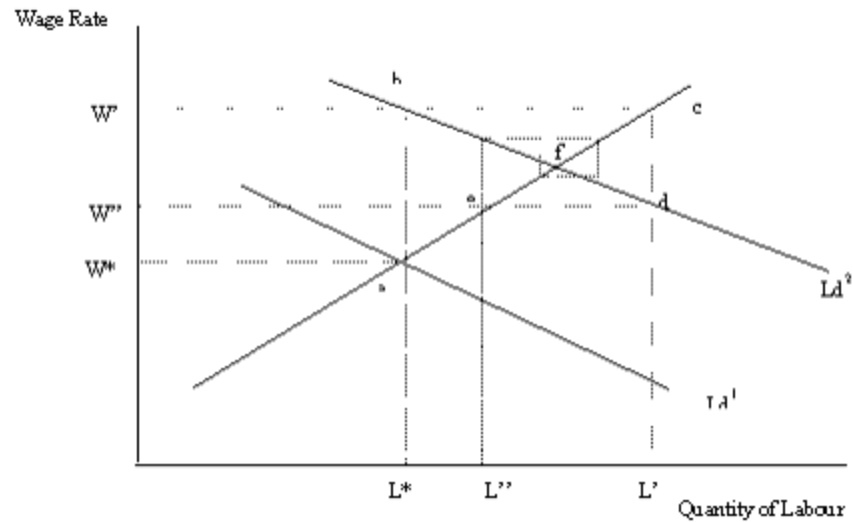


- - In the diagram is MRP constant over the person's worklife.
- - Firms and workers may enter into implicit contracts that increase pay as years of service rise.
- - Younger workers receive pay that is less than their MRP, while older workers are paid more than their MRP.
- - The prospect of high pay at the end of one's career may discourage shirking and reduce turnover.
- - Because of the increased productivity, workers may get higher lifetime earnings than if wages equaled MRP each year.

- 4.3 Critiques of Conventional Wage Theories
- Too often in economic analysis, labour market equilibrium is portrayed by a simple model of perfectly rational individual firms and individual workers coming together in a competitive market, where each economic agent is assumed to be motivated to maximise his own utility and arrive at a market clearing wage rate and employment level. Equally, it is too frequently assumed that any shocks to labour supply or demand necessitate adjustments in the wage rate until the labour market once again reaches competitive Walrasian equilibrium. Such a simple approach to the labour market, based upon marginal productivity theory, ignores a huge variety of specific traits associated with labour as a tradable economic commodity

- Institutionalism in Labour Markets
- The main area where orthodox theory has come under attack has been through the institutionalist view of labour markets. Its basic tenet is that collective bargaining on behalf of trade unions renders the assumption of individual 'atomised' wage bargaining unrealistic. A simple description offered by some advocates of collective bargaining is that employers' associations and trade unions shift supply and demand curves to their consciously determined wage rate to clear the market. Others consider it highly improbable that the wage rate resulting from collective bargaining will be the same as that which would emerge in a competitive setting, and consequently the level of unemployment would be different in the two scenarios.

The Cobweb Theory



- A more advanced method of assessing lagged labour supply adjustment, adapted by labour economists, is the cobweb theory. The theory describes how an initial unexpected outward shift in the labour demand curve leads to prices fluctuating in a "cobweb" pattern, until the long-run equilibrium is achieved. In figure above, the initial shift in demand leads to a shortage of labour, so that in effect there is a vertical short-run labour supply curve at L^* . The combination of the labour shortage and short-run inelastic supply raises the short-run equilibrium to point b on the demand curve, at a higher wage of w' . However, this high wage rate leads to an increase in the number of workers willing to work and L' workers will eventually fill the market. The supply response equates to a shift rightwards in the vertical short-run supply curve to L' . Yet, now that the quantity of workers employed is again temporarily fixed, a surplus of bc workers occurs at w' . To eliminate the surplus the wage rate drops to w'' , which leads to workers slowly exiting the market until only L'' workers are employed. The resulting labour shortage once again leads to a wage cycle until long-run equilibrium is finally attained at point f. This model disregards rational expectations and utilizes an adaptive expectations approach to describing labour supply adjustment.

- Internal Labour Markets and Dual Labour Markets
- A growing group of internal labour market theorists have criticised the neo-classical perspective of continuous open job competition. They have proposed that job ladders involving a sequential progression of jobs, based upon seniority and investment in human capital, exist in many large firms involving different activities. Consequently, jobs at the bottom rung of the job ladder (called a port of entry) are fought for competitively in an external labour market, since the firm must compete with other firms which are hiring the same kind of unskilled, unspecialised labour. On the other hand, an internal labour market which determines the recruits for higher positions on the job ladders is governed by administrative rules and seniority specific to the particular firm in question, and not the tâtonnement of external labour markets. Mulvey (1978) tried to illustrate this outcome as being Pareto sub-optimal by having a wage preference path not coincident with the output expansion path in a particular industry.

- The two basic reasons for internal labour markets are that the skills required for some more senior and specialized jobs are specific to each firm and can only be learned by people within that firm; and to get a return on human capital investment, firms must employ workers for a significant period of time. Although the resulting internal labour markets are said to allocate inefficiently, when viewed over time there are advantages which some authors refer to as dynamic efficiency. Such temporal efficiency results from the positive externalities associated with the reduced human capital investment for firms using internal labour markets and the job security and promotions which workers receive in internal labour markets, which are beneficial to both firms and workers (Salamon, 1992).
- Some labour economists even extend the internal labour markets idea into a theory of segmented dual labour markets, consisting of primary and secondary labour markets. Primary labour markets are characterised by employment stability, the presence of job ladders, strong and effective trade unions and efficient management, while secondary markets have the opposite characteristics (Brue and McConnell, 1995).

- Although some criticise dual labour market theory for being undeveloped and excessively Marxist or Galbraithian, there has been increasing evidence of empirical validity for the idea, due to certain industry structures and selective efficiency wage practices. Seasonal or cyclical variation in some industries will naturally lead to the development of secondary markets, while closed shop practices have proved an indomitable barrier to mobility in others. The central tenet of the argument is that, while predominantly classical wage characteristics can be seen in secondary markets, this is not the case in primary markets.

- Efficiency Wage Theories
- In internal labour market and dual labour market theory the source of the market imperfection and involuntary unemployment lies in human capital costs and the ability of 'insider' employees to influence wage levels. In efficiency wage models, the source is the asymmetric information between employees and firms regarding their profitability. Efficiency wage models explain how it is in the firm's best interests to pay a wage above the market clearing equilibrium rate in order to optimise the marginal revenue product of labour.

- A number of different microeconomic underpinnings have been found to explain the phenomenon (Akerlof and Yellen, 1986). The threat of shirking on the job may prompt firms to pay an additional wage to eliminate the 'moral hazard' of employee dishonesty and improve the general morale of workers in the firm. A corollary of this argument is Stoft's 'cheat-threat' theory which states that if the economy is nearing full employment, the threat of being fired for being caught cheating is less serious to employees, so they must be paid extra to dispel thoughts of dishonest behaviour (Stoft, 1982). Salop (1979) takes a different approach by emphasizing the lower labour turnover costs associated with the decreasing quit rates, which in turn result from the above market-clearing wage rates.

- A more sophisticated argument developed by some, like Weiss (1980), is that of adverse selection. Assuming that workers' abilities and dependability are positively correlated to their reservation wages, firms pay an efficiency wage in the hope of attracting the best workers. If workers underbid the efficiency wage, they stand the possibility of being regarded as 'lemons' by firms and will become involuntarily unemployed. Nevertheless, the lack of willingness to fire unproductive workers by some American firms prompted Akerlof to even suggest a sociological model based upon raising the standards of group work norms by above equilibrium wages that defies neo-classical wage theory.
- An explanation for efficiency wages in less developed countries, described by Leibenstein (1986), especially relevant in the agricultural sector, says that workers need to be paid enough to be well-fed and clothed even if this means paying them a wage above their marginal product of labour, which in many sectors is even zero. All these efficiency wage models account for Pareto sub-optimal equilibria even in the absence of institutionalist or insider/outsider structures.

- 4.4 Minimum Wage Theory
- • we analyze and discuss the effects of minimum wages under two different labour market assumptions: (i) a competitive labour market and (ii) monopsony in the labour market
- 4.4.1 The Competitive Model
- • the traditional economic argument against minimum wages assumes a competitive labour market
- • each profit-maximizing firm is assumed to be able to hire all the labour it wishes at the competitive market wage rate
- • The main purpose of the min wage is redistribution concern; however a trade-off between equity and efficiency arises.
- • in Figure 7-2 the market-clearing wage rate is W_0 with L_0 workers employed

- • suppose that the government imposes a minimum wage of $W_{min} > W_o$
- • at the wage W_{min} , profit-maximizing firms will only employ L_{min} workers and $(L_o - L_{min})$ workers will lose their jobs
- o the number of workers who lose their jobs following the imposition of a minimum wage depends on the elasticity of the labour demand curve for minimum wage workers and the size of the $(W_{min} - W_o)$ differential
- o for example, if the elasticity of labour demand curve is $-1/3$ and there are 1,000,000 Canadian workers paid the minimum wage, an increase in the minimum wage of 3% (about \$.20) will result in 10,000 Canadian workers losing their 'minimum wage' jobs; the vast majority of 'minimum wage' workers (990,000 out of 1,000,000) keep their jobs and receive a 3% wage increase

- - a modest increase in the minimum wage creates a few big losers (minimum wage workers who lose their jobs and their entire annual labour income) and many small winners (workers who keep their jobs and receive a modest increase in wages and annual income)
- - o most of the 'losers' tend to be young people with few skills and little work experience
- - it should also be noted that raising the minimum wage does little to alleviate poverty because relatively few low-income households have minimum wage workers (most poor individuals do not work) and relatively few minimum wage workers live in low-income households
- - o many 'minimum wage' workers (particularly young singles) are part of a family and have access to family income well above poverty levels

- • Shifts in labor market equilibrium:
- o an increase in labor demand results in an increase in both the equilibrium wage and the equilibrium level of employment,
- o a reduction in labor demand results in a decrease in both the equilibrium wage and the equilibrium level of employment,
- o an increase in labor supply results in a lower equilibrium wage, but a higher equilibrium level of employment, and
- o a reduction in labor supply results in a higher equilibrium wage, but a lower equilibrium level of employment.

- 4.4.2 The monopsony model
- • Figure 7-4 illustrates the effects of a minimum wage in a monopsony labour market
- • the profit-maximizing position for a monopsonist is again point A (where $MC = VMPL$)
- o the monopsonist hires L_m workers, pays each worker W_m , and makes a profit equal to the profit triangle AED
- • now suppose that the government sets the minimum wage at W_{min}
- • the minimum wage replaces that segment of the original labour supply curve to the left of point J in Figure 7-4; the minimum wage law prevents firms from hiring labour at a wage less than W_{min}
- o the new kinked LS curve is the horizontal line HJ (at the minimum wage) and the original LS curve for wage rates greater than the minimum wage

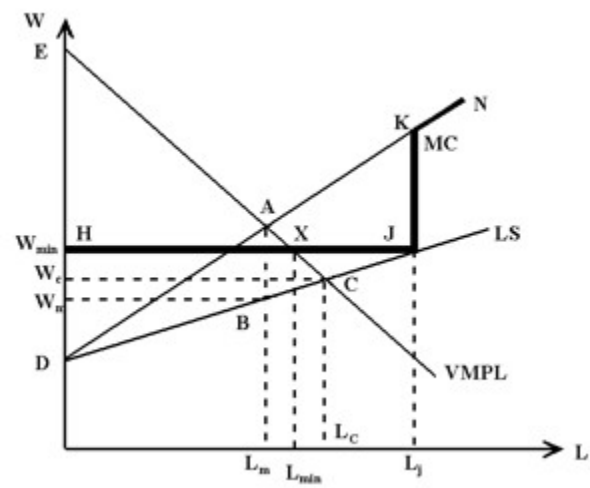


Figure 7-4

- - given a new kinked LS curve, there will be a new kinked marginal cost (MC) of hiring labour curve
- - along the new HJ horizontal section of the LS curve, the marginal cost of hiring an additional worker is the minimum wage
- - o the HJ horizontal section of the LS curve is also the monopsonist's marginal cost (MC) of labour curve (for labour quantities up to L_j)
- - if the firm hires more than L_j workers, it will have to pay all its existing workers a wage higher than the minimum wage; the marginal cost for hiring an additional worker past point J is considerably higher than the minimum wage
- - o past point J the marginal cost curve corresponds to the original marginal cost curve
- - the new 'minimum wage' marginal cost (MC) of labour curve is discontinuous at point J and is represented by the kinked line HJKN

- • the monopsonist will maximize profits at point X, where the new kinked MC curve intersects the VMPL curve
- o given a minimum wage W_{min} , the monopsonist will hire L_{min} workers and make a profit equal to the new profit triangle HEX
- • imposing a minimum wage on a monopsonist has the following labour market effects:
 - o an increase in employees hired, from L_m to L_{min}
 - o an increase in the wage rate paid to employees, from W_m to W_{min}
 - o a decrease in monopsonist profits (in Figure 7-4, the new profit triangle HEX is smaller than the original profit triangle ADE)
 - • in effect, the imposition of a minimum wage redistributes some of the monopsonist's profits to workers in the form of higher wages and/or increased employment levels

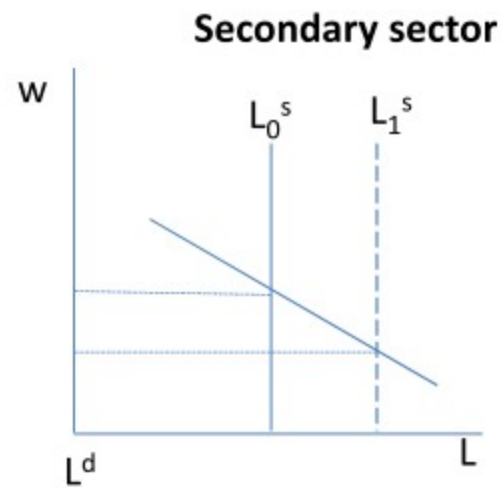
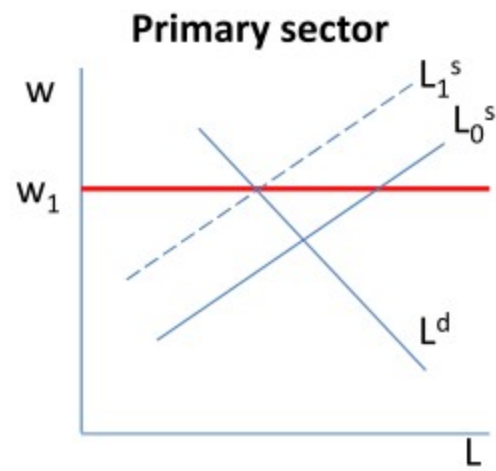
- - if the government wanted to achieve the maximum positive wage effect, it should set the minimum wage at point A; the monopsonist would hire no new employees but would pay all of its existing workers a much higher wage (point A compared to point B)
- - if the government wanted to achieve the maximum employment effect, it should set the minimum wage at point C (which corresponds to the competitive labour market outcome)
- - o the monopsonist would hire ($L_c - L_m$) additional employees and pay all workers a higher wage (W_c versus W_m)
- - the AC segment of the VMPL curve represents a 'trade-off' curve for a government imposing a minimum wage on a monopsonist
- - o moving from A towards C increases the positive employment effect of the minimum wage but decreases the wage gain for all existing workers

- - in conclusion, the imposition of a minimum wage in a labour market where employers have monopsony power can result in an increase in employment
- - o in a perfectly competitive labour market the imposition of a minimum wage will result in a decrease in employment
- - the radically different employment effects of a minimum wage under different assumptions about the competitive structure of the labour market raises an obvious question: are there monopsonists in the real world, or only in labour economics textbooks?
- - most firms have some monopsony power in the short run; most firms can lower the wage rate without immediately losing all of their workers and a firm may have to raise the wage rate to attract more workers in a local labour market
- - however, most firms do not have monopsony power in the long run; if the firm cuts its wage rate, eventually its employees will move to other employers (in the long run workers are mobile)

- - the obvious exception to the 'no monopsony in the long run' argument is a 'one company' town in an isolated labour market (for example, a mining company or pulp/paper mill in Northern Ontario)
- - o there are few other employers in the immediate area, existing workers are reluctant to leave their hometown, and new workers are reluctant to move to an isolated area (there is limited labour mobility)
- - o however, in most cases workers in the 'one company' have unionized and bargained for wage rates higher than the government would set as the minimum wage (see Chapter 15); the imposition of a minimum wage less than the union wage would have no effect on labour demand
- - are there any examples of labour markets where employers have monopsony power and employ low wage workers?

- • perhaps the McJob market, low-wage jobs in fast-food outlets
- o unskilled, low-wage workers tend to be immobile: there are costs to moving and little expected gain from a move (what difference does it make if you flip hamburgers here or there?)
- o a fast-food outlet may have a well-defined local labour market (within walking distance or a short bus ride)
- o it may also be the case that high welfare benefits reduce the net gain from working and the firm may have to offer a higher wage to attract welfare recipients
- • given a local labour market with limited labour mobility of low skilled workers, the labour supply curve for McJobs may slope up; the firm may have to increase the wage rate (or offer recruitment bonuses) to attract additional workers
- o under such circumstances, a minimum wage could increase employment levels

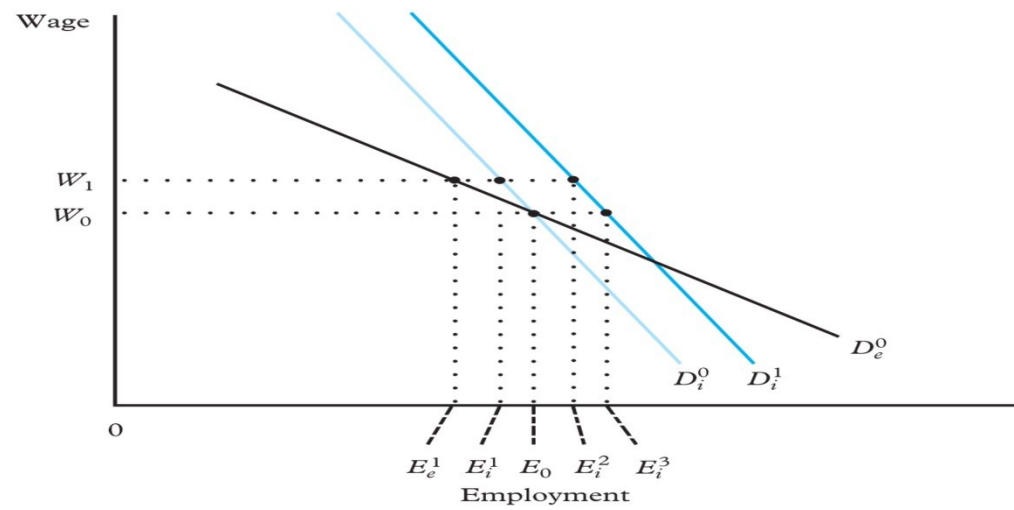
- Minimum wage in a dual labour market
- Two sectors: primary (formal) with a min wage, and secondary (informal) sector with no min wage.
- A minimum wage applied to the primary sector causes unemployment and a subsequent outflow of labour from this sector (leftward shift of the L_s). This excess of labour relocates to the secondary sector (rightward shift of the L_s). As wage flexibility characterises this sector, the increased labour supply is fully absorbed through a reduction of the wage. The min wage causes a reallocation of jobs and a larger wage differential between sectors with no loss of employment insofar as there is perfect inter industry mobility sectors, with no loss of employment insofar as there is perfect inter-industry mobility.



- 4.5 Labour union and labour market
- - Labor (trade) unions are institutions representing workers interest in the labor market and responsible for the improvement in working conditions. They are aimed at maximizing the wellbeing of their members.
 - As a result, the labor market impact of unions depends not only on the political and institutional environment that regulates the employer-union relationship, but also on the factor that motivates union to pursue certain strategies (such as making wage demand that may led to a strike).
 - Unions influence every aspect of the employment contract, including hours of work, wage, fringe benefits, labor turnover, job satisfaction, workers' productivity, firm's profitability, hiring, overtime, job assignment, promotion, and layoff. Unions can flourish only when firms earn above-normal profits.
 - Workers choose whether to join a union or not. A worker joins a union if the union offers him a wage employment package that provides more utility than the wage employment package offered by a non- union employer.

- The demand for and supply of union jobs
- In general, workers are more likely to support unionization when the union organizer can promise a high wage and small employment loss. Moreover, because there are additional costs to joining a union (such as union dues), the worker will be more likely to support unions when these costs are small. These factors generate the 'demand for union job'.
- The demand for union job is not the sole determinant of the extent of unionization in the labor market. The ability of union organizers to deliver union jobs depends on the cost of organizing the work force, on the legal environment that permits certain type of union activities and prohibits others, on the resistance of management to the introduction of collective bargaining. These forces, in effect, determine the supply of union jobs.
- The extent of unionization observed in the labor market is determined by the interaction of these two forces. As a result, the unionization rate will be higher the more workers have to gain by becoming unionized and will be lower the harder it is to convert jobs from non-union to union status. The cost-benefit approach helps as understand differences in unionization rate across demographic groups and industries.

- 4.5.1 Constraints in the achievement of union's objectives
- The fundamental market constraints on the ability of unions to accomplish their objectives are:
 - the shape of the downward-sloping labor demand curve, and
 - the elasticity of the labor demand curve.



Labor demand curves D_i^0 and D_e^0 show different degrees of elasticity of labor demand. Both demand curves intersect at the wage rate W_0 and the employment level is at E_0 .

Attempts by the union to seek a wage increase to W_1 will cause employment to decrease to E_e^1 for D_e^0 , which is a relatively elastic labor demand curve; and to E_i^1 for D_i^0 , which is a relatively inelastic labor demand curve.

If D_i^0 shifts out to D_i^1 (while negotiations are still underway about the wage increase) due to an increase in product demand, there will be no decrease in employment – unions will always be delighted with higher wages W_1 and higher employment such as E_i^2 (more membership). If the wage still remained at W_0 when labor demand increased due to the increase in product demand, employment rises to E_i^3 .

- 4.5.2 The effect of unions on wages and unemployment
-
- Unions wage effect
- By how much do unions increase the wage of their members? We began our analysis of this important question by specifying precisely what we mean by a 'union wage effect'. Suppose a particular worker 'i' earns W_{ni} if he works at a nonunion job but would earn W_{ui} if the firm becomes unionized.
- The percentage gain for this worker is defined as;
- $$\Delta t = \frac{\text{union wage gain for a particular worker} = W_{ui} - W_{ni}}{W_{ni}}$$
- Assume there are N workers in the labor market, we could then calculate how much each of the worker would gain if the worker become unionized and define the union wage gain as;
-
- $$\text{Union wage gain} = \sum \Delta_i$$
- $$N$$

- The union wage gain thus measures what the average worker in the economy would gain (in % terms) if he or she become a union member.
- Although we are interested in knowing the size of the union wage gain, this statistics is very difficult to calculate. We need to know how much the worker would earn if he were employed in a non union job and how much he would earn if the job suddenly becomes unionized. Typically we observed only one of these two wages (that is, either the job is unionized or it is not)
- As a result, we instead calculate a very different sort of union-non-union wage differential. In particular, assume that the average wage in union job is given W_u and that the average wage in non-union job is given by W_n .
- The union wage gain is then defined by
- $$D = \frac{W_u - W_n}{W_n}$$
- Which is the percentage wage differential between union job and non-union job. Estimates of the union wage gap typically adjust for different in socio economic characteristics between workers who are in union jobs and workers who are in non-union jobs.

- Unions and wage dispersion
- The wage distribution of unionized workers has less dispersion than that of non union workers. Unionization reduces wage dispersion in aggregate economy by as much as 10%. The compression of the wage distribution in the union sector arises partly because union workers are a more homogenous group (In terms of education and other observable skill measures) than non-union workers.
- Unionized firms, however, also offer their worker a lower payoff for skills than non-union firms. The rate of return to education among non-union workers is perhaps twice as high as the rate of return among union workers. The lower the payoff to skill found in non union firm might occur because union stresses pay equity considerations in collective bargaining negotiations. This negotiation prohibits employers from making wage setting decisions that reward very productive workers and penalize less productive workers.

- Unions and fringe benefits
- Unions also affect the value of the fringe benefit package offered by firms. These fringe benefits include health and life insurance, sick days, pension and bonuses. Because union wages are higher than the non-wages, the package of fringe benefits received by union worker is worth more than package received by non-union workers.
- Unions, productivity and profit
- The greater stability of employment in unionized firm provides a channel through which can have a favorable impact on the firm's productivity. Labor turnover, after all, is quite costly. It disrupts the production process, requires substantial expenditure and increases the cost of training the work force.
- Workers in unionized firms are more productive than non unionized firms. This is because an increase in wage due to unionization leads to an upward movement in the demand curve. As a result employment falls and the value of marginal product of labor rises. Moreover, the union wage increase may "shock" the firm into more diligent hiring practices. Because unions often impose restrictive rules on the dismissal of their members, unionized firms will be much more selective in their hiring decisions, and a better-screened workforce will typically be more productive. The favorable impact of unions on productivity, however, is not sufficiently large to compensate the firm for its larger payroll costs. As a result, unionized firms have lower profits.

- As we noted earlier, there is evidence that the market value of a firm (that is, the wealth of its shareholders) decreases as the rents are redistributed to union workers. In View of the negative impact that unions have on profits and shareholder wealth, it is not surprising that a firm's management often makes ingenious attempts to keep the Unions out. A recent study suggests that firms will incur large amounts of debt to reduce the threat of impending unionization.
- Pros
 - Trade unions maintain and improve workers' terms and conditions through bargaining with
 - employers.
 - Workers organized in trade unions benefit from higher wages—the so-called union wage premium.
 - Union bargaining also results in a fringe benefits premium for covered workers.
 - Trade unions reduce wage inequality.
 - The counter-cyclical wage premium helps to maintain the real wages of covered workers.

- Cons
- • Trade unions restrict employment flexibility.
- • Trade unions prevent markets from clearing.
- • By standardizing wages across regions, unions distort labor supply.
- • Trade unions harm businesses if the return for additional wages is low.
- • If the premium comes at the expense of normal profits, this can damage firms and employment growth.
- Negative Effects:
 - • Restrictive Work Rules (output, way of production, workers to be hired, works to be performed)
 - • Strikes
 - • Labor Misallocation (The higher wage in the union sector causes workers to be displaced to the nonunion sector.)

Chapter Five

Labour Markets

- Basic Concepts of Labour Markets
- - Labour market may be defined as a process by which supplies of particular type of labour and demands for that type of labour balance seek to obtain a balance.
 - The labour market is the place for operation of this process. It is the only device for sorting out workers with varying skills and interests among the multitude of different jobs in the economy. The labour market is also an area or place where relative wage levels for different plants, industries, occupations and regions are determined.

-
- Main characteristics of labour market are as follows:-
- 1. Unlike a commodity market, relationship between a buyer and seller in a labour market (expecting a casual labour market) is not temporary and is expected to continue for some limit.
- 2. Labour market is essentially local in character: - The concept of labour market stands for the buyer and sellers of labour who are in contact with each other for the purchase and sale of workers services. Thus, labour market is normally local markets in the sense of the demand for and the supply of labour being confined to a particular locality.
- 3. Lack of mobility: -Lack of mobility is an important characteristic of a labour market owing to various reasons. Labour obviously cannot move with the same ease and facility with which commodities are transported from place to place and the buyers of labour as well.
- 4. A labour market is essentially an imperfect market: - Labour market is an imperfect market where one does not find a normal wage rate to which the market rate naturally tends. There is a diversity of wage rate in a labour market; for the same types of work different wage are paid: a feature, which may last for a long period and may even be regularized.
- 5. Monopoly:- Monopoly in labour market is extremely rare excepting when the skill of a particular type of extremely rare or when a very powerful trade union exercise a complete control over the supply of labour in a given industry or in a given area.
- 6. Most of the labour markets in the world do not justice to the workers in the matter of the division of the proceeds of industry as between the employers and workers. In other words, the labour markets are usually the buyers' market.

- • Imperfections in the labour market:-
- There are certain peculiarities of labour market, which distinguish it from the market of other commodity. These peculiarities also make labour market imperfect
- 1. No equilibrium between demand for and supply of labour: -
- In the markets of other commodities prices keeps on changing till demand is equal to supply. In other words, the other markets are normally cleared. However, in labour market wages are not freely flexible. Therefore, as Hicks pointed out, the effect of increase in demand for labour on wage rate is felt only after a long passage of time.
- 2. Wage differentials:-
- In a perfectly competitive market for a commodity, all units of the commodity are homogeneous and price tends to be uniform. In fact, George Stigler defined the market of a commodity as all those places where the price of that commodity tends to be uniform after allowance for transport cost. However, in the actual market the wage differentials are often due to imperfections in the labour market.

- a) No perfect knowledge: - Worker and employers do not have perfect knowledge about the conditions in the labour market. Therefore, the wage differences prevail.
- b) Lack of perfect mobility of labour: - There is no perfect mobility of labour from one industry to another or from one locality to another.
- c) Barriers to the entry of labour: - There are barriers to the entry of new workers in specific occupations and so the artificial storage of such workers is maintained and so the wages of such workers continue.
- d) Employer's monopsony: - The labour market may have monopsony employer of labour power and he can follow the policy of wage discrimination by paying different wages to different workers for the same work.
- e) Institutional forces in determining wages: - In the market of the other commodities, mainly the market forces of demand and supply determine the price. Therefore, there cannot be an economic explanation of wage rates based purely on demand and supply conditions.

- Internal and external labour markets
- Internal labor markets are those where workers are hired in to lower level jobs and higher levels are filled from within. Wages are determined internally and may be quite free of market pressure.
- Internal labour markets refer to the institutional rules and procedures, such as recruitment, training and the price of labour, which govern the employment relationship.
- bureaucratic arrangements 'where employee management is regulated inside the firm through a series of rules
- The employment relationship for most individuals has been an important organizational tie that goes beyond a simple market transaction.
- The external labor market represents a heterogeneous collection of employment opportunities that might be available as an alternative to any particular person's current job. The internal labor market represents a heterogeneous collection of compensation and human resource management policies that describe the career possibilities for an individual who does not change employers. There is now a general consensus within labor economics that symmetric modeling of the employee and employer outcomes and detailed information on both the employer and employee are essential to distinguish internal and external labor market factors.

- External labor markets imply that workers move somewhat fluidly between firms and wages are determined by some aggregate process where firms do not have significant discretion over wage setting.
- In the external labour market recruitment, training and the price of labour are controlled directly by economic variables.
- 5.3. Segmentation Theory
- Segmentation is connected with segmentation of market into separate parts and identified by certain characteristics.
- Basic idea in segmentation theory is that all the labour market because of economic and political forces influencing it, is split into separate segments, which function and adjust separately. The advantage of segmentation theory lies in the fact that events in labour market are analyzed not only from economic point of view, but also social aspects are included into it.
- We define labor market segmentation as the historical process whereby political-economic forces encourage the division of the labor market into separate submarkets, or segments, distinguished by different labor market characteristics and behavioral rules. Segmented labor markets are thus the outcome of a segmentation process. Segments may cut horizontally across the occupational hierarchy as well as vertically.

- The present divisions of the labor market are best understood from an historical analysis of their origins. We argue that segmentation arose during the transition from competitive to monopoly capitalism. Our historical analysis focuses on the era of monopoly capitalism, from roughly 1890 to the present, with special emphasis on the earlier transitional years.
- During the preceding period of competitive capitalism, labor market developments pointed toward the progressive homogenization of the labor force, not toward segmentation. The factory system eliminated many skilled craft occupations, creating large pools of semiskilled jobs. Production for a mass market and increased mechanization forged standardized work requirements. Large establishments drew greater numbers of workers into common working environments.
- 1. Segmentation into Primary and Secondary Markets
- The primary and secondary segments, to use the terminology of dual labor market theory, are differentiated mainly by stability characteristics. Primary jobs require and develop stable working habits; This skills are often acquired on the job ; wages are relatively high; and job ladders exist. Secondary jobs do not require and often discourage stable working habits; wages are low; turnover is high; and job ladders are few. Secondary jobs are mainly (though not exclusively) filled by minority workers, women, and youth

- Segmentation Within the Primary Sector
- Within the primary sector we see a segmentation between what we call "sub-ordinate" and "independent" primary jobs. Subordinate primary jobs are routinized and encourage personality characteristics of dependability, discipline, responsiveness to rules and authority, and acceptance of a firm's goals. Both factory and office jobs are present in this segment.
- In contrast, independent primary jobs encourage and require creative, problem-solving, self-initiating characteristics and often have professional standards for work. Voluntary turnover is high and individual motivation and achievement are highly rewarded.
- 3. Segmentation by Race
- While minority workers are present in secondary, subordinate primary and independent primary segments they often face distinct segments within those sub-markets. Certain jobs are "race-typed," segregated by prejudice and by labor market institutions. Geographic separation plays an important role in maintaining divisions between race segments

- Segmentation by Sex
- Certain jobs have generally been restricted to men; others to women. Wages in the female segment are usually lower than in comparable male jobs; female jobs often require and encourage a "serving mentality—an orientation toward providing services to other people and particularly to men. These characteristics are encouraged by family and schooling institutions.
- Labor markets of all sectors have a single definite job: hiring and supplying of labor service and determination of wage to all types of jobs performed. But this does not mean that labor markets of an economy are perfectly homogeneous; there are some differences among sectors regarding wage determination. Many empirical studies have operationalized the concept of segments with particular reference to the wage determination mechanism.
- Jobs fall into either the primary or the secondary sector. Jobs in the primary sector are “good jobs” characterized by high wages, job security, substantial responsibility, and ladders where internal promotion is possible. Jobs in the secondary sector are characterized by low wages and casual attachments between workers and firms and are menial. Workers in the secondary sector envy those in the primary sector, who have both better jobs and higher wages.

- Most of the studies find that there is a difference between the wage determination mechanisms in the primary and secondary sectors. Descriptive literature argue that labor market segmentation theory provides a good description of the income distribution and is therefore of heuristic importance. The reasons for segmentation may be important for understanding the form of the income distribution, but segmentation as such does not imply any market failure.
- Two crucial elements of segmented labor market theory was identified
- First, the labor market can be usefully thought of as being made up of several distinct segments with different rules for wage determination and employment policies.
- Second, access to jobs in at least some sectors at some times is limited in the sense that more people want jobs than there are jobs offered. Thus there may be queuing for these jobs either in the form of unemployment or job queues among employed workers or both.
- All the early writers on labor market segmentation theory identified limited mobility among sectors as an important aspect of the theory. More significantly, they argued that there is a hierarchy of sectors with access to the highest paying being the most difficult. Generally, the essential elements of labor market segmentation theory are the existence of segments with different wage setting mechanisms and queues for high wage jobs.

- The Workings of Urban Labour Markets in Developing Economies
- Urbanisation is taking place on a massive scale, and around 60% of the world's people are expected to be living in cities by 2030. Growing urban centres are increasingly viewed as 'engines of growth' and as sites of opportunity – a welcome recognition of their role in national development. But enthusiasm should not mask the fact that high rates of economic growth do not always result in urban poverty reduction; and that rising urban inequality is a major concern.
- The urban poor are vulnerable to national growth downturns, and are affected disproportionately by macro-economic reforms, particularly in large urban centres. The poor engage in economic growth and adjust to economic change through their position in labour markets, but not all households adjust equally. The economic characteristics of the urban poor include: weak labour market positions; dependence on the informal sector; lack of employment; and casual employment. Generating demand for unskilled and semi-skilled labour is critical. The reality, however, is that these sectors often contract as urban economies restructure.
- Urban population growth does not always reflect urban economic growth or the expansion of employment opportunities, but population growth is propelling structural changes in urban economies, with major implications for urban poverty dynamics. This Briefing Paper examines the benefits and constraints from the labour market perspective, and offers a series of policy 'next steps' for municipal governments.

- Urban labour markets and poverty dynamics
- Labour may be the most important asset for urban poor households and it is through labour that they benefit most directly from economic growth. The clear 'pros and cons' to urbanisation are summarised in Table 1. The over-arching reality is that most urban poor people have to work until they die, in insecure livelihoods that pay poorly and that limit their chances to escape from poverty.
- The way in which people enter the labour market is key to urban poverty dynamics. Those who are not poor tend to be clustered in formal employment, while chronic poverty is associated with casual labour or female business activities. Informal work is a mixed blessing – depending on context – offering an escape from poverty where informality is the norm, but not where informal workers are more isolated.

Is urbanization good or bad for the poor?

The pros and cons

Pros	Cons
Greater economic opportunities create job opportunities	New opportunities may be confined to those with skills or from particular groups/classes
Population density means lower per-capita costs for basic services	Those who are unregistered or "illegal" may be overlooked in plans for basic services and face high prices for access
Easy access to mass media and other information channels	Loss of extended family links mean potential erosion of family cohesion
Large numbers of citizens can be mobilised around shared problems	Overcrowding, spread of disease, pollution
People can pool resources, ideas, creativity and innovation	Population density makes cities fertile ground for drugs trade, gang violence
More liberal attitudes make it easier for women to enter the labour market	Easy entry into the informal labour market may extend to child labour

- Increasing the number of economically active members in a household can increase the chances of such an escape, but this depends on the quality of labour acquired. Evidence from urban Peru and Madagascar indicates that the more household members that are involved in the informal sector, the poorer they are (Herrera and Rouband, 2003) indicating the poor returns from their work.
- Evidence from Ethiopia suggests that 27.5% of the heads of chronically poor urban households work as casual labourers or in female business activities, compared to only 7.7% of those who are never poor, who are far more likely to be wage workers. There are significant numbers of unemployed household heads among the poor, but the proportions are highest among the chronically poor (Kedir and McKay, 2003).
- Those who obtain and retain jobs in the formal sector certainly experience improved income levels, but these jobs are rare in most developing cities. In urban Ghana, for example, only one-fifth of households were engaged in waged employment in the 1990s. The urban informal sector created few waged jobs, and the lack of private sector investment at the time contracted formal wage employment (McKay and Aryeetey, 2004).

- Urban economic growth
- Urban areas differ in their economic diversity and their ability to respond to higher concentrations of consumers. It can be argued that costs are likely to be higher in cities that are experiencing economic growth, because growth entails higher monetisation of basic services and other non-food items such as housing, transport, and informal payments to maintain livelihoods. This is particularly true for the informal sector. The impact on expenditure and earnings is not entirely clear. While costs of living are higher in rapidly-growing cities, there may also be more income-earning opportunities. However, it is not necessarily clear that more opportunities translate into better working conditions or remuneration for the poor.
- Countries may have to choose between increasing employment in the unskilled sector to lift large numbers of people out of poverty, or upgrading skills and developing niche markets. Integration into global markets can create new opportunities but may reduce labour stability as a result of increased competition. Global buyers demand high quality at low prices with fast delivery (Jenkins, 2003). While growth in sectors that require a highly skilled labour force (e.g. software industry growth in Bangalore) can generate indirect employment elsewhere (with skilled workers needing shops and other services), this growth may, in fact, displace the informal sector.

- The development of sectors that support a large casual labour market may have a strong impact on poverty if it also ensures general well-being of workers, minimising the effects of hard physical labour that can damage health and inhibit future earnings (e.g. rickshaw pulling in urban Bangladesh, or port work in Vishakapatnam). These workers rarely have sickness benefits or employment contracts and are more likely to be laid off in times of low labour demand. Flexible employment to enable more women to join the labour force is also welcome if women are employed on the same terms as men. Women in the export sector in Bangladesh, for example, tend to be younger, single and new migrants to cities, and come from poorer rural households with no alternative employment, making them vulnerable to lower wages.
- Links between urban and rural markets
- There is a need to move beyond rural-urban stereo-types and to recognise that measuring the extent of poverty over-simplifies complex rates and patterns of change. The connections between different areas are strong, particularly during hard times. A strong domestic agriculture market, for example, may affect urban staple food prices, and increase the supply of goods that are produced locally. Prudent agricultural policies can make efficient use of rural unskilled labour and reduce the pressure of rural-urban migration on informal wages.
- While remittances from migrant workers to rural households are important, the flow of people into cities can be destabilizing and push urban wages down. Political deterrents to migration include denying migrants permanent registration. As long as the underlying economic incentives for migration remain strong, such deterrents will make migration more costly for migrants, but will not stop them coming. As a result, growing numbers of migrants live in the informal sector, confined to unskilled, low-paid, low-security work.

- Economic reforms and public spending
- Macro-economic reforms have disproportionate effects in urban centres, on urban labour and, in particular, on unskilled workers. In Peru the impact has been huge. Privatization has, in effect, abolished job stability and reduced dismissal costs, deregulation has increased job insecurity, and unionisation has plummeted (Vedera, 2000, in Herrera and Rouband, 2003). Liberalisation was a major policy thrust during the 1990s, creating both new income earning opportunities for some, but less stable working conditions for many others. The restructuring of the formal sector and increased competition in the informal sector has contributed to declines in some household incomes.
- Cuts in public spending may be inevitable in countries experiencing economic difficulties. Investment cutbacks in urban infrastructure have left cities in a general state of dilapidation, exactly where demand is greatest. Reductions in water supply, for example, can be particularly problematic, with the poor often resorting to ponds or streams, or buying drinking water at inflated prices from vendors. This can reduce household income and time resources for income generation.

- Informal, unregistered and illegal work
- With wage employment so limited in developing countries, there has been a trend towards informalisation of the labour market in recent years and there is potential for considerable wealth generation outside the formal sector. However, these profitable informal sectors are often controlled by particular groups, backed by caste, ethnicity or other social norms that make them more difficult for others to enter (Amis, 2004a:148). The drugs trade is the clearest example. Discounting criminal activity, the informal sector may be approaching saturation in some urban centres, reflecting the relative ease of entry for those with few alternatives.
- The informal sector has its own obstacles, particularly for those working illegally or without registration. Inadequate access to credit, bureaucratic licensing requirements and regulatory restrictions, as well as overzealous policing (e.g. removal of informal vendors, demolishing kiosks, confiscating stock, denying licences, etc.) can all destroy jobs and urban livelihoods (Rakodi, et al., 2000).
- The distinction between informal and formal labour markets (and economies generally) should not be exaggerated. The lines can be blurred and the linkages are important. Individuals can reside and work within both sectors and move between them (Amis, 2004). There are many situations where informal enterprises may be illegal in terms of health and safety or social security legislation, but may also be paying local taxes (Grindle, 1980). Relaxation of laws and regulations can have a positive effect in developing these linkages and strengthening local labour markets.

- Urban labour exploitation
- Urban work can be highly exploitative and urban labour markets are often skewed against the poorest – those who may be least able to respond to new opportunities. Maids, for example, often live in the homes of their employers and are vulnerable to abuse (Perlman, 2003:12), but they do not necessarily register among the urban poor as they ‘live’ within non-poor households. They may be overlooked by policy-makers.
- Labour markets can increase the vulnerability of certain people. Older people, particularly those on their own, are often dependent on wage labour and this is a feature of their poverty (Mitlin, 2003). It could be argued that they have to undercut their wages to remain in the informal labour market – barred by age restrictions from working in the formal sector.
- Self-employed women in the informal sector are concentrated in low-paying activities, such as petty trade, hawking or street-vending. Young mothers are particularly vulnerable, as they are also care-providers. Children in female-headed households are the children most likely to enter the labour market, along with children from households where adult members are ill and unable to earn a livelihood, or other households with falling incomes. Children may work in hazardous jobs, such as scavenging on garbage sites or performing tricks at traffic lights, or be exploited as beggars or commercial sex workers. The most vulnerable are older siblings and girls.

- Policy 'next steps' for municipal governments
- While some urban areas have expanded demand for labour, most municipal administrations have little understanding of the urban economy's strengths and weaknesses, and a city's economic policy tends to focus on attracting investment through such incentives as tax breaks with little regard for the true costs of such incentives (Amis 2004).
- There are a number of indirect ways in which governments can expand entry into the labour market, and improve the quality of jobs for the urban poor. These include support for health, education and other services as well as flexible migration controls.
- More direct measures include:

- The promotion of labour-intensive growth.
- Urban employment does not feature prominently in national Poverty Reduction Strategy Papers (PRSPs) and few cities pursue labour-intensive growth paths. Labour-intensive programmes are fairly common, but are not, in general, strategic economic policies. Most are programmes providing short term employment to the beneficiaries
- (not always the poorest) and building infrastructure for poor communities. They often benefit the targeted workers, but the benefits rarely out-live the programme itself. Even improvements in infrastructure can unravel without sustained maintenance. These programmes need better linkages to city-wide economic development, to generate longer term, sustainable employment

- Labour welfare legislation and social protection.
- The daily rate for casual labour can underpin an entire local labour market and can determine whether or not the benefits of growth pass to poor workers. Workers in the informal sector rarely have any labour protection and may be engaged in hazardous activities that damage their health in the long run or constitute abuse, but have limited channels for complaint. Evidence on the impact of regulation and legislation is mixed.
- They can lead to a decline in employment (for domestic workers, for example) and inhibit the development of a sector. It remains the case, however, that flexible workers need greater social protection – welfare measures that will protect them, and their families, against extreme poverty, ill-health and lack of education. Jenkins (2003) argues that labour protection does not have to undermine the competitive position of a sector, but can result in much enhanced worker loyalty, higher productivity, and – within global markets – a better ability to provide overseas buyers with quality products.

- Flexible trading laws and associated land use regulation.
- It is important that city governments do not destroy jobs by over-zealous regulation. Given the inherent difficulties in implementing diverse legislation, municipal (and national) governments tend to be selective (or perhaps arbitrary) in what they implement (Grindle, 1980).
- They often respond to non-poor demands for city beautification or master-planning approaches, but can damage the productivity of the urban poor. Relaxation of laws can have a positive effect and often occurs in response to organized protest by vendors. Municipal governments need to acknowledge the importance of informal labour markets to the urban poor, and operate more tolerant policies.

- Encourage private/commercial sector investment.
- Private sector development requires governments to provide appropriate regulatory frameworks, an attractive infrastructure investment, and a workforce with at least a basic education. They need to make the transition from informal to formal activities as simple as possible. Private sector investment requires careful design to respond appropriately to the reality of local labour markets. The encouragement of high-tech industry in Bangalore, for example, has not favoured local employment development, but has, instead, increased land values and forced out small enterprise, undermining the local economy (see Grant, 2004). There are important equity issues and governance issues (e.g. corruption and transparency) that need to be taken seriously.

- The monetary costs incurred in the purchase of education are of two general types:
- (1) Direct or out of pocket cost;
- E.g. expenditure for tuition, special fees and books and supplies (Room and food expenditure are not included)
- (2) Indirect or opportunity cost.
- E.g. earning you give up by joining that labor market when schooling.
- The economic benefits of investing in education are an enlarged future flow of earnings. For such investment decision, various types of cost-benefit techniques are used:
- 1. Cost-Benefit Ratio: a ratio of discounted future benefits of an investment to its discounted costs can be used to compare the economic yield of investment.
- 2. Net present Value: it is very clear that costs and benefits associated with investing in education accrue at different points in time. But because of the idea of time preference, dollars earned or expended have different value in the future than today.

Chapter six

Manpower Planning

- The Basic Model of Human Capital
- So far we have been making the implicit assumption that labor supply constitutes a homogeneous set, i.e. workers are all alike in their economic characteristic and perfect substitutes for each other in production. Here, we relax this assumption and introduce the component of human capital formation.
- Human capital is the stock of skills that the labor force possesses. Human capital investment or development refer to various input for human beings that serve the purpose of increasing their value as employees, citizens and people. To this end, human capital investment includes such factors as the impartation of skills, formal and informal education, health considerations and good morals.
- The human capital theory explains that the motive for human capital formation is provided by earning differentials. Individuals undergo additional schooling or on the job training in order to benefit from the relatively high future yields of the investment.
- The employed education-productivity-wage relation is defined recursively: labor productivity is assumed a positive function of “education” (or years of schooling) and labor productivity determines the wage rate.
- The benefits of human capital investment include
 - Economic- increased future income
 - Political- national defense and creative powers
 - Psychic – satisfaction from knowledge, etc
- Since our concern is with individual decision for human capital investment, the focus will be on the economic explanation for the investment. From purely economic stand point, a rational decision to whether to go to school will involve a comparison of the associated costs and benefits.

- Time Preference, in short, is the idea that people are impotent and subjectively prefer goods in the present over the same good in the future. Accordingly, to have a meaningful comparison, we better discount the future earnings or expenditures and obtain the net present values.
- The mathematical expression of the future value of a dollar is
 - $V_n = V_p (1+i)^n$ where, V_p = present or current value
 - V_n = future value (value at n years)
 - i = interest rate
- Decision Rule: the relevant investment criterion or decision rule based on this calculation is that the individual should make the investment if its net present value is greater than zero. $\Rightarrow NPV > 0$
- If NPV is positive, it means that the discounted value of benefits exceeds that of the cost (the benefit exceeds cost) and hence the decision to invest is economically rational.

- Internal Rate of Return:
- The internal rate of return is the rate of discount at which the net present value of human capital investment will be zero. In other words it is the rate that equates the discounted present value of benefits and costs. Here, instead of solving for VP, we solve for r given the values of E 's and assuming VP is zero. A moment's reflection makes clear that r indicates the maximum rate of interest which one could pay on borrowed funds to finance a human capital investment and still break even (because he could not get a return above r).
- Notice the difference between social (r_s) and private (r_p) rate of return. In general social costs of education include all costs incurred including government subsidies while the private calculation would include only the costs incurred by the private individual. On the benefit side, the usual practice is to define private benefits as gross incomes less direct taxes and to approximate social benefits by gross income.
- Decision Rule: the human capital investment decision rule or criteria appropriate to this approach involves a comparison of the internal rate of return, r (r_s or r_p) with the average yield rate of capital market which can be provided by the average interest rate(i) applicable at the banking sector.
- If $r > i$, the human capital model is profitable and should be undertaken.
- If $r < i$, the yield from the investment does not even cover the loan bearing rate. i.e., the investment is unprofitable and should not be undertaken.
- In this case, given i , it will be profitable to invest in all human opportunities up to the point where $i = r$.

- Limitations of the rate of return on education
- 1. Problem in measuring correctly and comprehensively the benefits and costs of the education or training program for which yield estimates are being made.
- On the cost side, although direct costs and forgone earnings could be measured, there is also another form of human capital formation that is not measured by the above two costs such as psychic loss, which includes the disadvantage of education. Some people may find education boring, difficult and nerve-racking, and sense of deprivation, loss of friends when go to other place.
- On the benefit side there are many “non-market returns” such as satisfaction, the pursuit of knowledge and the improvement in the capacity to enjoy leisure. These are sometimes called psychic returns to education, or externalities.
- Hence, the rate of return calculations from education is concerned only with economic, monetary or pecuniary yields of the investment. It is often replaced even by a narrowly defined proxy of “earning differentials” which is inapplicable where earning differentials are not based on productivity or educational differences among workers.
- 2. Rate of return calculation does not adjust for the probability of serious unemployment problems.
- 3. The assumed education-earning relationship which was driven by the human capital theory and base for the rate of return calculation is complex especially in developing countries. Some argued that employers are more interested in the affective than cognitive skills while others argued education is used as screening device than for productivity.
- 4. It is based on the extensive use of cross sectional data. Rate of return calculations may have predictive merits, *ceteris paribus*, if income and cost structures remained stable in the future or the plan period (which is hardly possible to occur). It is reasonable to assume that the longer the plan period, the less useful are the rate of return calculation as guide to educational policy because of the higher probability of instability.

- Generalizations and implications
- 1. Length of income stream: other things being equal, the larger the stream of post investment incremental earnings, the more likely the net present value of an investment in human capital will be positive. The longer the earnings stream, the higher the internal rate of return.
- 2. Costs: other things being equal, the lower the cost of human capital investment, the larger the number of people who will find that investment to be profitable.
- 3. Earnings differential: other things being equal, the higher the college-high school earning differentials, the larger the number of people who will invest in a college education. The college wage premium is the ratio of the earnings of college graduates to the earning of high school graduates.
- Why peoples invest in different amounts of human capital and realize different earnings?
- (i) Differences in ability
- (ii) Differing degrees of uncertainty concerning the capacity to transform skills and knowledge in to enhanced earnings due to discrimination;
- (iii) Differing accesses to borrowed funds for human capital investment (this is due to capital market imperfections).
- Note: the first two factors work through the demand side of human capital market where as the third works through the supply side. Imperfections in the capital market may bias investment toward physical capital than human capital.

- 6.2. The Wage-Gap among Workers
- • Wages can vary systematically across:
 - o jobs with different workplace amenities (compensating wage differentials)
 - o people with different levels of education or skill
 - o occupations, industries and regions
 - o the public versus private sectors
- Compensating Wage Differential
- Higher wage rates compensate workers for undesirable working conditions, such as an unsafe or unhealthy work environment or undesirable working hours (the midnight shift), and for incurring additional costs of employment. Wages will be higher for some individuals to compensate them for doing unpleasant jobs, while others will willingly accept a lower wage for jobs with more amenities
- Workplace safety can be used to illustrate the theory of compensating wage differentials; the risk of injury or illness that can result from an unsafe or unhealthy work environment gives rise to compensating wage differentials. Obviously, workers would like more of both wages and safety, but at any level of utility workers are willing to accept some additional risk in exchange for a higher wage rate
- But not all workers have the same attitudes toward workplace risk and different workers will put different values on workplace safety. A more risk-averse individual requires a larger compensating wage differential in return for giving up some safety and accepting a riskier work environment.
- Firms can choose their production technology to offer workers greater safety or they can economize on safety and offer the savings to workers in the form of higher wages; for any firm, there will generally be a tradeoff in offering more safety or higher wages, holding constant the level of profits

- Efficiency Wages and Inter industry Wage Differences
- - while barriers to mobility can often explain persistent wage gaps across sectors, inter industry wage differentials may also be generated by firms rationally paying efficiency wages above the market-clearing levels so as to improve morale, reduce turnover and absenteeism, elicit effort, reduce shirking, reduce the threat of unionization, and to have a queue of applicants so as to reduce recruiting and hiring costs
 - o it may be efficient for firms to pay wages above the market-clearing wage because of the positive effects of higher wages on labour productivity; such profit-maximizing wage differentials will not be eliminated by the forces of competition
 - given that the desirability of paying efficiency wages will differ by industry, persistent efficiency wage differentials can arise
 - o industries that utilize expensive capital equipment which requires lengthy labour training programs may find it profitable to pay efficiency wages to have a trustworthy work force and low turnover
 - some industries consistently pay higher wages than appear necessary; despite queues of workers who would like to switch out of low-wage industries, firms in high-wage industries are unwilling to cut wages and increase employment because it would reduce worker productivity

- Public-Private Sector Wage Differentials
- - the public sector, which includes the government as well as government-owned or funded firms and agencies (such as hospitals, schools, colleges, universities)
 - an important policy question is how much public employees should be paid, and whether through their unions and political pressure public sector employees are able to extract excessive wages
 - given that non-pecuniary factors, such as job security and fringe benefits, are likely greater in the public sector than in the private sector, the public sector might be expected to have lower wages to compensate for these positive non-pecuniary benefits
 - unions are able to negotiate higher wages for their members and thus wages in the highly unionized public sector likely will exceed wages in the less unionized private sector
 - in addition, there may be a substantial positive union impact on wages in the public sector because the demand for labour in the public sector may be more inelastic (since most public sector services are not provided in the private sector) than in the private sector and public sector unions can bargain for substantial wage increases without worrying about large reductions in employment

- • the political constraint in the public sector may be less binding than the profit constraint in the private sector
- o there may not be much pressure on public sector managers to limit the size of wage increases; taxpayers are diffuse and can exert pressure only infrequently and with respect to a package of issues, and future taxpayers have little or no representation in today's political process
- o in contrast to the private sector, during a strike in the public sector tax revenues keep rolling even though wage expenditures are reduced (there is a fiscal surplus available to finance union wage demands)
- o the forces of competition ensure a floor wage in the public sector (to recruit additional labour) but not an effective ceiling on wages in the public sector; if public sector wages exceed private sector wages (resulting in a queue of applicants for public sector jobs), public sector managers are not under a profit constraint to respond to these disequilibrium signals and lower the relative wage

- On the Job Training
- Training: the process of teaching new employees the basic skills they need to perform their jobs.
- There are two methods for imparting training, i.e. on-the-job training and off-the-job training. As their name suggests, the former refers to the training imparted at the actual job location involving 'hands-on' training, whereas the latter method involves giving training to the employees at a place other than the real job location.
- The basic differences between on-the-job and off-the-job training are indicated below:
 - The training method used to impart training to employees while he is at the workplace performing the job is known as On-the-job training. Off-the-job training involves the training of employees outside the job location.
 - On the job training has a practical approach, whereas off the job training is theoretical.
 - On the job training involves the hands-on experience of the trainees which is not with the case of off the job training.
 - On the job training is more time consuming as compared to off the job training.
 - 'On the job' training involves learning by doing the task while 'off the job' training involves learning by acquiring knowledge.
 - In on the job training, there is no work disruption as the training and production go hand in hand. Conversely, in off the job training, the work is hampered because first of all training is provided which is followed by the actual performance.
 - On-the-job training is carried out by the experienced employees. Unlike off-the-job training which is provided by the experts.
 - On the job training is less expensive than off the job training.
 - On the job training is appropriate for manufacturing entities while off the job training is suitable for non-manufacturing entities.

- General training refers to the creation of skill or characteristics which are equally usable in all firms and industries.
- - enhances the productivity of workers to all firms
 - the worker will pay for general training through the lower wages during the training period
- Specific training refers to a training which can be used only in the particular firm which provides that training.
- - Increases the worker productivity only in the firm providing that training.
 - The firm must bear the cost of specific training.
- A. Firm Specific Job Training
- - Assume that an untrained individual can always obtain a job which pays a wage of W_a based on a value of the marginal product of labour equal to VMP_a for an untrained worker.
 - a firm hiring an untrained worker has to decide whether to train this worker to increase the worker's value of the marginal product from VMP_a to VMP^*

- - there are obvious fixed (sunk) costs to training a worker: during the training period the worker's marginal product VM_{Pt} will be quite low, probably lower than the wage paid W_a
- - o the lower VM_{Pt} of a job trainee also reflects the costs of the supervisory personnel conducting the training sessions
- - But once the worker is trained, the value of the marginal product will be much higher. the firm incurs a cost during the training period ($VM_{Pt} < W_a$) to obtain future benefits ($VMP^* > \text{wage paid to a trained employee}$)
- - to ensure that the employee does not quit after the firm has incurred the sunk costs of job training, the firm will likely pay a higher wage W^* than W_a .
- - o the trained employee also benefits from the specific job training, receiving a wage premium over the alternative wage W_a available at other firms (since specific job training is not useful at other firms, the employee's value to other firms is only VMP_a)
- - a profit-maximizing firm will provide specific on-the-job training, with the fixed costs of training financed out of the future profits from the trained employee ($VMP^* > W^*$)
- - o $VMP^* > W^*$ provides a buffer which reduces the probability that a trained employee will be laid-off during a recession

- General Job Training
- - if a worker receives general job training which increases the value of the marginal product of labour from VMP_a to VMP^* for all firms, other firms will be willing to pay the trained worker a wage equal to VMP^*
 - to retain a worker with general job training, the firm must pay the worker a wage equal to VMP^* (otherwise some other firm will poach the worker away) and the firm will not be able to finance the cost of general job training out of future profits from a trained employee (the firm has to pay a wage equal to VMP^* to retain its trained workers)
 - competition in the labour market ensures that the worker reaps all of the benefits of general on-the-job training in the form of higher earnings
 - thus, general on-the-the job training will only occur if the employee pays for it; a firm will only provide general job training if the worker receiving the general training pays for it in the form of a low wage during the training period
 - institutional arrangements (such as minimum wage laws and union contracts) may prevent firms from paying a low wage equal to $VMPT$ during a training period and no general on-the-job training may occur
 - the market system may fail to provide the optimum amount of general on-the-job training and government-financed on-the-job training programs may be required
 - the evaluation of public programs requires a carefully constructed control group since a comparison of trainees earnings before and after the training program may overstate the benefits of training

- Manpower Planning Techniques
- How to determine the future training needs of the labor market in developing countries is a question that has confronted manpower analysts and educational planners for decades. There is no easy solution simply because no-one can forecast the future and, therefore, what labor demands are likely anymore than one can predict stock market movements or future economic growth rates. This has not stopped people from trying. However, models to perform manpower analyses have been subject to such scathing criticism that, as will be seen in this chapter, manpower practitioners have shied away from modelling techniques and as such there is a gap to be filled. As now a combination of techniques under the general heading of "labour market signalling" have become the accepted methods, in recent years, to assess manpower needs. However, few countries have created a system to do this and there is much theorizing but little action.
- Employment planning is concerned with the macro policy instruments that create employment. Manpower planning, on the other hand, is largely concerned with labour supply. Thus it is interested in such questions as how many people are coming onto the labour market, what are their education and training levels, what is their age etc. It is largely concerned in determining what training needs there are so that the labour supply can be shaped to meet the demands of the economy.
- In this chapter, techniques that look at both the supply and demand side of the employment and manpower planning puzzle are presented.

- The manpower requirements approach (MRA)
- The dominant model of manpower planning (according to Youdi, 1985) is what is known as the 'manpower-requirements' approach or model. It first came to widespread prominence in the OECD's Mediterranean Regional Project (MRP) in the early 1960s. The three major steps in manpower forecasting are: (a) projecting the demand for educated manpower, (b) projecting the supply of educated manpower, and (c) balancing supply and demand.
- 2. Rate of return approach
- This is based on the calculation of the net returns on educational expenditure (ILO, 1984), measured as the increase in net income that an individual will be able to command throughout his/her life in relation to the income he/she would have received if he/she had not reached a given educational level.
- For each specific educational programme, the present value of the flow of future net income is calculated on the basis of the above definition. Those programmes which show positive returns should be promoted, while those showing zero or negative net present value should be reduced or possibly abandoned.
- If the flow of net income is calculated as the difference between the income of the individuals benefited after tax, and the costs include both the direct costs paid for the education and the indirect in terms of income not earned because of participation in educational programmes, for a given discount rate this gives the private rate of return. If the income is calculated before payment of tax and the costs include all the resources utilised to implement the education programme, for a given discount rate this gives the social rate of return.

- Labour market information systems
- The seeming failure of both former approaches has led some authors to concentrate on the preparation and organisation of labour in Labour Market Information Systems (LMIS) as an 'alternative' to forecasting - see for example Mason (1979). And as Richter (1989) notes 'labour market information means nothing more nor less than what it says - information about labour markets'. Clearly, the mere collection of data sets without the sort of guide provided by a model is ridiculous. The publications are potentially useful, in a developing country context, to delineate the main variables of interest for manpower planning and to arrive at consistent definitions. Most of them do not do this. Indeed, at best they present a shopping list of items to be collected without providing an analytical framework within which to collect and then to analyze data for planning or policy formulation. They suffer, in a mirror image, the criticisms of manpower planning forecasts in that manpower forecasts suffer from lack of appropriate data whereas labour market information systems suffer from the lack of an analytical framework.
- 4. Cybernetic and pragmatic approaches
- The concerns expressed above about the rate of return approach, manpower requirements, LMIS etc. have left a space yet to be filled. As Hough (undated) notes, "there is a fundamental problem with any attempt to combine cost-benefit analysis with manpower planning" but observes that Dougherty (1971) showed that when the standard cost-benefit approach is modified to allow for relative wage levels to change over time, it is possible to incorporate the effects of the growth of the education system on the growth of each category of labour and thence on future wage rates. Hough continues "one of the rare attempts to embody RoR and manpower planning took place in Cyprus, using an eclectic approach which focuses more on particular forms of education and particular occupational and industrial employment categories, rather than combining all relevant factors into a single model

- Key informants
- Large scale statistical systems with regular (monthly or quarterly) surveys of labour market phenomena are essential for LMIS's. Except in a very few rich countries such systems hardly exist in the world and this is particularly so for the developing world. Richter (1982) has suggested an innovative scheme to provide labour market information through the use of key informants. The idea is to collect selected community level data through key informants using the local knowledge of particular categories of respondents public officials, teachers, businessmen, large farmers etc. This is much simpler and cheaper than a household survey and vastly cheaper than a census. Once reliable key informants have been identified the informants are questioned on a panel basis. The information includes both general purpose questions about the overall situation and longer term trends of labour markets plus specific questions related to short term movements and fluctuations in labour supply and demand.
- 6. Labor market signaling
- For short-term assessment of training needs labor market signaling (LMS) is recommended as a tool that could both help, and improve, training centers' ability to respond quickly to changes in market circumstances and, thereby, reduce inefficiencies(see Middleton,1993).
- "Labor market signaling requires planners to focus on education and training qualifications rather than on occupational classifications. The reasons relate to the quality of occupational statistics, the effect of technology on the concept of an occupation, and the practical link between academic specialization and occupational placements." (Middleton, 1993, p. 140)
- This quote of is important because it underlines the basic theoretical model of the World Bank, and most international agencies who tend to take World Bank pronouncements as the status quo. Particularly revealing is the emphasis on education and training qualifications rather than occupations. However, nowhere in the Bank work is a list of skills defined that can be used as a basis for analysis. This lacuna means that no real alternative to an occupational classification has been suggested. Obviously, a fixed occupational classification over the years will ignore technological changes, however there is no reason why these cannot be factored in - this is what the most advanced country, the USA, does in practice.

- Labour accounting matrices (LAM's)
- Grootaert (1986) describes in some detail how to build up a LAM and in his matrix one component of a LAM is that showing headcounts and/or hours worked. In the rows are given 'Types of Individuals' and these correspond to the population and labour supply. 'Types of Individuals' can be disaggregated into In the labour force/out of labour force, each further divided into sex, education and age group. The total, of course, sums to the total population of the nation. Clearly it could be further disaggregated into urban/rural areas.
- Next, households are grouped in columns under 'Types of Households' to allow the construction of socioeconomic groups that have relevance with respect to economic and social policy issues. There is quite a range of criteria to describe households and to group them. For some purposes characteristics of the household as a unit are appropriate, such as race or ethnic background, ownership of productive inputs, income level, household size and composition, dependency condition of the household etc. As an alternative, or jointly with household characteristics, some features of the household head or main income earner can be used to describe the household, viz. sex, educational attainment, occupation, industry, employment status, earnings etc. Most censuses will allow the construction of such a block.

- The second building block describes the basic economic status of the individual members of the population. Under the heading 'Productive Activities' a first major dividing line could be whether or not an activity results in economic output, i.e. contributes to GDP according to the UN definition. It is possible, of course, to deviate from the UN guidelines to include all nonmarket activities the UN only includes selected nonmarket activities. This could then be subdivided into a classification scheme such as the ISIC scheme, then further subdivided into formal/nonformal activities and/or public/private. Individuals could also be distinguished by the number of hours they work or some other definition of employed/underemployed. To turn the LAM into a manpower matrix individuals could be further divided into educational or occupational categories. The LAM is then a manpower matrix because it indicates the number (or number of hours) that each type of person is, on the average, available for work. The matrix of activities can either be a supply or a demand for labour statement and shows the numbers each type of individual spends being engaged in productive activities and the number of hours that are lost due to unemployment.
- 'Other activities' could include unemployment, nonmarket activities, schooling, retirement, homemaking etc. These could be relocated under the heading of 'productive activities' should it be so desired. Note that the left hand side of the matrix 'Types of Individuals' provides a number of difficulties for the allocations in 'Productive (or Other) Activities'. For example, identifying those in the informal sector is difficult enough without the added complication of age/sex disaggregations.

- Under 'productive activities' the seasonality dimension could be added to the labour force matrices. It could be available as both a head count table and a table with number of hours, this set of tables would permit the analysis of the seasonal pattern of job availability. It could indicate which type of jobs open up during which season, and how many hours of work they provide and which types of individuals (age/sex/education) obtain all year round jobs versus seasonal jobs. It could give information about hours of work during peak and slack seasons which is useful to estimate the opportunity cost of time.
- In order to examine the relations between economic variables and the LAM we might wish to link our LAM into a SAM. To do this we need to add a number of other items to the matrix. The remaining factors of production, land and capital goods need to be introduced. Since these factors are not only owned by households, but also by firms and by the government, columns for these institutions need to be added to the accounting scheme. Again the accounting can be accomplished in headcount, hours worked or value. To complete a LAM to a full SAM, links with the rest of the world, transfers within and across institutions, demand for commodities, an input output table and current and capital accounts must be added. An example of an application of a LAM in Kenya can be found in Vandemoortele (1986). He describes how, using a SAM in a comparative statics framework, an exogenous change in the distribution of income can be shown to affect employment and output. Since a LAM is a subset of a SAM the model also applies to LAM's.
- The first step in developing a SAM model is to separate the endogenous accounts from the exogenous ones so that the impact of the latter on the former can be measured. The endogenous accounts generally comprise the factor accounts, the accounts for households and companies (the endogenous institutions) and the accounts for the production activities. The exogenous accounts include the government account, the investment account, the accounts for indirect taxes and international transactions.
- There are, of course, overlaps between each of the techniques suggested but none of the approaches has come to dominate in practice.

Chapter Seven

Labour Policy